

GLOBAL CREDIT TRADER

State of Play for Leveraged Finance Defaults

As we highlighted recently, absent a sharp deterioration in the growth backdrop, we believe the peak in aggregate default activity in the syndicated credit market (leveraged loans and HY bonds) is likely behind us. That said, under the surface, there are several key default-related themes for corporate credit investors to monitor—consistent with the broader theme of market dispersion.

In this week's *Global Credit Trader*, we examine the state of play for defaults in the US and European leveraged finance markets and highlight several key themes.

1. **The direction:** The absolute level of defaults has been trending higher from the very low levels that persisted in 2022/23 but remains near historical median levels. The one exception is the floating rate broadly syndicated loan (BSL) market, where defaults peaked in late 2024 and have been declining.
2. **Structures:** Distressed exchanges, or liability management exercises (LMEs), remain the vehicle of choice for reorganizing the liability side of a firm's balance sheet. However, these are increasingly leading to a 'repeat defaulter' phenomenon.
3. **Ultimate losses:** Recovery rates have continued to trend lower—extending the longer-term pattern. We partially attribute this dynamic to the rise of repeat defaults and 'loan-only' capital structures.
4. **Sector patterns:** In contrast to the prevailing market narrative related to potential disruption of Software-focused firms, the sector composition mix still skews heavily towards capital industries and consumer-facing businesses.
5. **The view ahead:** The growth backdrop is the key signpost to watch, in our view. Looking ahead, none of the metrics we monitor signal a near-term uptick in expected defaults in the US. In Europe, we still envision a modest increase in defaults through year-end, owing to a more challenging growth, inflation, monetary policy mix. That said, we still expect defaults to remain below the local peak. The loan market is more vulnerable than HY, in our view.

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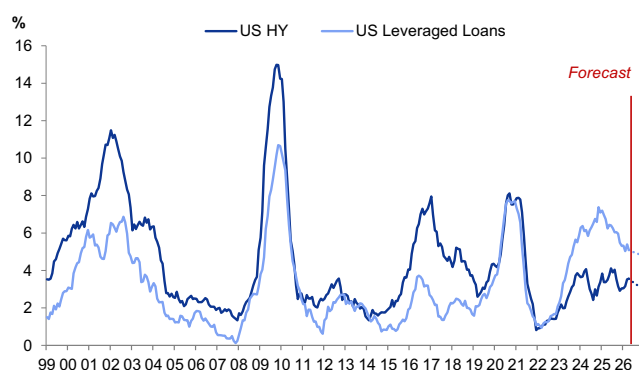
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State of Play for Leveraged Finance Defaults

Over the past several years, defaults have exhibited divergent trends across regions and asset classes. Exhibits 1 and 2 illustrate the 12-month trailing, issuer-weighted default rates in the US and European HY and leveraged loan markets (inclusive of distressed exchanges). While US HY defaults have been oscillating in a narrow band between 3% and 4% for the last couple of years, the US BSL market has demonstrated a discernible deceleration from the funding-cost-driven spike of 2023 and 2024, with the default rate falling to 5.1% from a peak of 7.4%. In Europe, both HY bond and BSL defaults show signs of trending higher, consistent with a more challenging growth, inflation, and monetary policy mix related to the ongoing commodity market disruptions.

Exhibit 1: US HY defaults have oscillated in a narrow band while US BSL defaults are still decelerating from their recent peak

12-month trailing issuer-weighted default rates in the US HY bond and leveraged loan markets

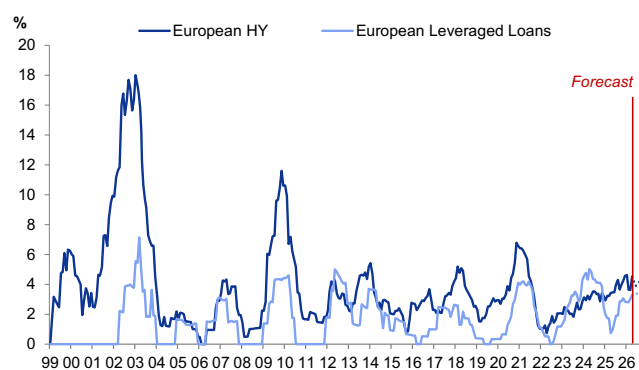


Note: As of April 2026 (most recent as of May 14, 2026).

Source: Moody's, Goldman Sachs Global investment Research

Exhibit 2: The path of European defaults is likely higher owing to a challenging growth, inflation and policy mix

12-month trailing issuer-weighted default rates in the European HY bond and leveraged loan markets

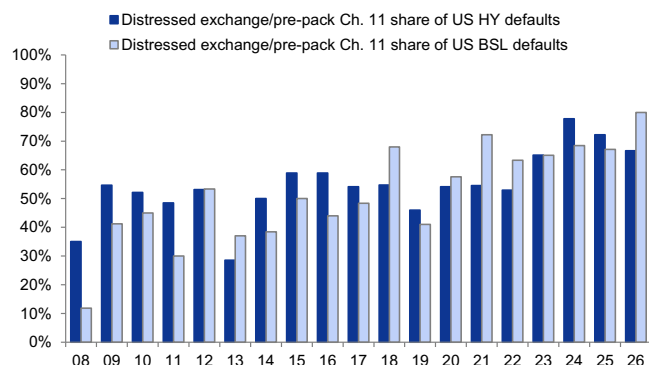


Note: As of April 2026 (most recent as of May 14, 2026).

Source: Moody's, Goldman Sachs Global Investment Research

A defining theme of the current cycle is the structural shift away from traditional in-court restructurings and towards distressed exchanges and, to a lesser extent, pre-packaged Chapter 11s. As Exhibits 3 and 4 highlight, the shares of defaults that were classified as distressed exchanges or pre-packaged Chapter 11 filings have surged in recent years and are currently hovering in the 70-80% range, depending on the market. On average, we find that LMEs tend to preserve recovery values better than in-court restructurings initially—which often makes such transactions the preferred path for private equity sponsors. For example, in 2025, US HY senior unsecured recovery rates for distressed exchanges averaged around 60% vs. roughly 10% for the handful of 'hard' defaults that occurred.

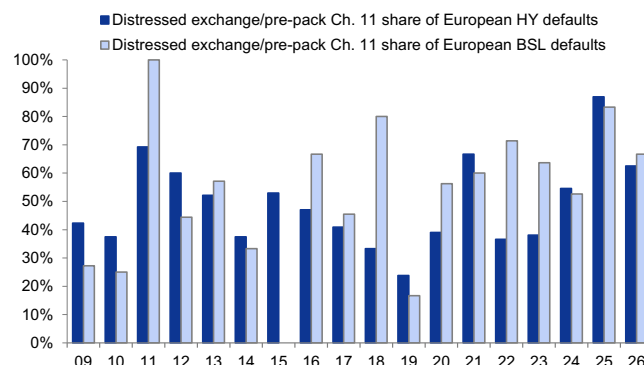
Exhibit 3: The LME share of US defaults continues to rise
Annual share of defaults in the US HY and BSL markets classified as distressed exchanges or 'pre-pack' Chapter 11 filings



Note: As of April 2026 (most recent as of May 14, 2026).

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 4: The distressed exchange share of European defaults is also elevated
Annual share of defaults in the European HY and BSL markets classified as distressed exchanges or 'pre-pack' Chapter 11 filings



Note: As of April 2026 (most recent as of May 14, 2026).

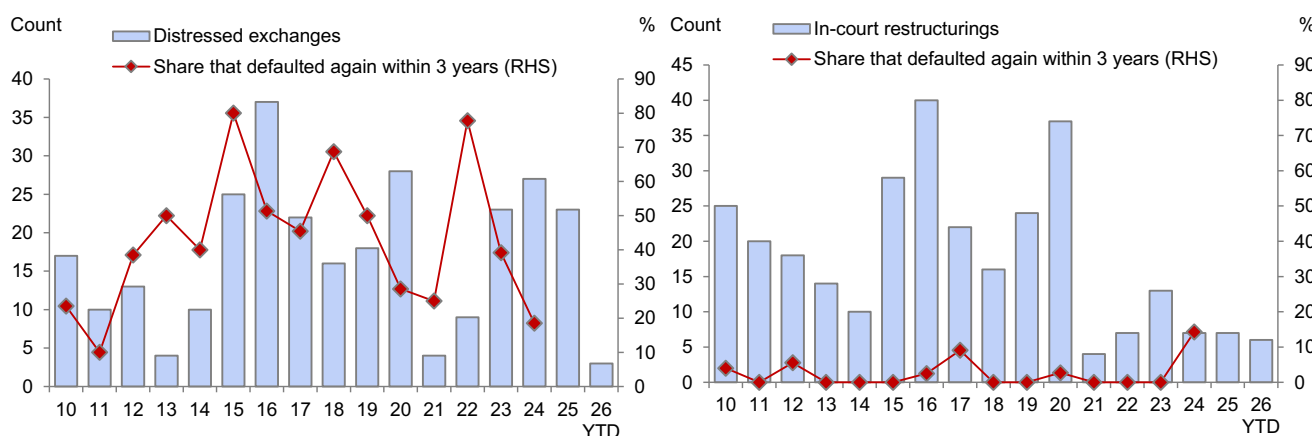
Source: Moody's, Goldman Sachs Global Investment Research

The dominance of LMEs comes with an important caveat: they are often an iterative process, and for many corporate borrowers a single distressed exchange fails to place their debt burden on a sustainable path. This 'incomplete' balance sheet relief results in a visible trend of 'repeat defaults.'

Exhibits 5 through 8 show the annual share of repeat defaults (across various markets) involving distressed exchanges vs. in-court restructurings, using a three-year window to define a repeat default. The main takeaway is that in-court restructurings are rarely followed by a repeat default, while distressed exchanges lead to repeat defaults roughly 50% of the time. As we have documented in previous research, these iterative defaults systematically erode recovery outcomes with each successive restructuring.

Exhibit 5: LMEs are often iterative and lead to 'repeat defaults' much more often than in-court restructurings...

Share of US HY distressed exchanges (left panel) and in-court restructurings (right panel) in each year that defaulted again within three years

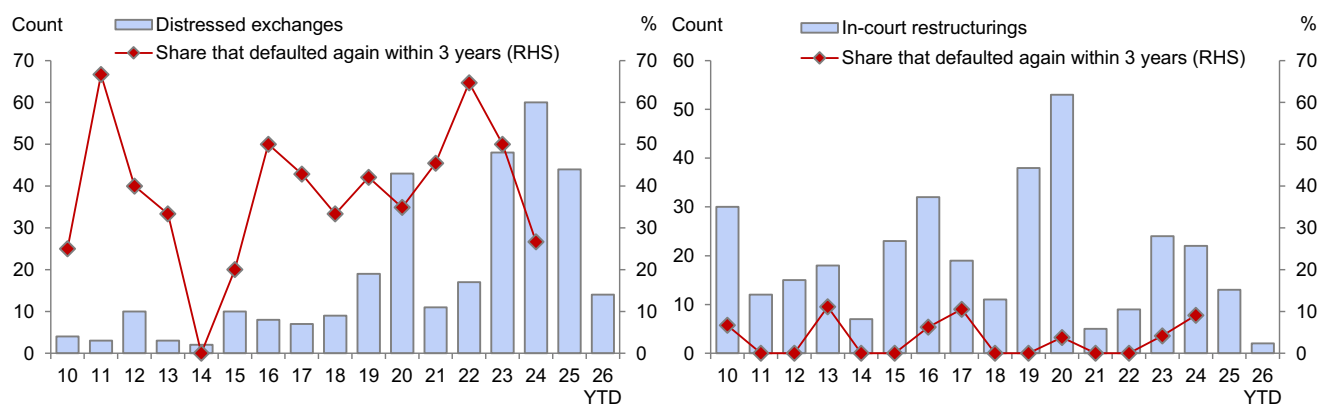


Note: As of April 2026 (most recent as of May 14, 2026).

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 6: ...the same is true in the broadly syndicated loan market

Share of US leveraged loan distressed exchanges (left panel) and in-court restructurings (right panel) in each year that defaulted again within three years

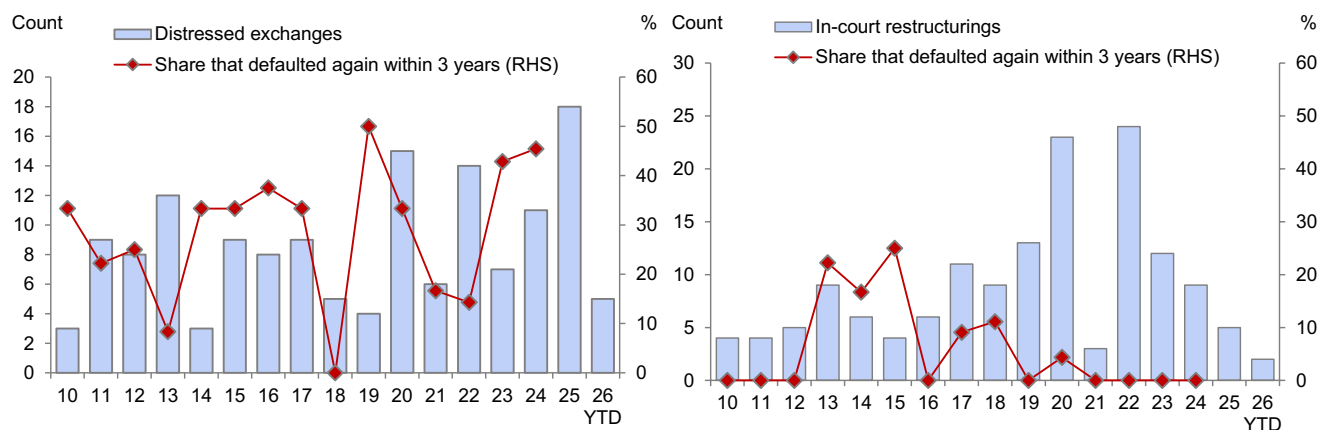


Note: As of April 2026 (most recent as of May 14, 2026).

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 7: The same pattern holds in European HY...

Share of European HY distressed exchanges (left panel) and in-court restructurings (right panel) in each year that defaulted again within three years

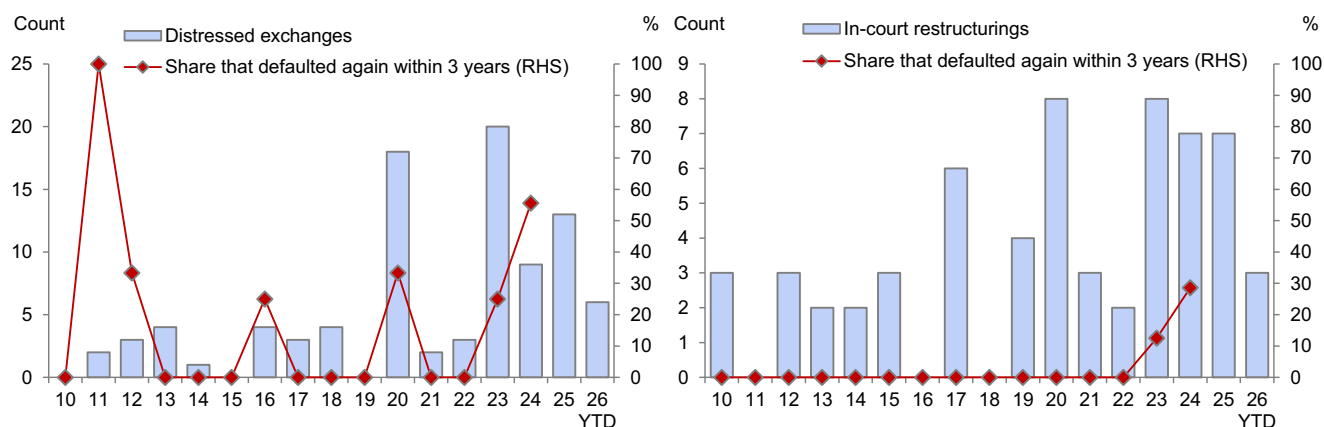


Note: As of April 2026 (most recent as of May 14, 2026).

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 8: ...and in the European leveraged loan market

Share of European leveraged loan distressed exchanges (left panel) and in-court restructurings (right panel) in each year that defaulted again within three years



Note: As of April 2026 (most recent as of May 14, 2026).

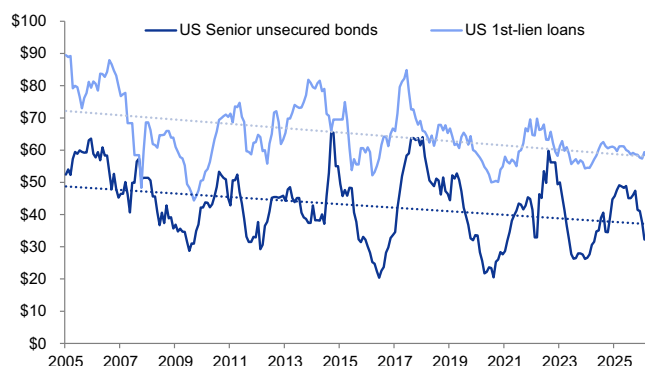
Source: Moody's, Goldman Sachs Global Investment Research

The rise of repeat default activity has coincided somewhat with a broader aggregate decline in recovery values in the USD market (in Europe, where the data is more sparse, we have not observed a similar trend). Exhibit 9 tracks the 12-month trailing recovery rate for US senior unsecured bonds vs. 1st lien loans over the past 20 years. While the acceleration in repeat defaults is likely one contributing factor to this erosion in average recovery values, we would also point to two other drivers.

First, the proliferation of 'loan-only' capital structures has reduced the subordination cushion that HY bonds used to provide in 'mixed' capital structures that had both loans and bonds outstanding. We estimate that over 70% of loans are now issued by loan-only capital structures, up from just 45% in 2010. Second, as shown in Exhibit 10, the US economy has structurally transitioned from asset-heavy industries (e.g., manufacturing) toward an asset-light, information and services-based economy. In this regime, there are fewer tangible assets available for liquidation in the case of bankruptcy—which leads to lower recovery values.

Exhibit 9: Recovery values have been steadily eroding, in aggregate

12-month trailing recovery given default for US senior unsecured bonds and 1st lien loans

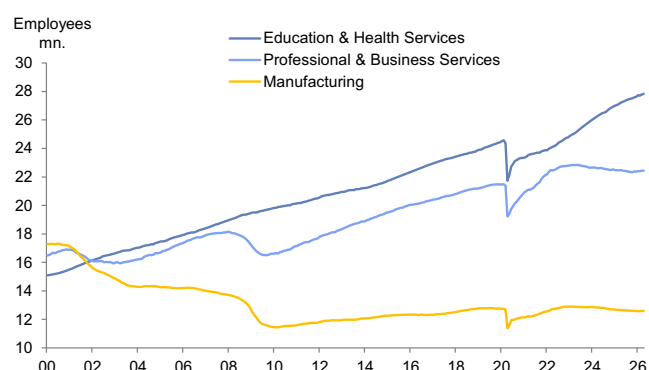


Note: As of March 2026 (most recent as of May 14, 2026).

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 10: Asset-light sectors have grown to dominate the US economy compared to asset-heavy sectors such as Manufacturing

Individuals employed (in millions) in the US economy, for select sectors



Note: As of April 2026 (most recent available as of May 14, 2026).

Source: Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

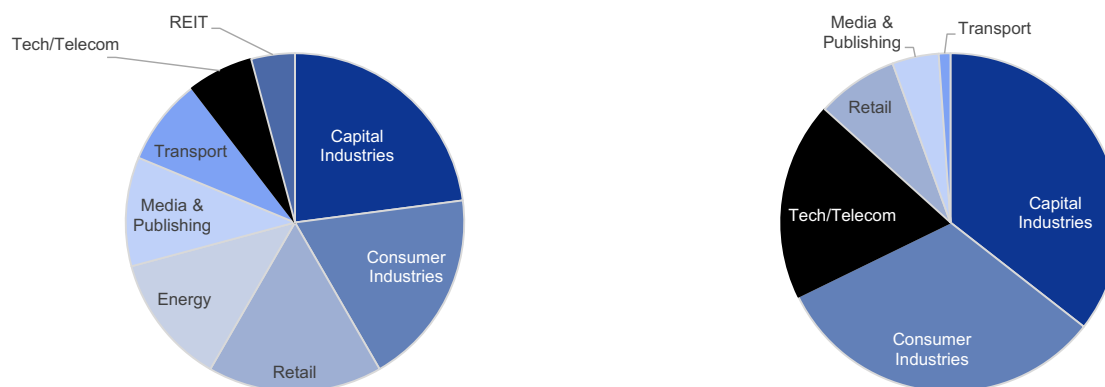
Many market participants are focused on the magnitude and timing of the potential disruption to Software business models from innovation in artificial intelligence (AI). So far, however, it has not yet become evident in the underlying default data. Exhibits 11 and 12 categorize defaults over the last 12 months by industry across the US and European HY and loan markets.

The clear theme, in our view, is that defaults have mostly been concentrated among Capital Industries (Chemicals, Packaging, Automotive, Equipment, etc.) and Consumer Industries (Healthcare, Food & Beverage, Lodging/Leisure, etc.). In our view, this pattern has been driven by a mix of factors, including structurally higher debt servicing costs (at least relative to much of the post-financial crisis era), shifts in consumer purchasing preferences, and cost input pressures.

Notably, however, we have not yet observed a significant uptick in Software-related defaults. Our Software equity research team has outlined that many software firms still have competitive moats and that the narrative around disruption in the sector is nuanced, depending on a range of factors. That said, we expect this to nonetheless remain an overhang on the sector, as uncertainty around the pace and scope of AI-related innovation will make it difficult to 'disprove' a thesis of disruption. Should significant, near-term disruption to the Software sector materialize, it would present upside risk to our loan default forecasts in the US and Europe given the large weight of the sector in both regions.

Exhibit 11: Recent defaults have been concentrated in capital industries and consumer-facing sectors...

Share of US HY (left panel) and leveraged loan (right panel) defaults over the last 12-months, by broad sector

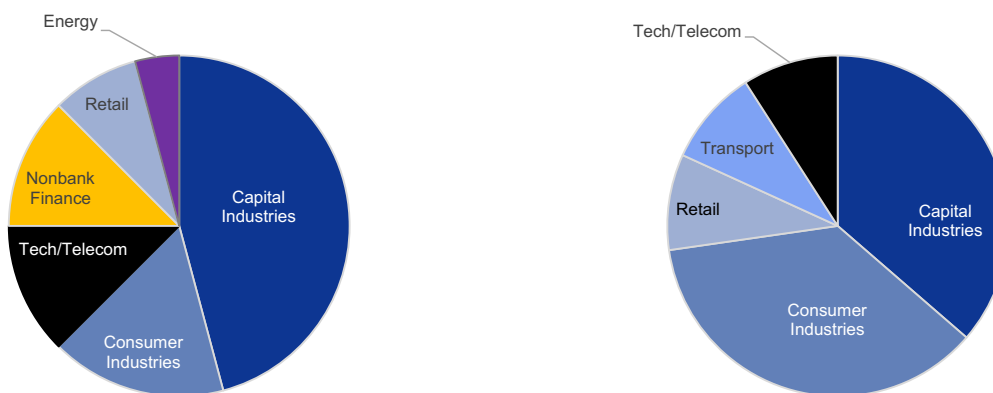


Note: As of April 2026 (most recent as of May 14, 2026).

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 12: ...and the same has been true in European defaults

Share of European HY (left panel) and leveraged loan (right panel) defaults over the last 12-months, by broad sector



Note: As of April 2026 (most recent as of May 14, 2026).

Source: Moody's, Goldman Sachs Global Investment Research

Looking ahead, three key factors inform our default expectations in the US and Europe: (1) the combination of a weaker (yet non-recessionary) growth backdrop—with particular softness in Europe, (2) scope for profit margin deterioration in commodity-reliant sectors, and (3) headwinds to borrowing costs for floating rate issuers (or as in the case of the US, little scope for further near-term improvement).

We expect US HY and BSL defaults to modestly decelerate into year-end from 3.6% and 5.1% currently, to 3.0% and 4.75%, respectively. In Europe, given the weaker growth backdrop and greater disruption from higher oil and gas prices, we expect HY and loan default rates to continue to rise modestly into year-end, from 4.5% and 3.3% currently, to 4.7% and 4.0%, respectively. While increases in defaults are unwelcome by investors, the context around these figures is important: across all forecasts, defaults are well-within a normal range compared to the historical long-term averages.

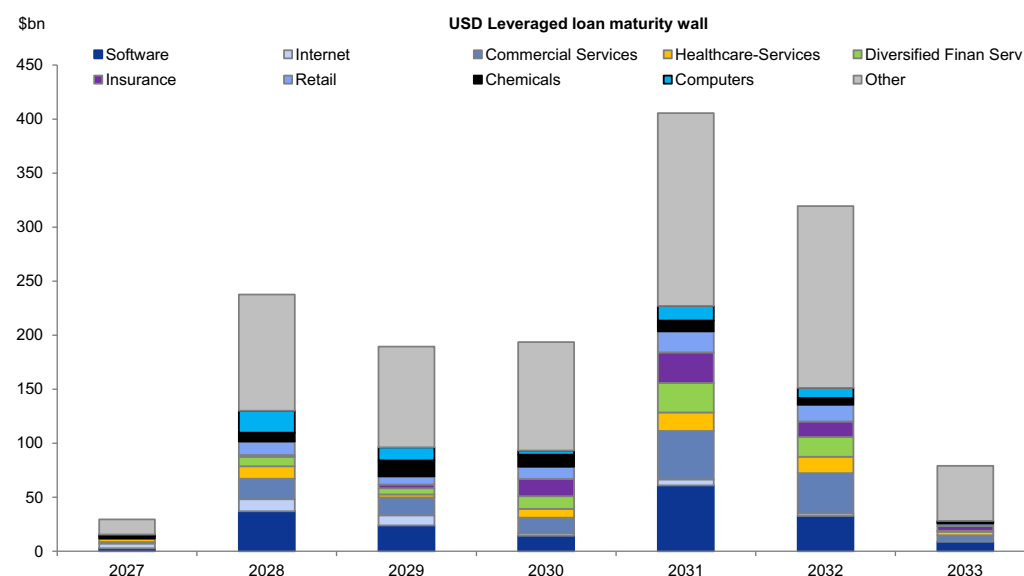
We see three risks to our baseline default outlook. First and foremost is the growth backdrop. Modestly below-trend growth is generally a 'good enough' environment for

credit—especially given the solid starting point for corporate fundamentals. A sharp deterioration in growth, however, would unquestionably place upward pressure on default rates (and, by extension, realized losses). Second, both geopolitical and policy risk remain elevated, which could lead to an increase in defaults across affected sectors. That said, in this scenario, we would expect more *dispersion*, as opposed to *widespread market disruption*. Third, as noted above, AI-driven disruption and structural headwinds in the Software sector pose a distinct upside risk to our default projections. However, this is likely to provide headwinds to some sectors while potentially offering tailwinds to other industries (underscoring our focus on dispersion).

While many market participants are focused on near-term maturity walls, as we recently highlighted, they are generally not outsized for the US and European HY bond markets. Moreover, a large portion of the upcoming maturities is represented by the higher-rated cohorts of the HY universes. We believe this reflects a choice by management teams to keep older, lower-cost debt on the balance sheet as long as possible before refinancing into a higher cost of debt—as opposed to an ‘inability’ to refinance.

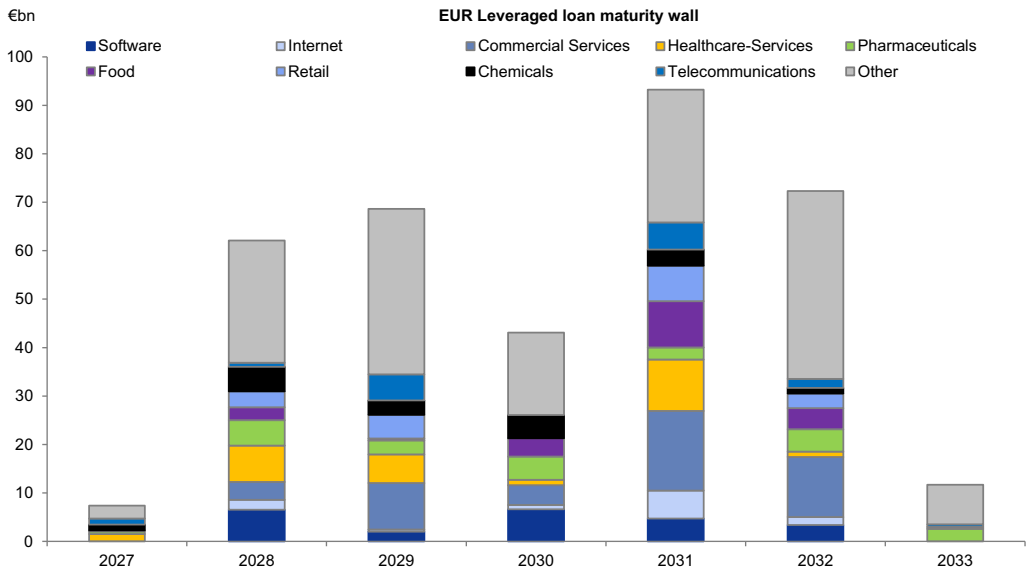
That said, the dynamics in the leveraged loan market are different, as borrowers have encountered higher debt service costs in tandem with the rate hikes over the course of 2022-2023. As a result, the growth in scheduled maturities over the next two years (Exhibits 13 and 14) warrants close monitoring—especially given the loan market’s weight toward Software and dependence on the demand from CLOs.

Exhibit 13: The near-term USD leveraged loan scheduled maturity wall is elevated...
USD leveraged loans notional maturing by sector



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 14: ...as is the near-term EUR leveraged loan maturity wall
EUR leveraged loans notional maturing by sector



Source: Bloomberg, Goldman Sachs Global Investment Research

Thank you to Patricia Pacheco for her contribution to this publication.

Forecast dashboard: Spreads, returns, issuance, defaults, and ratings migrations

Exhibit 15: USD and EUR spread forecasts

Index	Updated as of May. 13, 2026			
	Current	2026Q2	2026Q3	2026Q4
USD spreads				
USD IG	78	95	92	90
USD Fin	81	103	98	94
USD Non-Fin	75	92	89	86
USD HY	264	335	325	315
EUR spreads				
EUR IG	91	120	117	114
EUR Fin	96	127	122	118
EUR Non-Fin	88	116	113	111
EUR HY	273	375	368	360

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 17: Issuance forecasts

Market	2026 YTD (bn)	2026 full-year forecast (bn)
USD market		
Gross issuance		
IG	\$975	\$2,100
HY	\$136	\$400
Leveraged loans	\$148	--
Net issuance		
IG	\$440	\$850
HY	\$61	\$160
EUR market		
Gross issuance		
IG	€ 360	€ 850
HY	€ 43	€ 125
Net issuance		
IG	€ 82	€ 265
HY	€ 24	€ 60

Source: Dealogic, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 16: USD and EUR excess returns forecast

Index	Updated as of May. 13, 2026		
	Current duration	2026	2026 full-year forecast (%)
USD excess returns			
USD IG	6.8	0.4%	0.0%
USD HY	2.9	1.0%	1.3%
EUR excess returns			
EUR IG	4.6	0.3%	0.0%
EUR HY	2.7	0.2%	-0.4%

Source: Bloomberg, ICE-BAML, iBoxx, , Goldman Sachs Global Investment Research

Exhibit 18: Ratings migration and default forecasts

Default rates are issuer-weighted and 12-month trailing

Market	Updated as of May. 13, 2026	
	2026 YTD	2026 full-year forecast
USD market		
Rising stars	\$36bn	\$40bn
Fallen angels	\$31bn	\$60bn
HY defaults (12-month trailing)	3.5%	3.0%
Leveraged loan defaults (12-month trailing)	5.4%	4.75%
EUR market		
Rising stars	€ 5bn	€20bn
Fallen angels	€ 4bn	€30bn
HY defaults (12-month trailing)	3.6%	4.7%
Leveraged loan defaults (12-month trailing)	3.0%	4.0%

Source: Bloomberg, Moody's, Goldman Sachs Global Investment Research

Forecast dashboard: Relative value views

Exhibit 19: Relative value views

Relative value views		
Corporate Credit		
Regional	Overweight USD vs. EUR	The growth, inflation and monetary policy mix is somewhat more favorable in the US, relative to Europe. As a result, and relative to the USD market, we forecast a wider peak (and a slower recovery in 2H2026) in EUR credit spreads.
	Overweight IG vs. HY	We have a slight preference for IG vs. HY in the USD market, owing to valuations and our economists' forecasts for below-potential growth.
	Overweight BBBs vs. higher-rated cohorts	We prefer owning BBBs over the higher-rated segments of USD IG (As and AAs), as a way to capture additional spread. We also see scope for balance sheet deterioration at the highest end of the rating spectrum in IG.
USD market	Overweight BBs, Neutral CCCs, Underweight Bs	Current spread levels indicate that single-Bs offer the lowest credit risk premium relative to expected losses among all HY rating buckets.
	Overweight HY bonds vs. loans	HY bonds still provide a superior trade-off between carry and duration compared to leveraged loans. In addition, loans' substantial software sector exposure is likely to present a sentiment overhang, at least in the near-term.
	Overweight CLO mezz tranches vs. leveraged loans Neutral CLO vs. USD IG Neutral CLO vs. CMOs	For floating rate investors, we prefer CLO mezzanine tranches vs. leveraged loans. Historical data points to low impairment rates for BB and BBB CLO tranches, and current valuations offer attractive carry and a cushion against moderate to elevated default outcomes. Elsewhere, we are neutral between senior CLO tranches and USD IG; we would use spread widening to add exposure in CLOs. We are also neutral between CLO AAA and CMO floaters.
	Prefer Agency MBS vs. IG	Agency MBS offers an attractive diversification benefit vs. a portfolio of IG corporate bonds.
	Overweight IG vs. HY	We prefer IG over HY given the more challenging growth, inflation, and monetary policy mix in the Euro Area.
EUR market	Overweight BBBs vs. As	In the EUR market, we prefer As over BBBs, as the weaker growth backdrop warrants more of an up-in-quality stance. That said, we would not move too far up the quality spectrum, as we expect supply technicals will be a headwind in the EUR AA market.
	Overweight HY bonds vs. loans	Similar to the US, we prefer HY bonds given the loan market's weighting toward software (and the potential negative sentiment overhang). The reaction function of the ECB may also favor floating rate credit, given the central bank's focus on inflationary pressures.
	Overweight IG 10s vs. 2s	We have a slight preference for extending duration in EUR IG to optimize carry, but have scaled back this approach.
	Overweight BBs, Neutral Bs, Underweight CCCs	At current valuations and given our expectation for heightened dispersion, we prefer moving up in quality within EUR HY.
	Underweight AT1s vs. EUR HY	Given downside tail risks and a limited valuation cushion, we recommend an underweight on AT1s vs. EUR HY.

Source: Goldman Sachs Global Investment Research

Earnings calendar

Exhibit 20: USD IG earnings report dates

Upcoming earnings report dates for the Bloomberg USD IG Index (LUACTRUU). Minimum \$1 billion notional value and including up to the top 20 capital structures in the index reporting that day.

Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)
5/15/2026			5/18/2026			5/19/2026			5/20/2026		
MUFG	Mitsubishi UFJ Financial Group	41.8	BIDU	Baidu Inc	2.9	KEYS	Keysight Technologies Inc	1.9	NVDA	NVIDIA Corp	7.5
MIZUHO	Mizuho Financial Group Inc	24.7				EXP	Eagle Materials Inc	1.5	ADI	Analog Devices Inc	7.2
						TOL	Toll Brothers Inc	1.3	INTU	Intuit Inc	4.3
									TJX	TJX Cos Inc/The	1.9
									NDSN	Nordson Corp	1.5
5/21/2026			5/22/2026			5/25/2026			5/26/2026		
WMT	Walmart Inc	31.5	BAH	Booz Allen Hamilton Holding Cx	1.3				AZO	AutoZone Inc	7.3
DE	Deere & Co	31.2									
BRITEL	BT Group PLC	3.4									
TTWO	Take-Two Interactive Software	1.9									
FLO	Flowers Foods Inc	1.3									
RL	Ralph Lauren Corp	1.3									

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 21: USD HY earnings report dates

Upcoming earnings report dates for the Bloomberg USD HY index (LF98TRUU). Minimum \$500 million notional value and including up to the top 20 capital structures in the index reporting that day.

Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)
5/15/2026			5/18/2026			5/19/2026			5/20/2026		
KIOXIA	Kioxia Holdings Corp	2.2							VFC	VF Corp	1.3
GOLF	Misto Holdings Corp	0.5							ENS	EnerSys	0.6
RBC	RBC Bearings Inc	0.5									
5/21/2026			5/22/2026			5/25/2026			5/26/2026		
AAP	Advance Auto Parts Inc	3.4	HNDLIN	Hindalco Industries Ltd	3.9						
WMS	Advanced Drainage Systems In	1.0									
HOV	Hovnanian Enterprises Inc	0.9									
BRITEL	BT Group PLC	0.5									

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 22: EUR IG earnings report dates

Upcoming earnings report dates for the Bloomberg EUR IG Index (LECPTRUU). Minimum €1 billion notional value and including up to the top 20 capital structures in the index reporting that day.

Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)
5/15/2026			5/18/2026			5/19/2026			5/20/2026		
MIZUHO	Mizuho Financial Group Inc	13.2				ENXFP	Euronext NV	2.6	SVTLN	Severn Trent PLC	3.1
MUFG	Mitsubishi UFJ Financial Group	4.9							EXPNLN	Experian PLC	1.7
OTPHB	OTP Bank Nyrt	2.3							MSINS	MS&AD Insurance Group Holdi	1.4
SYENS	Syensqo SA	1.2									
MITHCC	Mitsubishi HC Capital Inc	1.0									
5/21/2026			5/22/2026			5/25/2026			5/26/2026		
GIM	Generali	10.1	CFRVX	Cie Financiere Richemont SA	4.5						
BRITEL	BT Group PLC	7.8	TVLRO	Banca Transilvania SA	1.5						
DE	Deere & Co	3.9									
EZJLN	easyJet PLC	2.1									
INVPLN	Investec PLC	1.0									
SGELN	Sage Group PLC/The	1.0									
WMT	Walmart Inc	1.0									

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 23: EUR HY earnings report dates

Upcoming earnings report dates for the Bloomberg Pan-Euro HY index (LP01TREU). Minimum €100 million notional value and including up to the top 20 capital structures in the index reporting that day.

Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)
<u>5/20/2026</u>			<u>5/21/2026</u>			<u>5/22/2026</u>			<u>5/25/2026</u>		
VFC	VF Corp	1.5	BRITEL	BT Group PLC	2.0	HNDLIN	Hindalco Industries Ltd	0.5			
ELIOR	Elior Group SA	0.7	ASMDEE	Asmodee Group AB	0.6						
			TATTGA	CrediaBank SA	0.2						
<u>5/26/2026</u>			<u>5/27/2026</u>			<u>5/28/2026</u>			<u>5/29/2026</u>		
			ARNDTN	Aroundtown SA	2.1	ALPHA	Alpha Bank SA	1.0			
			GYCGR	Aroundtown SA	1.2						

Source: Bloomberg, Goldman Sachs Global Investment Research

Performance dashboard: Cash and synthetic markets**Exhibit 24: Performance of USD and EUR markets**

Market	Current*		Total return			Spread change (YTD,bp)	Yield change (YTD, %)
	Spread	Yield	MTD	QTD	YTD		
USD Corporate bonds							
IG	75	5.2	0.11	0.56	0.02	-3	0.36
AA	49	4.9	-0.09	0.14	-0.17	0	0.33
A	63	5.0	0.04	0.53	0.13	-2	0.31
BBB	96	5.4	0.13	0.85	0.37	-5	0.30
HY	267	7.1	-0.05	1.65	1.14	1	0.52
BB	171	6.3	0.02	1.49	1.10	2	0.31
B	310	7.8	0.00	1.82	1.43	9	0.44
CCC	931	13.7	-0.71	2.03	-0.22	46	0.72
USD Leveraged Loans							
Leveraged loan index	437	8.2	0.00	0.09	-0.46	42	0.4
BB	257	6.3	0.00	0.00	0.70	2	-0.1
B	448	8.3	0.00	0.00	-0.93	57	0.5
CCC	1877	23.9	0.00	0.00	-5.09	314	3.3
USD Securitized products							
Agency MBS	21	5.0	-0.11	-0.04	0.36	-1	0.3
ABS (Autos)	63	4.5	0.07	0.40	1.07	-3	0.3
ABS (Credit cards)	32	4.2	0.02	0.31	0.83	-1	0.3
ABS (Student loans)	100	4.9	0.16	0.43	1.52	127	-0.1
USD Macro indices							
CDX IG	53	--	0.07	0.52	0.29	3	--
CDX HY	329	--	0.20	2.92	1.06	13	--
EUR Corporate bonds							
IG	93	3.8	-0.10	0.85	-0.17	3	0.4
AA	66	3.7	-0.22	0.62	-0.32	5	0.4
A	85	3.8	-0.11	0.82	-0.17	1	0.3
BBB	106	4.0	-0.07	0.92	-0.13	5	0.4
HY	262	5.9	0.49	2.44	0.68	-8	0.3
AT1	274	5.6	0.42	2.82	1.43	-12	0.3
EUR Leveraged loans							
Leveraged loan index	497	7.2	0.00	0.18	-0.87	47	0.6
BB	282	5.0	0.00	0.00	0.75	-70	-0.6
B	471	6.9	0.00	0.00	-1.30	56	0.7
EUR Macro indices							
iTraxx Main	57	--	0.12	0.77	0.2	6	--
iTraxx Xover	279	--	0.76	3.85	1.2	34	--
iTraxx SenFin	60	--	0.15	0.95	0.4	6	--
iTraxx SubFin	97	--	0.31	1.85	0.9	4	--

*Data as of previous day close.

For USD IG and HY we use the Bloomberg indices with history back to Jun-89 and Jan-94, respectively. Corp rating level, Agency MBS and ABS we use ICE-BAML indices with history back to Dec-96, EUR IG we use the iBoxx QW5A index with history back to Jan-99, EUR HY we use the ICE-BAML index with history back to Dec-97, and USD and EUR leveraged loans we use the Morningstar LSTA LLI and ELLI indices with data going back to Jan-97 and Jan-02, respectively.

Source: Bloomberg, iBoxx, ICE-BAML, Pitchbook LCD, Goldman Sachs Global Investment Research

Performance dashboard: Sectors

Exhibit 25: Sector performance: USD IG

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	2%	175.3	77		5.3	7.4	BAA1/BAA2	-4	-5	-19	-12	-0.1	0.2	1.0	0.6
Chemicals	1%	77.3	84		5.4	7.7	BAA1/BAA2	-5	-6	-18	-14	0.0	0.3	1.0	1.0
Metals & Mining	1%	71.6	67		5.2	7.7	A3/BAA1	-3	-4	-22	-11	-0.2	0.1	1.1	0.4
Paper	0%	26.4	87		5.2	6.0	BAA2/BAA3	-2	-5	-19	-9	-0.1	0.2	0.9	0.1
Capital Goods	5%	409.5	58		5.0	6.8	A3/BAA1	-4	-5	-18	-9	-0.2	0.1	0.5	0.1
Aerospace & Defense	2%	150.7	65		5.2	8.3	BAA1/BAA2	-3	-5	-17	-9	-0.2	0.1	0.5	0.1
Building Materials	1%	40.0	72		5.1	6.6	BAA1/BAA2	-3	-5	-16	-5	-0.2	0.1	0.6	0.0
Diversified Manufacturing	2%	115.1	58		5.0	6.6	A3/BAA1	-4	-5	-17	-6	-0.2	0.1	0.4	0.1
Construction Machinery	1%	58.4	35		4.6	4.7	A1/A2	-4	-5	-22	-15	-0.2	0.0	0.3	0.3
Packaging	0%	14.3	73		4.9	4.6	BAA2/BAA3	-4	-5	-18	-7	-0.1	0.0	0.5	0.3
Environmental	0%	31.1	43		4.7	5.8	A3/A3	-4	-6	-19	-13	-0.2	0.0	0.2	0.1
Communications	8%	567.0	92		5.5	8.6	A3/BAA1	-3	-2	-8	-2	-0.2	0.0	0.1	-0.5
Cable & Satellite	1%	107.9	134		6.0	9.3	BAA1/BAA2	-1	0	-4	2	-0.5	-0.3	-0.2	-0.5
Media & Entertainment	2%	137.7	81		5.4	8.6	A1/A2	-5	5	-1	-4	-0.2	-0.2	-0.3	-1.9
Wireless	2%	125.0	78		5.2	7.3	BAA1/BAA2	-3	-5	-14	-4	-0.2	0.1	0.5	0.1
Wirelines	3%	196.4	87		5.5	9.0	BAA1/BAA2	-2	-3	-9	-3	-0.2	0.2	0.2	-0.2
Consumer Cyclical	7%	549.1	67		5.0	6.2	A2/A3	-4	-5	-18	-5	-0.1	0.1	0.5	0.0
Automotive	2%	169.5	85		5.0	4.3	A3/BAA1	-4	-6	-19	-4	-0.1	0.1	0.8	0.5
Leisure	0%	4.7	106		5.4	6.4	BAA2/BAA2	1	-2	-11	4	-0.5	-0.1	0.5	-0.9
Gaming	0%	23.6	105		5.3	4.7	BAA2/BAA3	-3	-6	-19	-5	0.0	0.2	1.1	0.5
Home Construction	0%	10.4	83		5.1	4.9	BAA1/BAA2	-2	-2	-15	-1	-0.2	0.0	0.6	0.2
Lodging	0%	20.0	72		4.9	4.8	BAA2/BAA3	-3	-6	-18	-2	-0.1	0.1	0.6	0.1
Retailers	3%	256.2	51		5.0	7.7	A1/A2	-5	-5	-17	-3	-0.2	0.0	0.2	-0.3
Restaurants	0%	35.0	57		5.1	7.5	BAA1/BAA2	-4	-7	-20	-10	-0.3	0.1	0.5	0.1
Consumer Cyc Services	0%	29.6	74		5.1	6.1	A3/BAA1	-6	-5	-13	4	-0.1	0.1	0.5	-0.2
Consumer Non-Cyclical	15%	1125.4	63		5.1	7.8	A2/A3	-3	-5	-16	-6	-0.2	0.1	0.4	-0.1
Consumer Products	1%	59.8	45		4.8	6.0	A2/A3	-3	-3	-18	-10	-0.2	0.0	0.4	0.2
Food & Beverage	3%	235.0	68		5.2	7.6	A3/BAA1	-3	-6	-18	-2	-0.3	0.2	0.7	-0.2
Healthcare	4%	331.2	72		5.2	7.8	A3/BAA1	-3	-5	-15	-6	-0.2	0.1	0.4	-0.1
Pharmaceuticals	5%	400.5	54		5.1	8.5	A1/A2	-4	-6	-16	-8	-0.2	0.1	0.3	-0.1
Supermarkets	0%	13.4	83		5.5	9.6	BAA1/BAA2	-2	-5	-13	-4	-0.4	0.2	0.5	-0.4
Tobacco	1%	85.5	70		5.1	6.6	A3/BAA1	-4	-6	-17	-10	-0.2	0.1	0.6	0.2
Energy	7%	534.1	76		5.3	7.6	A3/BAA1	-4	-6	-17	-15	-0.2	0.2	0.7	0.9
Independent Energy	1%	93.9	77		5.3	7.4	BAA1/BAA2	-5	-7	-18	-19	-0.1	0.2	0.9	1.5
Integrated Energy	1%	103.2	50		5.0	8.3	AA3/A1	-5	-5	-18	-15	-0.3	0.0	0.4	0.3
Oil Field Services	0%	25.2	73		5.2	7.6	A3/BAA1	-3	-7	-21	-24	-0.2	0.2	1.0	1.2
Refining	0%	30.3	86		5.3	7.2	BAA1/BAA2	-4	-5	-16	-19	-0.1	0.2	0.8	1.2
Midstream	4%	281.4	84		5.3	7.4	BAA1/BAA2	-4	-6	-16	-13	-0.1	0.2	0.8	0.8
Technology	10%	762.2	79		5.2	7.3	A2/A3	-4	-5	-14	4	-0.1	0.2	0.5	-0.2
Transportation	2%	150.5	68		5.3	9.0	A3/BAA1	-3	-4	-16	-6	-0.3	0.0	0.5	0.0
Airlines	0%	18.3	87		5.0	3.5	A3/BAA1	-3	-4	-15	-4	-0.1	0.0	0.6	0.5
Railroads	1%	90.2	62		5.4	10.8	A3/BAA1	-3	-4	-18	-8	-0.4	0.0	0.5	-0.2
Transportation Services	1%	42.0	72		5.2	7.8	A3/BAA1	-3	-1	-14	-3	-0.2	-0.1	0.4	0.1
Other Industrial	1%	38.7	61		5.3	10.3	AA3/A1	-3	-3	-13	-4	-0.4	-0.1	0.0	-0.3
Utility	10%	750.0	79		5.3	8.1	A3/BAA1	-3	-5	-15	-7	-0.2	0.1	0.5	0.0
Electric	9%	675.0	78		5.3	8.1	A3/BAA1	-3	-5	-15	-7	-0.2	0.1	0.5	0.0
Natural Gas	1%	58.7	88		5.4	7.8	A3/BAA1	-3	-4	-17	-9	-0.2	0.1	0.7	0.2
Other Utility	0%	16.3	67		5.2	8.2	BAA1/BAA2	-3	-5	-16	-4	-0.3	0.0	0.4	-0.3
Financial Institutions	32%	2394.5	78		5.1	5.4	A2/A3	-4	-5	-17	0	-0.1	0.1	0.7	0.0
Banking	22%	1636.0	72		5.0	4.9	A2/A3	-4	-5	-16	2	-0.1	0.1	0.6	0.0
Brokerage Asset Managers Exchanges	2%	117.4	77		5.1	6.2	A3/BAA1	-3	-7	-18	5	-0.2	0.2	0.9	-0.1
Finance Companies	1%	106.7	153		5.7	3.6	BAA2/BAA3	-1	-4	-16	8	-0.1	0.1	1.2	0.2
Insurance	4%	330.4	84		5.4	8.2	A3/BAA1	-4	-6	-21	-6	-0.1	0.2	1.0	-0.1
REITS	3%	196.5	72		5.0	5.4	A3/BAA1	-3	-6	-18	-5	-0.1	0.1	0.6	0.2
Other Financial	0%	7.5	92		5.1	4.7	A3/BAA1	-1	-4	-18	-9	-0.1	0.1	1.1	1.0
US Corporate IG		7456.3	75		5.2	6.9	A3/BAA1	-4	-5	-16	-4	-0.2	0.1	0.6	0.0

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 26: Sector performance: USD HY

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	6%	92.6	262		7.0	3.4	BA3/B1	-2	-3	-19	-10	-0.1	0.1	2.0	1.8
Chemicals	3%	47.5	237		6.8	3.4	BA3/B1	-2	-3	-21	-26	-0.2	0.0	2.0	3.4
Metals & Mining	3%	37.9	208		6.5	3.4	BA3/B1	-4	-6	-25	0	-0.1	0.2	2.3	1.0
Paper	0%	7.2	705		11.3	2.6	B1/B2	0	2	-1	41	0.1	0.3	0.9	-3.6
Capital Goods	10%	151.3	232		6.7	3.1	BA3/B1	1	1	-10	9	-0.2	-0.2	1.2	-0.6
Aerospace & Defense	2%	31.7	108		5.6	2.9	BA3/B1	-2	-2	-22	8	-0.1	0.1	1.2	1.0
Building Materials	2%	36.2	390		8.4	3.4	BA3/B1	3	4	-7	37	-0.5	-0.8	1.0	-2.5
Diversified Manufacturing	2%	27.9	171		6.1	2.5	BA2/BA3	5	2	-13	63	-0.1	-0.1	1.0	0.8
Construction Machinery	1%	21.7	128		5.7	3.0	BA3/B1	-3	-1	-30	-3	-0.1	0.0	1.7	1.4
Packaging	2%	25.1	349		7.9	3.5	B1/B2	-2	1	3	-16	-0.2	0.1	1.3	-3.5
Environmental	1%	8.8	146		5.9	3.1	BA3/B1	-1	0	-15	-1	-0.1	0.0	1.0	1.1
Communications	15%	219.2	402		8.4	3.7	B1/B2	2	5	-7	16	-0.5	-0.7	1.2	1.1
Cable & Satellite	5%	81.2	536		9.6	2.9	B1/B2	4	7	-1	34	-0.7	-1.4	-0.7	-0.2
Media & Entertainment	5%	80.1	321		7.6	4.2	B1/B2	3	5	-9	-4	-0.5	-0.5	2.3	1.1
Wireless	1%	19.9	259		7.0	4.1	BA3/B1	1	6	-19	3	-0.2	-0.4	2.2	0.9
Wirelines	3%	38.0	360		8.1	4.2	B1/B2	-1	-2	-13	15	0.0	0.2	2.4	3.5
Consumer Cyclical	18%	275.7	239		6.7	3.2	BA3/B1	2	0	-20	-5	-0.3	0.0	1.8	1.0
Automotive	3%	45.9	253		6.8	3.0	BA3/B1	1	0	-27	-8	-0.3	0.2	2.3	1.2
Leisure	2%	29.1	209		6.4	3.4	BA3/B1	5	4	-10	16	-0.6	-0.3	0.8	0.2
Gaming	3%	45.7	248		6.9	2.9	BA3/B1	0	-4	-19	2	-0.1	0.3	2.0	1.2
Home Construction	1%	18.9	200		6.3	3.0	BA3/B1	0	-3	-21	-1	0.0	0.1	1.8	1.2
Lodging	1%	16.4	134		5.7	3.9	BA2/BA3	6	2	-19	1	-0.3	-0.1	1.3	0.6
Retailers	4%	60.6	256		6.8	3.4	BA3/B1	6	4	-24	-22	-0.4	0.0	2.0	1.1
Restaurants	1%	16.5	149		5.7	3.3	BA3/B1	1	-3	-21	-9	-0.1	0.2	1.4	1.1
Consumer Cyc Services	3%	42.7	301		7.4	2.7	B1/B2	2	0	-14	9	-0.3	-0.1	1.9	0.9
Consumer Non-Cyclical	11%	158.2	247		6.8	3.3	B1/B2	-1	-1	-12	0	-0.2	0.0	1.6	1.5
Consumer Products	2%	22.8	332		7.6	4.0	BA3/B1	3	7	-7	2	-0.9	-1.2	0.7	-0.4
Food & Beverage	2%	28.9	173		6.1	3.4	B1/B2	3	2	-5	1	-0.3	-0.1	1.1	1.1
Healthcare	5%	70.9	220		6.6	2.9	B1/B2	-5	-5	-12	6	0.0	0.3	1.5	1.8
Pharmaceuticals	2%	25.4	365		8.0	3.1	B2/B3	-2	-2	-21	-17	0.0	0.2	3.8	3.7
Supermarkets	1%	9.8	160		5.9	3.6	BA2/BA3	2	4	-4	132	-0.2	-0.3	0.3	0.3
Tobacco	0%	0.3	197		6.5	3.4	BA3/BA3	-8	-13	-20	15	0.3	0.9	1.9	0.0
Energy	11%	165.6	150		6.0	3.0	BA2/BA3	-5	-1	-24	-44	0.1	0.1	1.4	4.0
Independent Energy	3%	45.1	163		6.1	2.6	BA3/B1	-5	6	-20	-35	0.1	-0.1	1.5	4.2
Oil Field Services	2%	26.7	151		6.3	2.7	BA3/B1	-4	7	-27	-50	0.0	-0.1	1.2	3.8
Refining	0%	5.6	228		6.9	2.3	BA3/B1	-4	-1	-8	-37	0.2	0.3	1.1	5.4
Midstream	6%	88.2	139		5.8	3.3	BA2/BA3	-6	-7	-26	-48	0.0	0.2	1.5	3.8
Technology	9%	140.6	344		7.8	3.1	BA3/B1	-1	-7	-16	9	0.0	0.7	1.9	0.2
Transportation	2%	31.6	444		8.8	2.8	BA3/B1	1	3	-12	23	-0.6	-0.2	1.5	-0.9
Airlines	1%	13.4	390		8.1	2.6	BA3/B1	3	2	-13	41	-0.8	-0.4	1.4	-1.0
Railroads	0%	1.9	1180		16.4	2.7	B2/B3	0	3	7	23	0.8	0.4	-6.6	-6.8
Transportation Services	1%	16.3	404		8.5	2.9	B1/B2	0	3	-15	15	-0.5	-0.2	2.7	0.0
Other Industrial	1%	15.3	216		6.6	3.1	BA3/B1	17	1	-17	-17	-0.6	-0.2	1.7	0.8
Utility	3%	44.7	172		6.1	4.2	BA2/BA3	4	3	-16	17	-0.3	0.0	1.6	1.7
Financial Institutions	13%	196.1	246		6.8	3.2	BA3/B1	0	-1	-24	10	-0.1	0.0	2.2	0.4
Banking	0%	3.1	187		6.2	4.4	BA1/BA2	-1	-1	-31	-13	-0.3	-0.1	2.6	-0.2
Brokerage Asset Managers Exchanges	1%	18.1	181		6.3	2.9	BA3/B1	0	-7	-32	-4	0.0	0.4	2.8	1.5
Finance Companies	4%	63.0	271		7.1	3.2	BA2/BA3	0	-1	-21	30	0.0	0.0	2.4	-0.6
Insurance	4%	64.4	222		6.6	3.5	BA3/B1	1	1	-24	16	-0.2	-0.2	1.7	0.1
REITS	2%	29.5	249		6.8	2.9	B1/B2	2	-2	-25	-8	-0.1	0.1	2.6	1.9
Other Financial	1%	18.0	308		7.4	2.9	BA3/B1	-4	-1	-17	-3	0.1	0.1	2.2	1.4
US Corp High Yield		1490.9	267		7.1	3.2	BA3/B1	1	0	-16	0	-0.2	0.0	1.6	1.1

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 27: Sector performance: EUR IG

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumer Goods	12.5	€ 386.4	88.1		3.8	4.6	A	-1	-1	-15	4	-0.43	-0.11	0.84	-0.25
Automobiles & Parts	5.1	€ 157.6	106.4		3.9	3.7	A	-1	-2	-20	5	-0.32	-0.02	0.97	-0.13
Beverages	2.9	€ 90.4	83.9		3.9	5.9	A	0	-1	-11	3	-0.55	-0.21	0.78	-0.37
Food Producers	1.9	€ 59.7	71.9		3.7	5.2	A	0	1	-11	2	-0.50	-0.16	0.70	-0.33
Household Goods	0.2	€ 7.2	53.6		3.6	5.4	AA	0	-2	-14	2	-0.52	-0.21	0.64	-0.35
Personal Goods	1.7	€ 51.7	68.1		3.6	4.4	AA	0	0	-15	5	-0.44	-0.12	0.76	-0.39
Tobacco	0.6	€ 19.9	114.0		4.0	4.8	BBB	-2	-1	-14	2	-0.38	-0.05	0.85	-0.12
Consumer Services	4.6	€ 142.0	89.6		3.9	5.2	A	-1	0	-10	7	-0.47	-0.14	0.64	-0.41
Food & Drug Retailers	0.7	€ 20.2	88.5		3.8	4.3	BBB	0	0	-14	3	-0.41	-0.09	0.81	-0.16
General Retailers	0.6	€ 18.6	94.6		4.1	6.9	AA	-1	1	-7	16	-0.57	-0.26	0.43	-0.67
Media	1.2	€ 35.8	94.5		3.8	4.0	BBB	0	-1	-16	11	-0.40	-0.09	0.85	-0.31
Travel & Leisure	2.2	€ 67.4	86.4		3.9	5.6	A	-1	-1	-9	5	-0.50	-0.15	0.54	-0.47
Energy	3.2	€ 97.4	96.4		3.9	4.9	A	-2	-2	-15	-1	-0.40	-0.03	0.93	0.15
Oil & Gas Producers	2.8	€ 87.2	96.7		3.9	4.8	A	-2	-3	-16	-2	-0.38	-0.02	0.94	0.15
Oil Equipment / Services & Distribution	0.2	€ 7.7	93.6		4.0	5.7	A	0	3	5	17	-0.54	-0.21	0.78	0.13
Renewable Energy	0.1	€ 2.5	95.3		3.8	5.0	BBB	1	1	-11	-6	-0.54	-0.11	0.72	0.30
Industrials	9.7	€ 298.0	83.3		3.7	4.5	A	0	0	-15	3	-0.46	-0.13	0.81	-0.24
Aerospace & Defense	0.4	€ 13.5	72.6		3.7	4.4	A	0	0	-13	8	-0.45	-0.21	0.69	-0.45
Construction & Materials	1.6	€ 49.9	85.7		3.8	4.3	BBB	0	0	-16	2	-0.43	-0.13	0.88	-0.14
Electronic & Electrical Equipment	0.9	€ 28.2	72.4		3.6	4.9	A	0	0	-15	2	-0.48	-0.12	0.82	-0.28
General Industrials	1.3	€ 39.2	74.4		3.7	5.3	A	0	0	-17	0	-0.53	-0.19	0.86	-0.32
Industrial Engineering	1.1	€ 34.8	80.4		3.6	3.4	A	0	-2	-17	3	-0.33	-0.04	0.86	-0.04
Industrial Transportation	3.3	€ 102.4	87.3		3.8	4.8	BBB	0	0	-13	3	-0.49	-0.15	0.73	-0.35
Support Services	1.0	€ 30.0	98.3		3.8	4.1	BBB	0	-1	-18	3	-0.41	-0.09	0.90	-0.05
Health Care	5.5	€ 169.3	80.4		3.9	5.8	A	0	0	-13	1	-0.56	-0.24	0.72	-0.25
Health Care Equipment & Services	2.4	€ 74.7	84.7		3.9	5.7	A	0	-2	-13	1	-0.55	-0.23	0.71	-0.30
Pharmaceuticals & Biotechnology	3.1	€ 94.5	77.1		3.8	5.9	A	0	1	-13	1	-0.56	-0.24	0.73	-0.20
Basic Materials	2.7	€ 84.0	86.9		3.8	4.8	A	0	0	-17	1	-0.47	-0.13	0.91	-0.15
Chemicals	2.0	€ 63.2	85.6		3.8	5.0	A	0	0	-17	2	-0.48	-0.13	0.90	-0.18
Forestry & Paper	0.2	€ 5.6	94.9		3.8	4.2	BBB	-1	0	-21	8	-0.38	-0.04	1.06	-0.35
Industrial Metals	0.1	€ 3.7	91.0		3.8	5.1	BBB	1	1	-18	-1	-0.56	-0.20	0.98	0.00
Mining	0.4	€ 11.5	90.2		3.8	4.1	BBB	1	0	-19	-6	-0.44	-0.12	0.90	0.10
Technology	2.5	€ 76.5	96.7		4.0	5.3	A	0	2	-13	10	-0.49	-0.28	0.77	-0.61
Software & Computer Services	2.0	€ 63.1	100.6		4.1	5.7	A	0	3	-12	11	-0.53	-0.33	0.73	-0.67
Technology Hardware & Equipment	0.4	€ 13.4	67.1		3.5	3.5	A	-1	-1	-21	7	-0.33	-0.04	0.94	-0.32
Telecommunications	5.5	€ 169.6	94.5		3.9	5.5	BBB	0	0	-13	3	-0.51	-0.19	0.76	-0.30
Fixed Line Telecommunications	4.4	€ 135.5	92.8		3.9	5.6	BBB	0	-1	-13	2	-0.51	-0.18	0.73	-0.30
Mobile Telecommunications	1.1	€ 34.0	101.5		4.0	5.4	BBB	0	4	-13	6	-0.50	-0.22	0.86	-0.31
Financials	42.5	€ 1,310.8	96.6		3.8	4.0	A	0	-2	-19	2	-0.40	-0.04	0.95	-0.06
Banks Sen	26.5	€ 819.5	85.7		3.7	3.9	A	0	-2	-17	2	-0.39	-0.03	0.85	-0.06
Banks Sub	4.1	€ 126.1	129.4		4.1	3.5	BBB	0	-4	-25	3	-0.36	0.05	1.13	0.14
Financial Services Sen	2.9	€ 89.0	85.0		3.8	4.6	A	0	0	-15	2	-0.44	-0.09	0.88	-0.04
Financial Services Sub	0.0	€ 1.0	117.3		3.8	2.1	A	-10	-11	-31	-21	0.00	0.27	0.98	0.84
Insurance Sen	1.5	€ 44.9	78.4		3.7	4.2	A	1	-1	-21	3	-0.46	-0.10	0.98	-0.21
Insurance Sub	2.6	€ 81.0	136.1		4.3	4.8	BBB	1	-3	-20	2	-0.51	-0.03	1.09	-0.09
Real Estate Sen	4.8	€ 148.9	114.7		4.1	4.7	BBB	0	0	-25	6	-0.44	-0.12	1.30	-0.26
Real Estate Sub	0.0	€ 0.5	156.1		4.5	5.8	A	-13	-25	-50	0	0.19	0.91	3.02	0.00
Utilities	11.4	€ 353.3	97.9		4.0	5.3	BBB	0	0	-11	4	-0.48	-0.15	0.74	-0.18
Electricity	6.9	€ 212.2	100.8		4.0	5.3	BBB	-1	-1	-12	4	-0.48	-0.13	0.75	-0.17
Gas / Water & Multiutilities	4.6	€ 141.2	93.5		3.9	5.2	BBB	0	1	-11	3	-0.50	-0.17	0.72	-0.18
Index	100.0	€ 3,087.3	92.5		3.9	4.6	A	0	-1	-16	3	-0.44	-0.10	0.85	-0.17

Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 28: Sector performance: EUR HY

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumers	27.9	€ 123.7	294		5.7	2.7	BB	-14	-25	-63	18	0.14	0.63	2.46	0.71
Consumer Goods	15.6	€ 68.9	264		5.4	2.7	BB	-12	-23	-56	28	0.09	0.54	2.44	0.86
Consumer Services	12.4	€ 54.8	332		6.1	2.7	BB	-15	-23	-61	13	0.20	0.74	2.48	0.54
Energy	9.9	€ 44.0	243		5.2	3.1	BB	-13	-23	-46	-15	0.09	0.36	1.51	0.63
Energy	2.3	€ 10.3	250		5.2	2.7	BB	-20	-21	-60	-51	0.24	0.46	1.88	1.96
Utilities	7.6	€ 33.8	241		5.2	3.2	BB	-11	-24	-43	-6	0.04	0.33	1.40	0.25
Industrials	30.5	€ 135.1	323		6.0	2.7	BB	-16	-24	-70	-21	0.32	0.85	2.72	1.45
Basic Materials	5.8	€ 25.7	404		6.8	2.7	B	-23	-54	-169	-159	0.76	1.72	5.62	5.45
Construction & Materials	1.2	€ 5.5	661		9.3	2.6	BB	-24	-28	-53	53	0.74	1.32	2.74	-0.71
Health Care	8.8	€ 38.8	282		5.6	2.4	B	-11	-16	-59	16	0.14	0.50	2.10	1.14
Industrials	14.7	€ 65.1	287		5.6	2.8	BB	-15	-20	-53	1	0.23	0.67	2.03	0.41
TMT	20.1	€ 88.9	314		6.0	3.4	BB	-13	-14	-52	28	0.19	0.57	2.30	0.23
Technology	3.9	€ 17.4	424		7.1	3.7	B	-17	-34	-62	46	0.38	0.68	2.27	-1.81
Telecommunications	16.1	€ 71.5	284		5.7	3.3	BB	-13	-11	-52	11	0.14	0.54	2.31	0.68
Financials	11.6	€ 51.3	361		6.4	3.2	BB	-2	-2	-51	34	-0.24	0.00	1.89	-0.25
Core Financials	4.3	€ 18.9	191		4.6	3.1	BB	-6	-15	-48	12	-0.11	0.25	1.59	0.11
Financial Services	1.6	€ 7.1	802		10.8	2.5	B	74	90	121	200	-1.95	-2.14	-1.31	-2.67
Real Estate	5.7	€ 25.3	386		6.6	3.4	BB	-14	-15	-79	8	0.16	0.42	3.05	0.21
Index	100.0	€ 443.0	310		5.9	2.9	BB	-12	-19	-60	7	0.16	0.58	2.35	0.73

0 50 100

Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: USD Credit ETFs

Exhibit 29: USD ETF performance

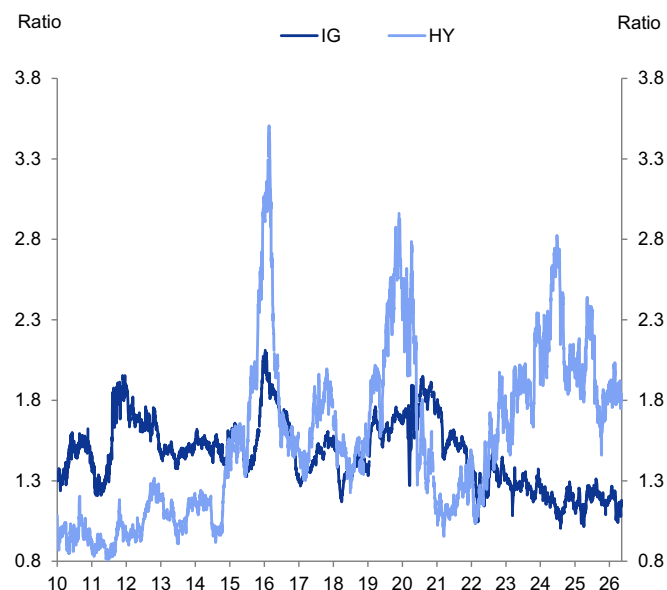
Sector	ETF Ticker	AUM	Weekly Flow (\$Mn.)	YTD cum flow (% of AUM)	Shares Outstanding (Mn.)	Short Interest (%)	Price	% NAV Premium	YTD Total Return	QTD Total Return	MTD Total Return	Benchmark Index
Investment Grade	LQD	\$29.57B	358	0.6%	272	15.25	108.55	0.07%	0.0	0.4	0.1	iBoxx USD Liquid Investment Grade Index
	VCSH	\$42.7B	126	22.0%	541	1.24	78.96	0.05%	0.5	0.4	0.0	Bloomberg Barclays US Corporate 1-5 Yrs Total Return Index Value Unhedged
	VCIT	\$65.86B	264	32.7%	798	2.34	82.46	0.08%	0.0	0.5	-0.1	Bloomberg Barclays US Credit Corporate 5-10 Yrs Total Return Index Value Unhedged
	FLOT	\$9.29B	31	22.5%	183	--	50.95	0.06%	1.6	0.7	0.2	Bloomberg Barclays US FRN < 5 yr Total Return Index Value Unhedged
	IGSB	\$22.06B	10	4.5%	421	0.63	52.36	0.00%	0.5	0.4	-0.1	ICE BofAML 1-5 Year US Corporate Bond Index
	IGIB	\$18.03B	98	19.7%	340	2.22	53.04	0.04%	0.0	0.5	-0.1	Bloomberg Barclays U.S. Interim
	SPSB	\$10.19B	84	22.5%	340	1.58	29.98	0.04%	0.7	0.4	0.0	Bloomberg Barclays US Corporate 1-3 Yrs Total Return Index Value Unhedged
	SPIB	\$11.28B	13	12.7%	338	2.24	33.41	0.02%	0.3	0.3	-0.1	Bloomberg Barclays Intermediate Corporate Total Return Index Value Unhedged
	VCLT	\$8.35B	-90	-38.5%	112	25.22	74.53	0.17%	0.0	0.7	0.3	Bloomberg Barclays Long U.S. Corporate Total Return Index Value Unhedged
High Yield	HYG	\$16.88B	95	13.4%	212	40.19	79.86	0.15%	1.0	1.4	-0.1	iBoxx USD Liquid High Yield Index
	JNK	\$7.69B	346	0.4%	80	9.98	96.12	0.11%	1.1	1.5	-0.2	Bloomberg Barclays VLI: High Yield Total Return Index Value Unhedged USD
	SJNK	\$4.76B	-37	2.7%	190	2.46	25.02	0.15%	1.1	1.3	-0.2	Bloomberg Barclays US High Yield 350M Cash 0-5yr 2% Cap Total Return Index
	SHYG	\$7.55B	30	22.9%	178	0.39	42.40	0.20%	1.2	1.4	-0.1	Markit iBoxx USD Liquid High Yield 0-5 Total Return Index
	HYS	\$1.72B	28	31.9%	18	0.06	93.29	-0.09%	0.8	1.2	-0.1	ICE BofAML 0-5 Year US High Yield Constrained Index
	HYLS	\$1.64B	2	0.7%	40	0.07	40.94	0.05%	0.0	1.5	0.0	ICE BofAML US High Yield Constrained Index
	ANGL	\$3.08B	0	-0.4%	106	0.42	29.01	0.00%	0.9	2.2	-0.3	ICE BofAML US Fallen Angel High Yield Index
	BKLN	\$7.15B	178	-20.8%	346	23.34	20.66	0.00%	0.4	1.7	0.4	S&P/LSTA U.S. Leveraged Loan 100 Index
Loans	SRLN	\$5.28B	0	-30.4%	130	7.91	40.59	-0.06%	0.7	2.3	0.5	Markit iBoxx USD Liquid Leverage Loans Total Return Index
	FTSL	\$2.3B	9	6.3%	51	0.87	45.20	0.10%	0.6	1.4	0.3	S&P/LSTA US Leveraged Loan 100 Market Value Index
EM	EMB	\$14.08B	-76	-4.4%	148	9.65	95.53	0.13%	1.0	2.6	0.1	J.P. Morgan EMBI Global Core Index
	EMLC	\$4.86B	0	78.0%	191	1.62	25.47	-0.01%	0.6	2.5	0.2	J.P. Morgan Government Bond Index Emerging Markets Global Core Index
CLOs	JAAA	\$27.12B	-127	63.8%	536	21.76	50.64	0.02%	1.7	0.9	0.2	
	PAAA	\$9.3B	123	447.6%	181	--	51.42	-0.01%	1.8	0.8	0.2	
	JBBB	\$1.18B	33	-13.1%	25	--	47.42	0.23%	1.4	2.3	0.1	
	CLOI	\$1.32B	0	66.8%	25	13.03	52.83	0.05%	1.7	1.1	0.4	
	CLOA	\$2.14B	47	228.3%	41	--	51.90	-0.04%	1.8	0.9	0.2	

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Performance dashboard: Bond-level dispersion

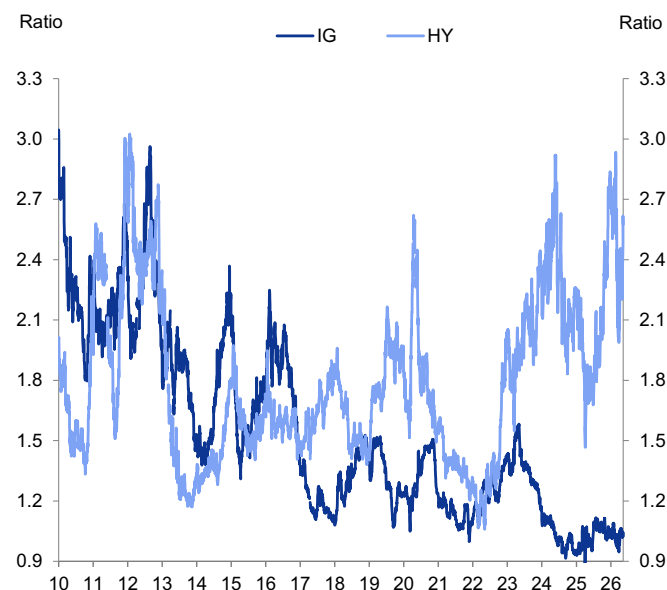
Note: Spread dispersion is the inter-decile range divided by the median across the bond index constituents and return dispersion is the inter-decile range of 10-day returns divided by the median across the bond index constituents.

Exhibit 30: USD spread dispersion



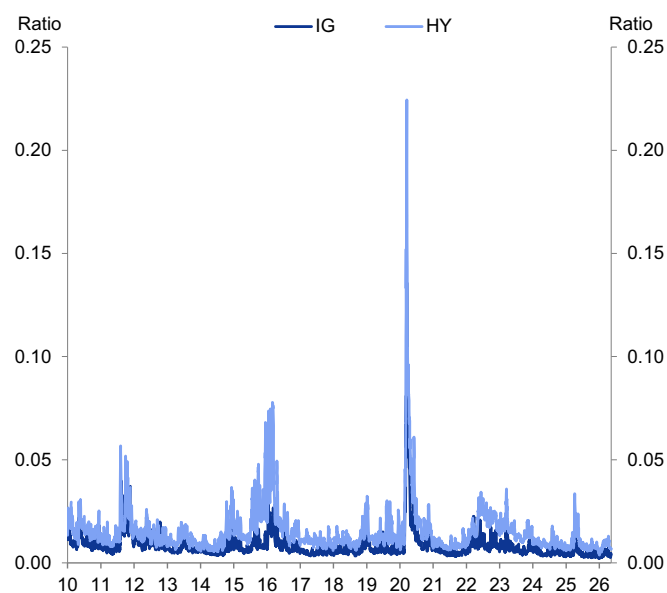
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 31: EUR spread dispersion



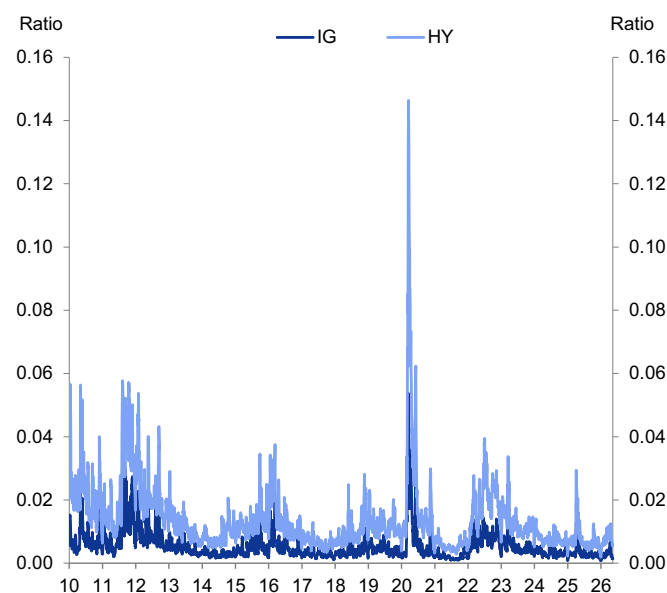
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 32: USD return dispersion



Source: iBoxx, Goldman Sachs Global Investment Research

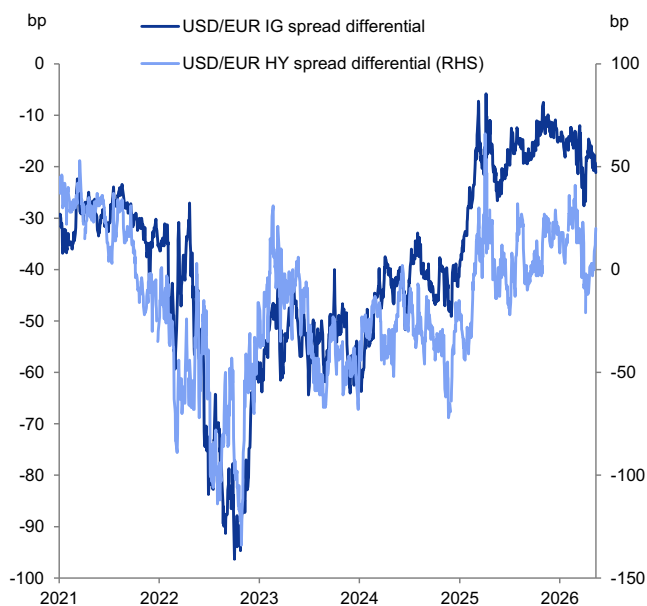
Exhibit 33: EUR return dispersion



Source: iBoxx, Goldman Sachs Global Investment Research

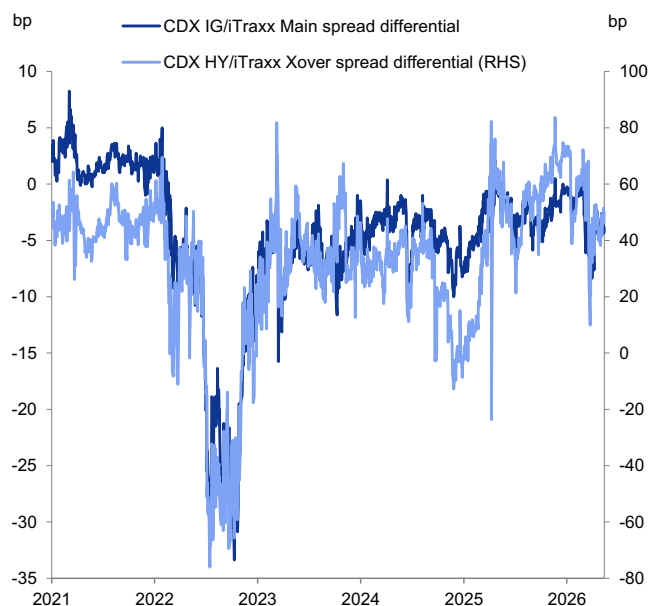
Relative value dashboard: Excess spreads in the USD and EUR markets

Exhibit 34: USD/EUR IG and HY spread differential



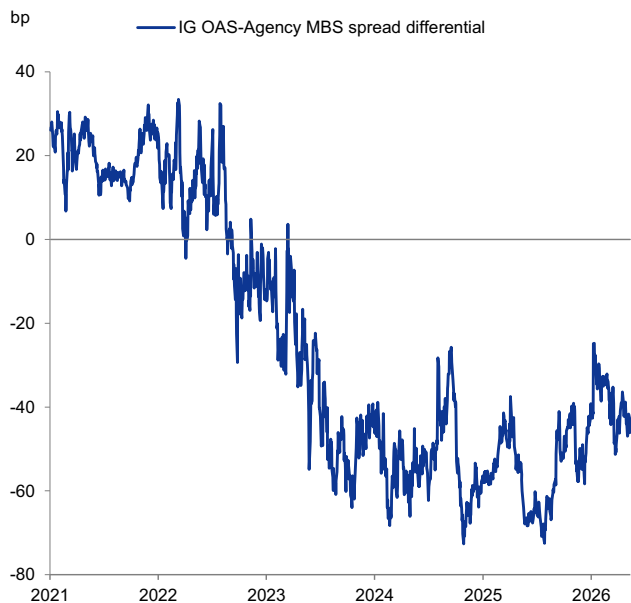
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 35: CDX IG/iTraxx Main and CDX HY/iTraxx Xover spread differential



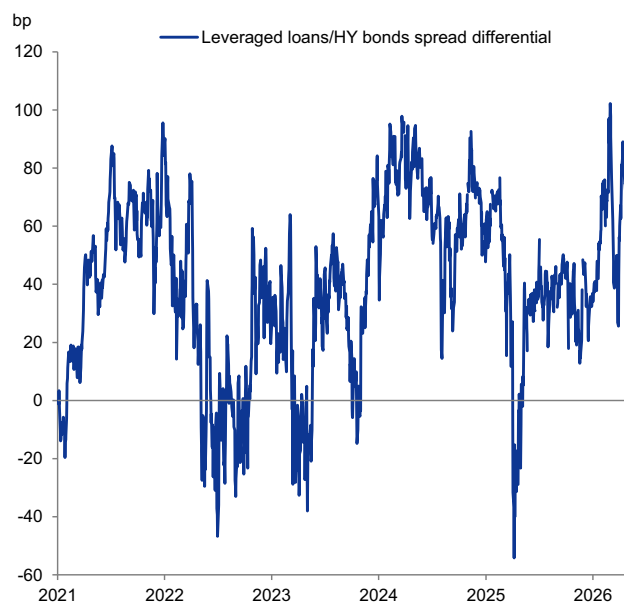
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 36: IG OAS/Agency MBS spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 37: Leveraged loans/HY bonds spread differential



Source: Bloomberg, Morningstar LCD, Goldman Sachs Global Investment Research

Performance dashboard: Biggest movers in the USD bond market

Exhibit 38: Top and bottom 20 performers in the USD IG market

Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Meta Platforms Inc	Internet	META 5 3/4 11/15/65	4,500	Aa3/AA/--	1.6%	1.5	92.00
Old Natl Bancorp	Banks	ONB 5.768 02/15/36	450	Baa1/--/--	1.6%	1.6	100.11
Boeing Co	Aerospace/Defense	BA 3.6 05/01/34	850	Baa3/BBB-/BBB-	1.6%	1.4	90.29
State Street Corp	Banks	STT 5.159 05/18/34	1,000	Aa3/A/AA-	1.5%	1.5	101.67
American International	Insurance	AIG 5 1/8 03/27/33	750	Baa1/A-/A-	1.5%	1.5	101.61
Campbells Company/The	Food	CPB 5.4 03/21/34	1,000	Baa2/BBB-/BBB-	1.4%	1.4	98.09
Salesforce Inc	Software	CRM 6.55 03/15/56	3,750	A2/A+/-	1.4%	1.4	100.10
Mondelez International	Food	MDLZ 5 1/8 05/06/35	400	Baa1/BBB/--	1.4%	1.4	101.18
Kraft Heinz Foods Co	Food	KHC 5.4 03/15/35	500	Baa2/BBB/BBB	1.3%	1.4	101.48
Polaris Inc	Leisure Time	PII 5.6 03/01/31	500	Baa3/BBB-/BBB-	1.3%	1.3	99.89
Oracle Corp	Software	ORCL 5 1/2 08/03/35	1,750	Baa2/BBB/BBB	1.3%	1.2	96.34
American Express Co	Diversified Finan Serv	AXP 5.043 05/01/34	1,250	A2/A-/A-	1.3%	1.3	100.86
Estee Lauder Co Inc	Cosmetics/Personal Care	EL 5 02/14/34	650	A3/A/--	1.2%	1.2	100.43
Cigna Group/The	Healthcare-Services	CI 5 1/4 02/15/34	1,250	Baa1/A-/BBB+	1.2%	1.2	101.84
At&T Inc	Telecommunications	T 5.7 11/01/54	1,500	Baa2/BBB/BBB+	1.2%	1.1	93.73
Comcast Corp	Media	CMCSA 6.05 05/15/55	1,000	A3/A-/A-	1.1%	1.1	98.75
Ebay Inc	Internet	EBAY 4 07/15/42	750	Baa1/BBB+/-	1.1%	0.9	79.32
American Honda Finance	Auto Manufacturers	HNDA 4.9 01/10/34	750	A3/BBB+/A-	1.0%	1.0	98.48
Bristol-Myers Squibb Co	Pharmaceuticals	BMJ 5.9 11/15/33	750	A2/A/--	1.0%	1.1	107.45
Union Pacific Corp	Transportation	UNP 3.6 09/15/37	500	A3/A-/A-	1.0%	0.9	88.29
Bottom 20 performers							
Hsbc Holdings Plc	Banks	HSBC 6 1/2 09/15/37	985	Baa1/BBB+/A-	-2.7%	-3.0	108.80
Oracle Corp	Software	ORCL 6 1/8 07/08/39	1,250	Baa2/BBB/BBB	-2.7%	-2.6	99.38
Wells Fargo & Company	Banks	WFC 5 3/8 02/07/35	470	A1/BBB+/A+	-2.7%	-2.8	105.17
Altria Group Inc	Agriculture	MO 4 1/4 08/09/42	900	A3/BBB+/BBB+	-2.5%	-2.1	83.93
Ibm Internat Capital	Computers	IBM 5 1/4 02/05/44	1,000	A3/A-/A-	-2.4%	-2.2	93.80
Abbvie Inc	Pharmaceuticals	ABBV 4.55 03/15/35	1,681	A2/A/--	-2.3%	-2.3	99.19
T-Mobile Usa Inc	Telecommunications	TMUS 5 7/8 11/15/55	1,250	Baa1/BBB+/BBB+	-2.2%	-2.2	98.91
Phillips 66 Co	Oil&Gas	PSX 5 7/8 03/15/56	1,000	Baa2/BBB/--	-2.0%	-2.0	99.63
Charter Comm Opt Llc/Cap	Media	CHTR 4.8 03/01/50	2,800	Ba1/BBB-/BBB-	-2.0%	-1.5	73.89
Brighthouse Financial In	Insurance	BHF 4.7 06/22/47	1,014	Baa3/BBB/BBB-	-2.0%	-1.4	72.45
Oracle Corp	Software	ORCL 6 1/2 04/15/38	1,250	Baa2/BBB/BBB	-1.9%	-2.0	103.20
Campbells Company/The	Food	CPB 4 3/4 03/23/35	800	Baa2/BBB-/BBB-	-1.9%	-1.8	94.43
Rtx Corp	Aerospace/Defense	RTX 3.03 03/15/52	1,100	Baa1/BBB+/-	-1.8%	-1.2	64.43
Home Depot Inc	Retail	HD 5 7/8 12/16/36	3,000	A2/A/A	-1.8%	-1.9	108.05
Johnson & Johnson	Pharmaceuticals	JNJ 4.9 06/01/31	1,150	Aaa/AAA/--	-1.8%	-1.8	104.69
Intel Corp	Semiconductors	INTC 4 1/4 12/15/42	567	Baa2/BBB/BBB	-1.7%	-1.4	83.50
Citigroup Inc	Banks	C 5.316 03/26/41	1,300	A3/BBB+/A	-1.7%	-1.7	98.37
General Dynamics Corp	Aerospace/Defense	GD 2.85 06/01/41	500	A1/A/--	-1.7%	-1.3	75.03
Paypal Holdings Inc	Commercial Services	PYPL 5.05 06/01/52	1,000	A3/A-/A-	-1.7%	-1.5	86.32
Charter Comm Opt Llc/Cap	Media	CHTR 5 1/4 04/01/53	1,500	Ba1/BBB-/BBB-	-1.7%	-1.3	77.90

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

Exhibit 39: Top and bottom 20 performers in the USD HY market

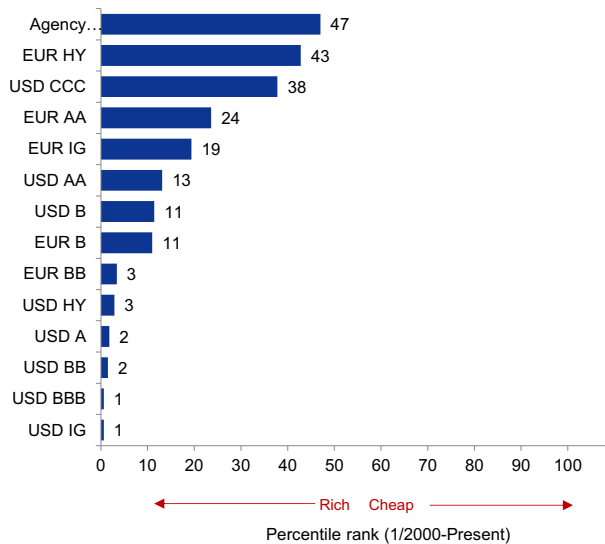
Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Liberty Interactive Llc	Retail	LINTA 8 1/4 02/01/30	504	WR/D/C	20.7%	1.1	5.18
Csc Holdings Llc	Media	CSCHLD 4 1/2 11/15/31	1,500	Caa2/CCC+/-	3.2%	1.8	54.25
Csc Holdings Llc	Media	CSCHLD 4 1/8 12/01/30	1,100	Caa2/CCC+/-	3.1%	1.7	54.63
Goeasy Ltd	Diversified Finan Serv	GSYCN 6 7/8 05/15/30	400	B2/B-/-	3.0%	2.5	84.50
Csc Holdings Llc	Media	CSCHLD 6 1/2 02/01/29	1,750	Caa2/CCC+/-	2.6%	1.5	58.50
Clydesdale Acquisition	Packaging&Containers	NOVHOL 8 3/4 04/15/30	1,110	Caa1/B-/BB	2.5%	2.3	91.47
Dye & Durham Limited	Software	DNDN 8 5/8 04/15/29	509	B2/B-/-	2.3%	1.8	81.19
Cntrl Parent/Cdk Glb Inc	Software	CDK 7 1/4 06/15/29	750	B3/B-/-	2.1%	1.1	50.01
Embecka Corp	Healthcare-Products	EMBC 6 3/4 02/15/30	200	B3/B-/-	1.9%	1.6	80.70
Mcafee Corp	Computers	MCFE 7 3/8 02/15/30	1,733	Caa1/CCC+/-	1.8%	1.5	81.50
Planet Financial Group	Diversified Finan Serv	PLNFIN 10 1/2 12/15/29	600	Caa1/B-/B-	1.8%	1.8	96.75
Cd&R Smokey / Radio Sys	Housewares	PETSAF 9 1/2 10/15/29	775	B3/B-/-	1.8%	1.2	67.06
Ziprecruiter Inc	Internet	ZIP 5 01/15/30	550	B3/B/CCC+	1.6%	1.1	70.00
Alta Equipment Group	Commercial Services	ALTG 9 06/01/29	500	Caa1/B-/-	1.6%	1.5	93.50
Clue Opco Llc	Holding Companies-Divers	FWRD 9 1/2 10/15/31	725	B2/B/B+	1.5%	1.4	90.75
Kennedy-Wilson Inc	Real Estate	KW 4 3/4 02/01/30	600	B2/B-/-	1.3%	1.3	99.60
Clydesdale Acquisition	Packaging&Containers	NOVHOL 6 3/4 04/15/32	1,415	B2/B+/BB+	1.1%	1.0	93.50
Kingpin Intermediate Ho	Leisure Time	LUCK 7 1/4 10/15/32	500	B3/B-/-	1.1%	0.9	84.23
Burford Capital Gbl Fin	Diversified Finan Serv	BURLN 9 1/4 07/01/31	675	Ba1/BB-/-	1.0%	1.0	93.25
Cp Atlas Buyer Inc	Building Materials	AMBATH 9 3/4 07/15/30	500	B2/CCC+/-	0.9%	0.8	92.25
Bottom 20 performers							
Csc Holdings Llc	Media	CSCHLD 7 1/2 04/01/28	1,046	Ca/CCC-/-	-13.7%	-6.2	45.05
Lbm Acquisition Llc	Retail	USLBMH 6 1/4 01/15/29	726	Caa3/CCC/CCC	-10.8%	-7.3	67.25
Wabash National Corp	Auto Manufacturers	WNC 4 1/2 10/15/28	400	Caa1/B-/-	-7.3%	-6.2	84.66
Jeldwen Holding	Building Materials	JELD 7 09/01/32	350	Caa2/CCC+/-	-6.7%	-4.0	59.50
Getty Images Inc	Internet	ABEGET 10 1/2 11/15/30	628	B3/BB-/-	-6.0%	-5.0	83.75
Lbm Acquisition Llc	Retail	USLBMH 9 1/2 06/15/31	950	Caa1/B-/B-	-5.6%	-4.8	85.50
Brand Industrial Service	Engineering&Construction	BRANDI 10 3/8 08/01/30	1,435	Caa1/CCC+/-	-4.9%	-4.5	92.00
Gray Media Inc	Media	GTN 5 3/8 11/15/31	1,220	Caa1/CCC-/-	-4.9%	-3.8	76.25
Goodyear Tire & Rubber	Auto Parts&Equipment	GT 5 5/8 04/30/33	450	B2/B+/BB-	-4.4%	-3.9	89.25
Iheartcommunications Inc	Media	IHRT 10 7/8 05/01/30	675	Caa3/CCC-/-	-4.2%	-4.0	94.50
Gray Media Inc	Media	GTN 4 3/4 10/15/30	790	Caa1/CCC-/-	-4.1%	-3.3	79.00
Goodyear Tire & Rubber	Auto Parts&Equipment	GT 5 1/4 07/15/31	600	B2/B+/BB-	-4.1%	-3.7	90.00
Whirlpool Corp	Home Furnishings	WHR 5 3/4 03/01/34	300	B1/BB/BB	-4.0%	-3.5	87.78
Cable One Inc	Media	CABO 4 11/15/30	548	B3/B-/-	-3.7%	-2.4	63.35
Angi Group Llc	Internet	ANGI 3 7/8 08/15/28	400	B2/B-/-	-3.6%	-3.2	89.20
Chc Group Llc	Oil&Gas Services	CHHCF 11 3/4 09/01/30	550	--/-/-	-3.5%	-3.0	84.99
Goodyear Tire & Rubber	Auto Parts&Equipment	GT 5 1/4 04/30/31	550	B2/B+/BB-	-3.5%	-3.1	90.89
Kohl'S Corporation	Retail	KSS 3 3/8 05/01/31	425	B3/BB-/BB-	-3.5%	-2.9	83.72
Whirlpool Corp	Home Furnishings	WHR 6 1/2 06/15/33	600	B1/BB/BB	-3.3%	-3.1	93.60
Cerdia Finanz GmbH	Chemicals	PLATIN 9 3/8 10/03/31	900	B2/B-/-	-3.2%	-3.0	93.25

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

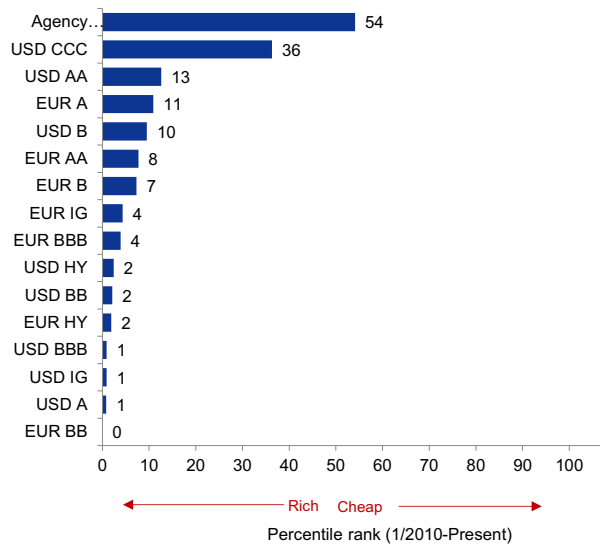
Valuation dashboard: Percentile ranks for spreads and yields

Exhibit 40: Percentile ranks for spreads across markets (2000-current)



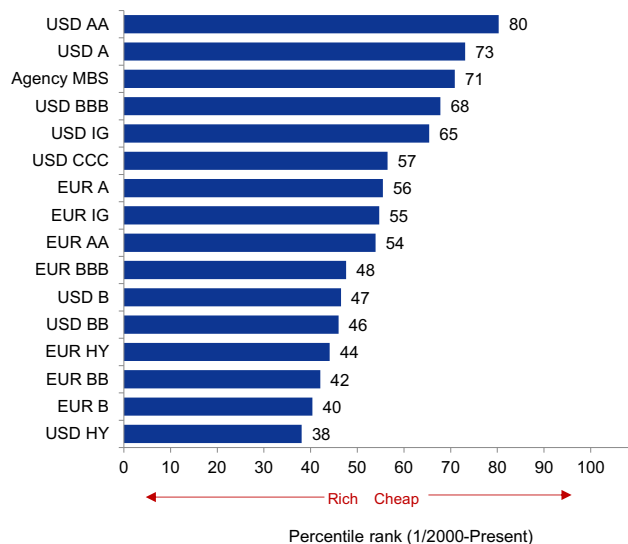
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 41: Percentile ranks for spreads across markets (2010-current)



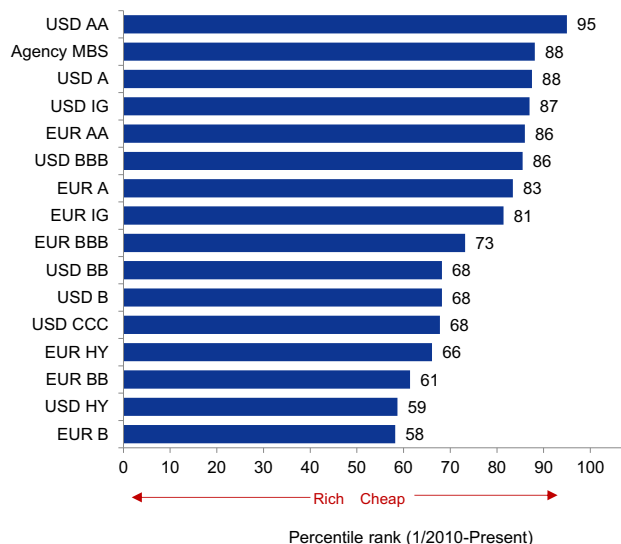
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 42: Percentile ranks for yields across markets (2000-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

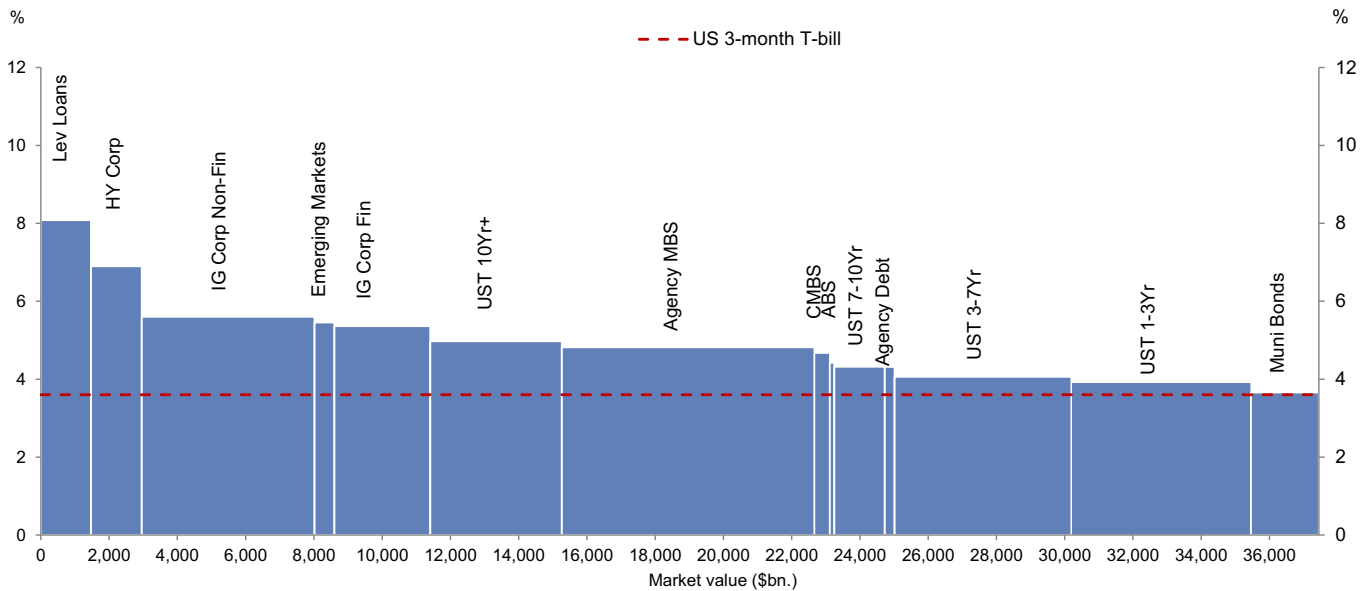
Exhibit 43: Percentile ranks for yields across markets (2010-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

Valuation dashboard: Sizing the carry in USD and EUR fixed income markets

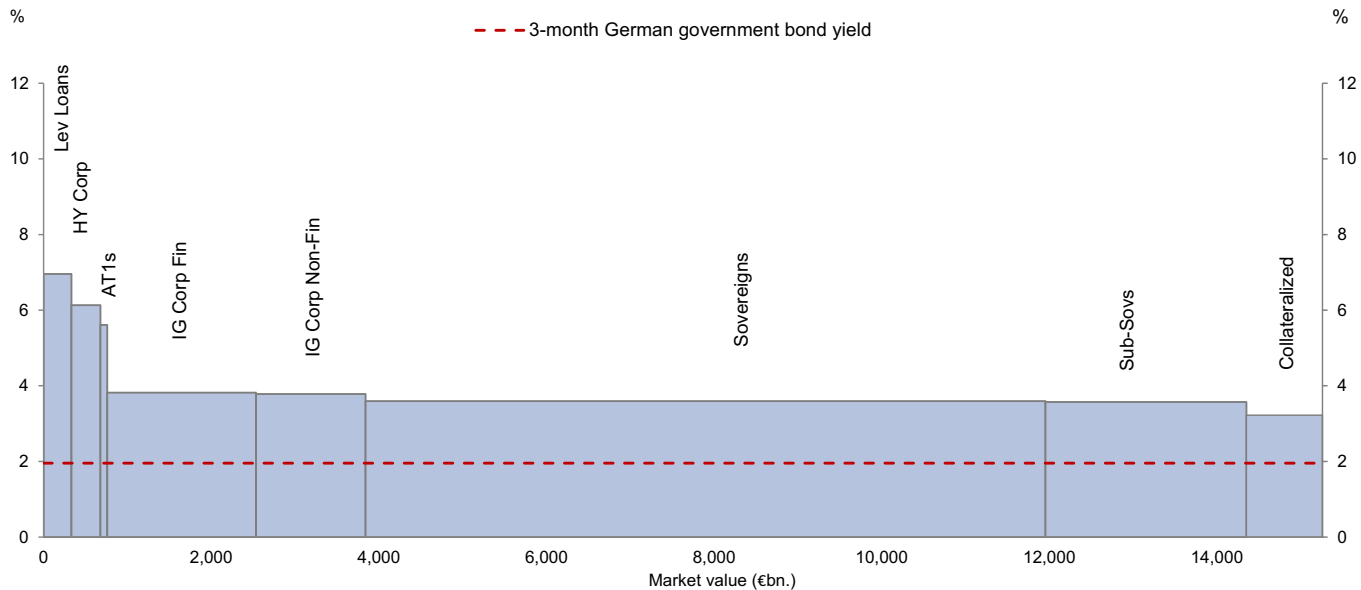
Exhibit 44: Average yield vs. total market value outstanding across the USD fixed income complex



Note: We use the iBoxx indices for the IG corporate credit and US Treasury markets, the Bloomberg indices for the HY, MBS, CMBS and ABS markets, and the Morningstar LSTA index for the leveraged loan market.

Source: iBoxx, Bloomberg, Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 45: Average yield vs. total market value outstanding for the EUR fixed income markets

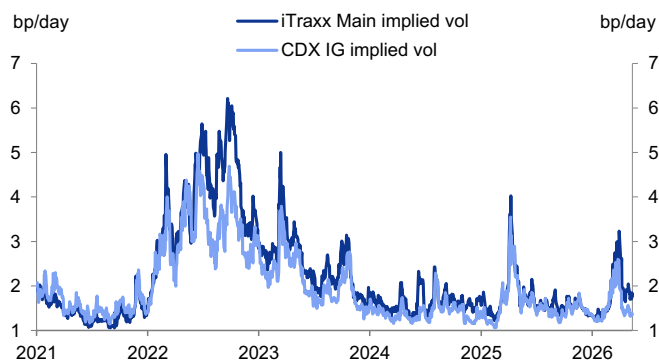


Source: iBoxx, Goldman Sachs Global Investment Research

Valuation dashboard: Macro index volatility

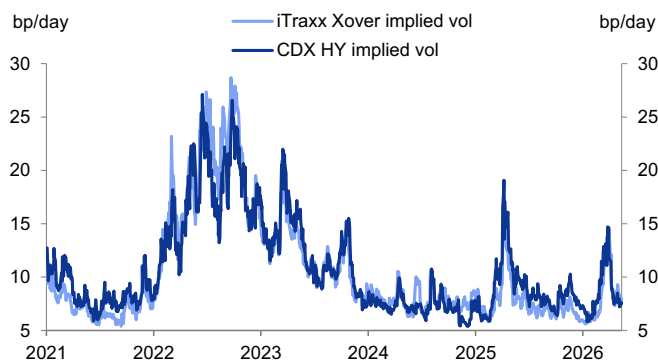
Note: Skew is calculated using difference in 3m implied volatility between 25-delta payers and receivers, divided by 3m ATM implied volatility.

Exhibit 46: CDX IG and iTraxx Main vol 3m ATM implied volatility



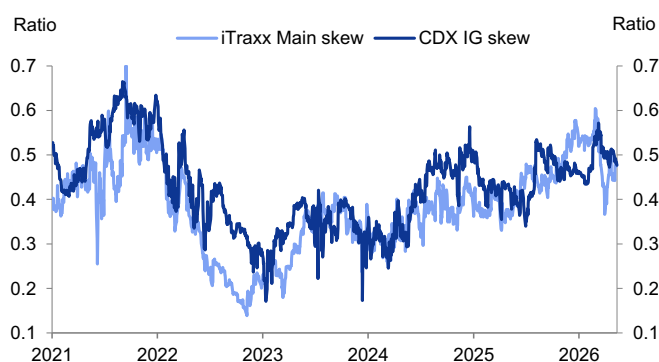
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 47: CDX HY and iTraxx Xover vol 3m ATM implied volatility



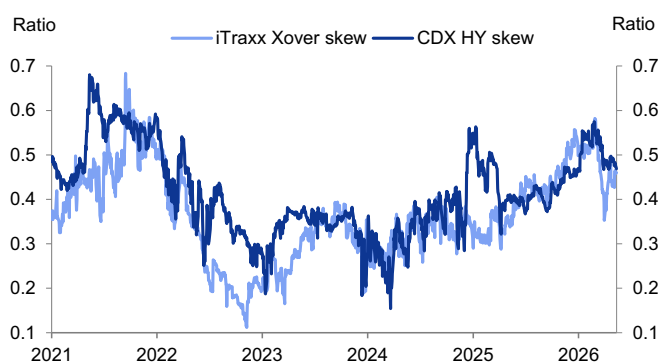
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 48: CDX IG and iTraxx Main skew



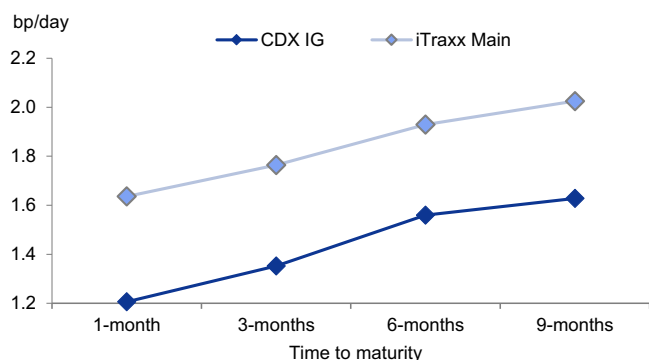
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 49: CDX HY and iTraxx Xover skew



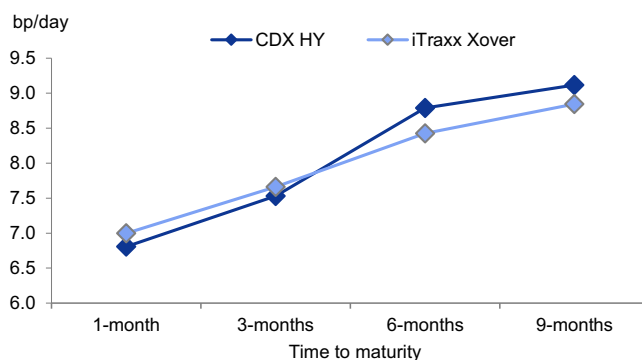
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 50: CDX IG and iTraxx Main ATM implied volatility term structure



Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 51: CDX HY and iTraxx Xover ATM implied volatility term structure



Source: Goldman Sachs, Goldman Sachs Global Investment Research

Valuation dashboard: CLOs

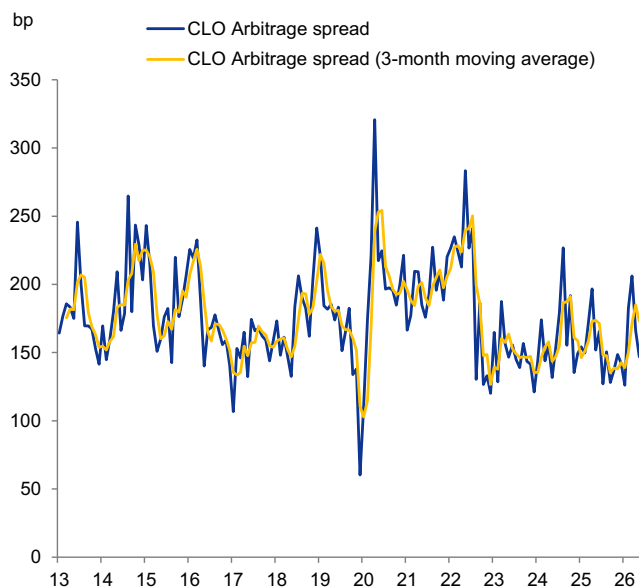
Exhibit 52: CLO spread percentile ranks

USD	Current spread (bp)	Percentile rank		
		3-year	5-year	10-year
AAA	104	5%	3%	2%
AA	144	5%	3%	2%
A	164	0%	0%	0%
BBB	281	30%	18%	12%
BB	754	62%	57%	63%

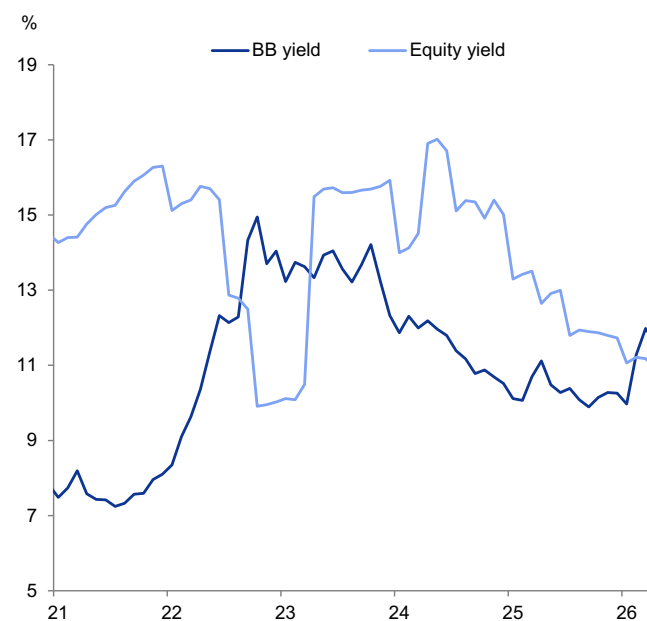
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 54: CLO new issue arbitrage spread

New issue CLO arbitrage spread, i.e., the difference between new issue loan spreads and new issue CLO spreads (across the capital stack)



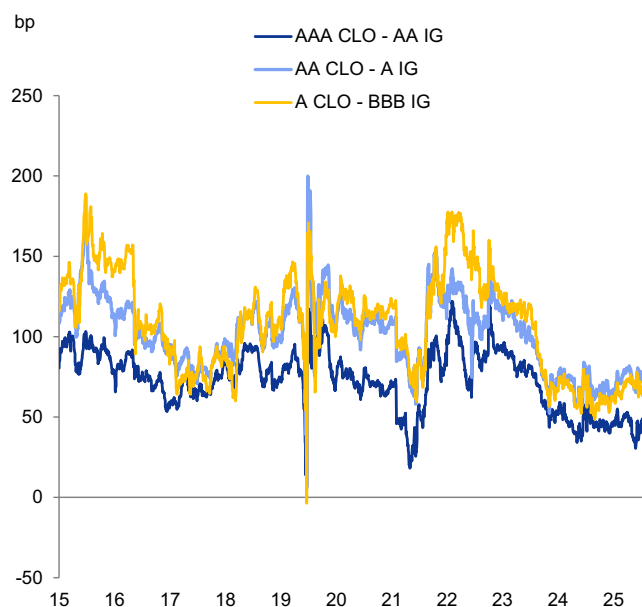
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 53: CLO equity vs. CLO BB yield

Source: Intex, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 55: CLO vs. USD IG spread differential

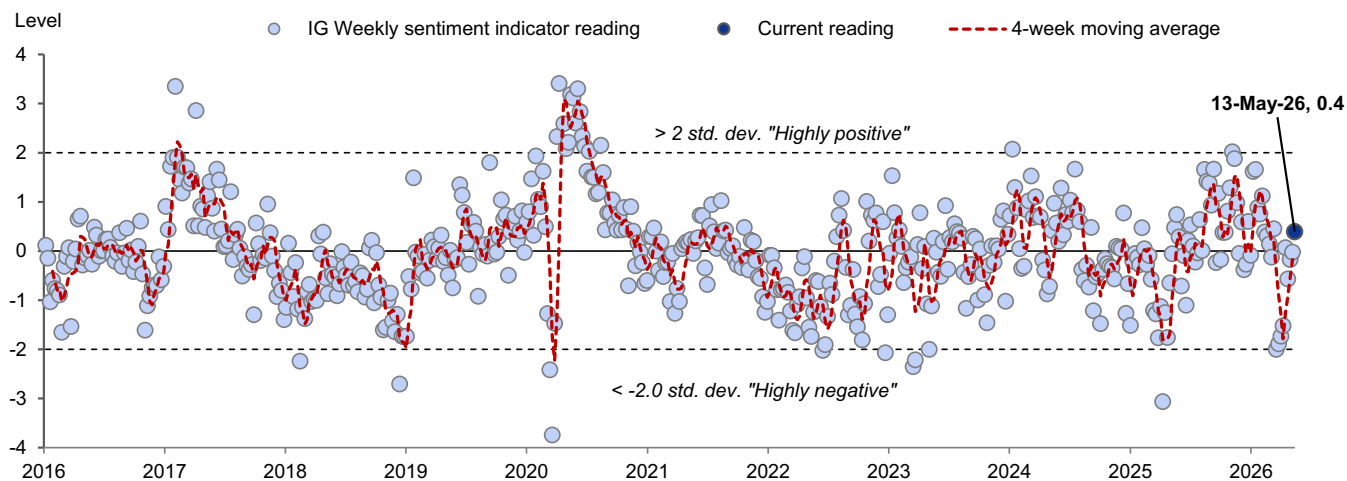
CLO by rating and USD IG bond by rating spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

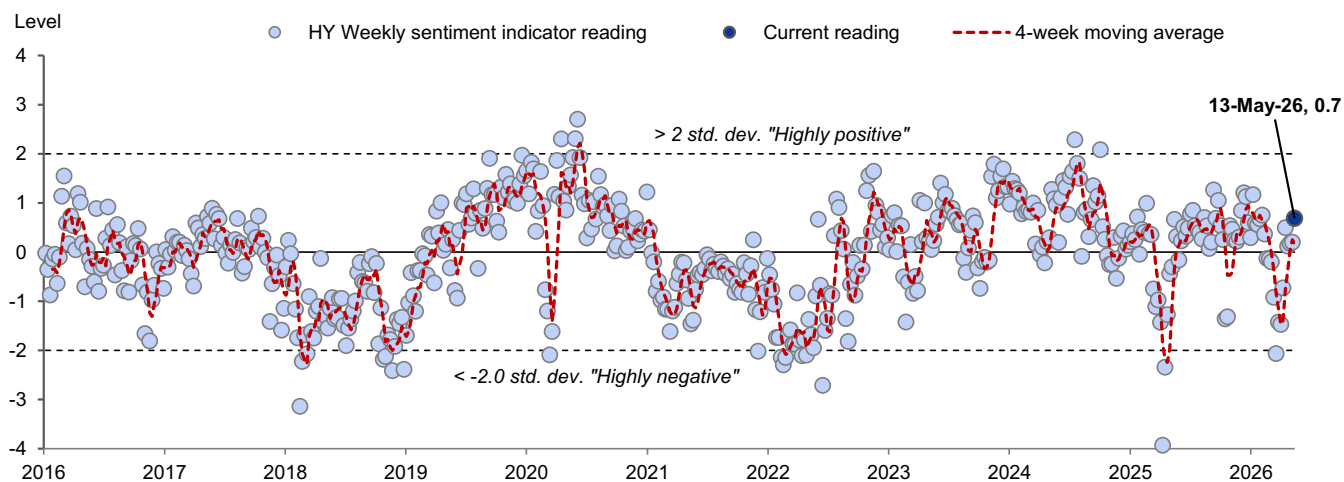
USD IG and HY Sentiment indicators

Exhibit 56: IG Weekly sentiment indicator



Source: See table below.

Exhibit 57: HY Weekly sentiment indicator



Source: See table below.

Exhibit 58: Variables included in the USD IG and HY weekly sentiment indicators

Measure	Frequency	Data Source
Synthetic index Positioning	Weekly	DTCC
Net Dealer Positioning	Weekly	FRBNY
Net Dealer Lift	Weekly	TRACE
Active US Credit Fund flows	Weekly	EPFR
Passive US Credit Fund flows	Weekly	EPFR
US Money Market Fund flows	Weekly	EPFR
Short interest on the largest credit ETFs	Semimonthly	GS FICC & Equities
Credit Mutual Fund cash positions	Monthly	ICI

Source: Goldman Sachs Global Investment Research

Microstructure dashboard: Top 20 most traded capital structures in the USD IG and HY markets

Exhibit 59: Top 20 most liquid issuers by weekly turnover in the USD IG market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
CRM	Salesforce Inc	33.5	691.0	2.1	A	1.5	7.3
HUM	Humana Inc	10.9	218.1	2.0	BBB	3.9	7.2
GOOGL	Alphabet Inc	51.0	1015.3	2.0	AA	1.4	9.9
ZTS	Zoetis Inc	6.8	119.3	1.8	BBB+	6.0	6.7
EXPE	Expedia Group Inc	5.5	96.5	1.8	BBB	4.2	4.3
CPB	The Campbell's Company	5.7	97.8	1.7	BBB-	4.4	6.4
LYB	LYB International Finance III LLC	10.8	178.8	1.7	BBB-	6.1	10.5
PYPL	PayPal Holdings Inc	8.0	128.0	1.6	A-	4.5	7.5
AMZN	Amazon.com Inc	97.5	1502.8	1.5	AA-	3.0	8.9
KHC	Kraft Heinz Foods Co	13.7	208.4	1.5	BBB	8.5	9.5
ABT	Abbott Laboratories	27.6	410.5	1.5	A+	3.4	9.0
ARE	Alexandria Real Estate Equities Inc	7.1	104.4	1.5	BBB+	4.0	7.4
BCRED	Blackstone Private Credit Fund	6.5	95.6	1.5	BBB-	1.2	4.1
SOLV	Solventum Corp	5.1	73.9	1.5	BBB	1.3	7.0
BBVASM	Banco Bilbao Vizcaya Argentaria SA	6.1	88.3	1.4	A-	1.7	4.3
ETN	Eaton Corp	13.9	196.5	1.4	A-	2.7	6.2
COR	Cencora Inc	8.6	119.1	1.4	BBB+	3.3	6.1
ARCC	Ares Capital Corp	7.5	101.9	1.4	BBB	2.4	3.4
FANG	Diamondback Energy Inc	9.3	125.5	1.4	BBB	2.9	8.3
JEF	Jefferies Financial Group Inc	8.1	108.0	1.3	BBB	4.0	5.2

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

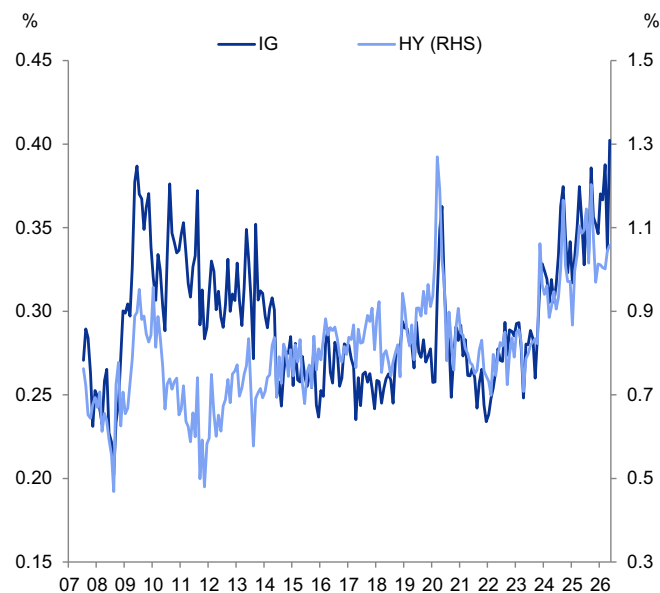
Exhibit 60: Top 20 most liquid issuers by weekly turnover in the USD HY market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
JBLU	JetBlue Airways Corp / JetBlue Loyalty LP	2.0	145.1	7.3	B	1.7	4.1
AXL	American Axle & Manufacturing Inc	2.7	107.9	4.0	B+	1.5	4.6
IHRT	iHeartCommunications Inc	2.1	61.6	3.0	CCC+	1.4	3.1
FUN	Six Flags Entertainment Corp/Canada's Wonderland Co/Millen	3.1	83.2	2.6	CCC+	2.3	3.4
CRWV	CoreWeave Inc	6.5	160.0	2.5	B+	0.5	3.2
BURLN	Burford Capital Global Finance LLC	2.1	48.8	2.4	BB-	2.2	4.3
SABHLD	Sabre GBL Inc	3.2	75.4	2.3	B-	0.7	2.7
GT	The Goodyear Tire & Rubber Company	3.0	68.1	2.3	B+	3.8	4.1
GTN	Gray Media Inc	4.8	109.4	2.3	CCC	2.8	3.3
NOVHOL	Clydesdale Acquisition Holdings Inc	3.5	77.2	2.2	B+	2.5	3.9
NWL	Newell Brands Inc	3.5	69.1	2.0	B	1.9	2.6
UAL	United Airlines Holdings Inc	2.0	38.5	1.9	BB+	0.2	3.4
W	Wayfair LLC	2.2	41.8	1.9	BB-	1.1	3.4
MINAU	Mineral Resources Ltd	3.7	66.1	1.8	BB-	1.5	2.9
FSK	FS KKR Capital Corp	2.9	48.5	1.7	BB+	2.4	2.9
GSYN	goeasy Ltd	2.4	37.4	1.6	B-	1.7	3.1
NCLH	NCL Corp Ltd	5.0	74.4	1.5	B-	1.8	4.1
NAVI	Navient Corp	2.8	39.3	1.4	BB-	3.3	2.9
CC	The Chemours Company	2.5	33.8	1.3	B+	2.8	3.2
SVC	Service Properties Trust	2.7	35.3	1.3	B-	4.7	2.5

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

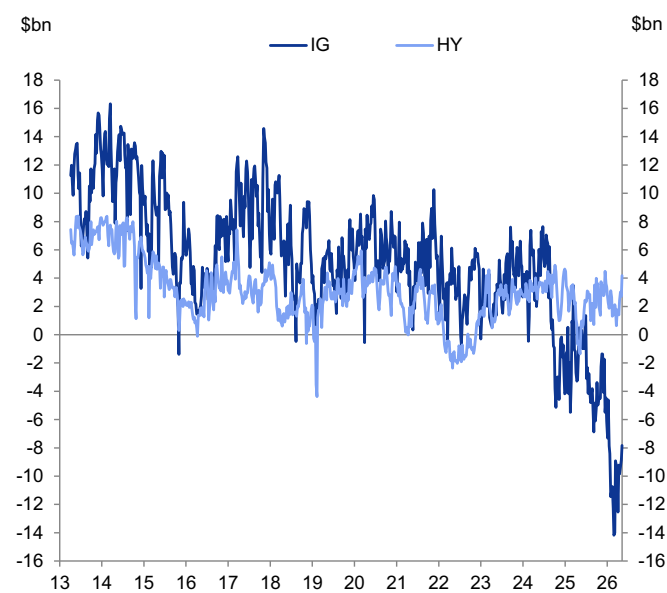
Microstructure dashboard: Liquidity conditions in the USD secondary bond market

Exhibit 61: Seasonally adjusted daily average turnover (trading volume/amount outstanding)



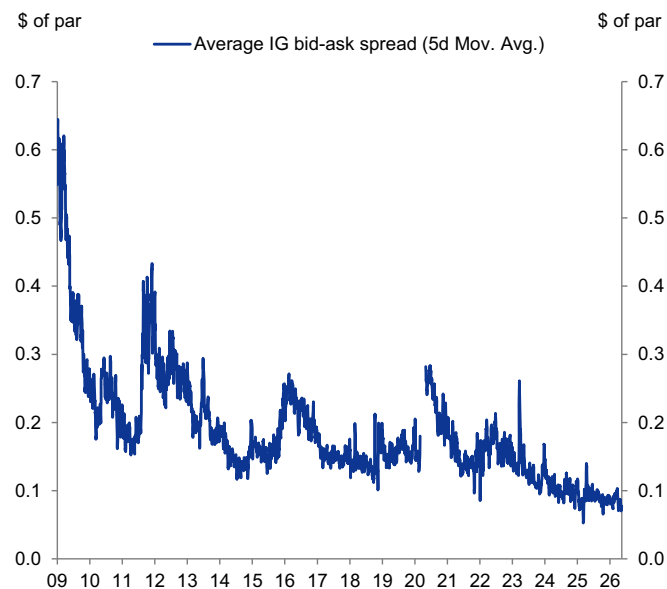
Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 62: Primary dealer net positioning in USD corporate bonds



Source: FRBNY, Goldman Sachs Global Investment Research

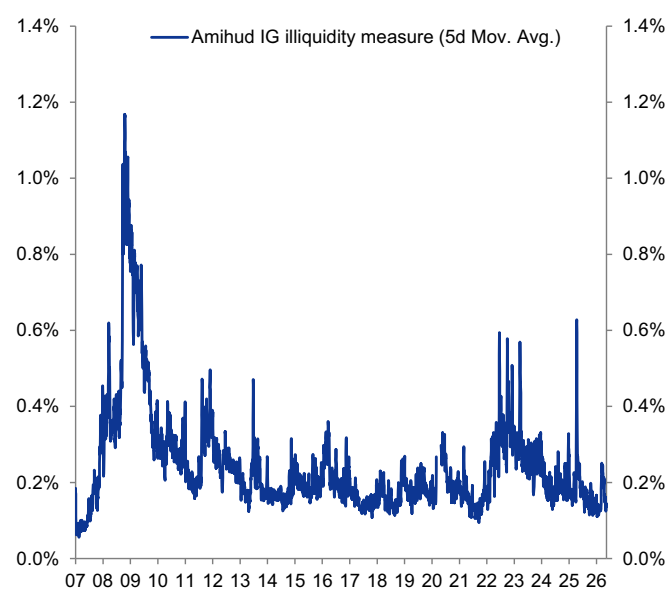
Exhibit 63: Average IG bid-ask spread 5-day moving average



Note: March and April 2020 are omitted.

Source: Trace, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 64: Amihud IG illiquidity measure 5-day moving average

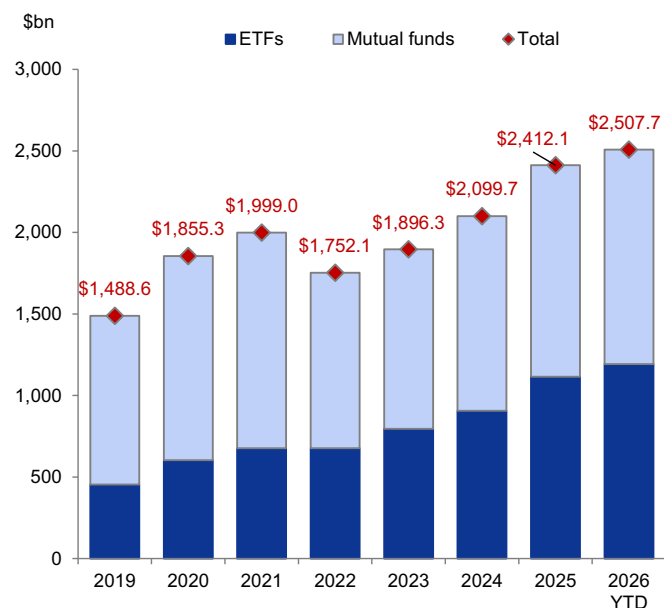


Note: March and April 2020 are omitted.

Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

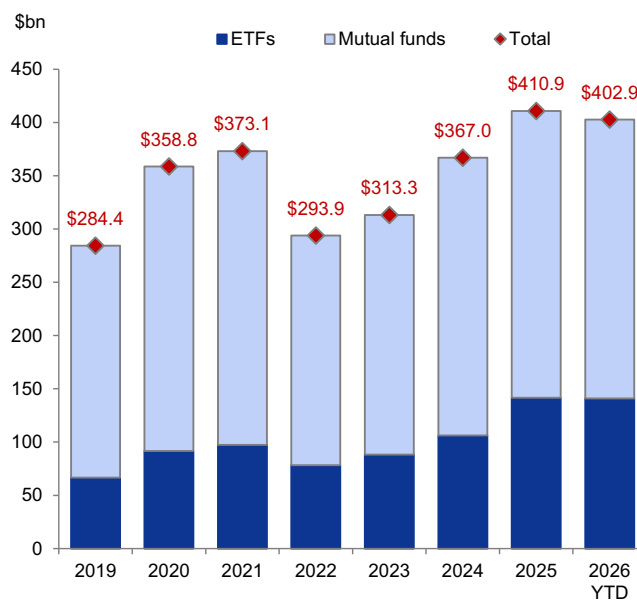
Flow dashboard: Mutual fund and ETF AUM

Exhibit 65: AUM in USD IG-focused Mutual Funds and ETFs



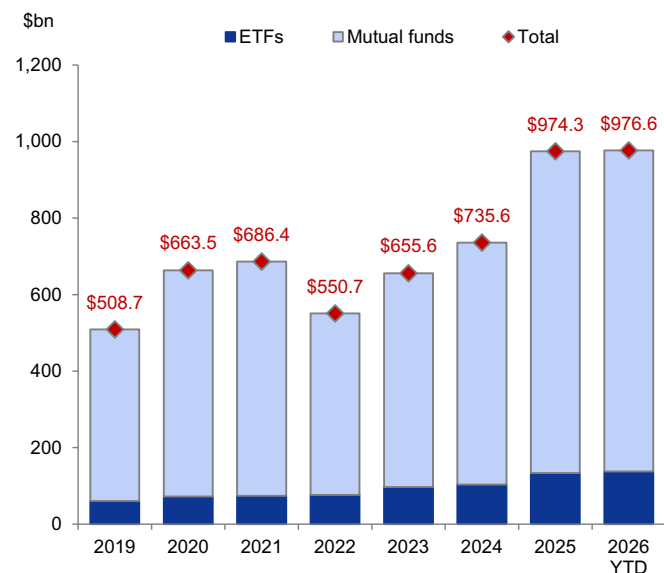
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 66: AUM in USD HY-focused Mutual Funds and ETFs



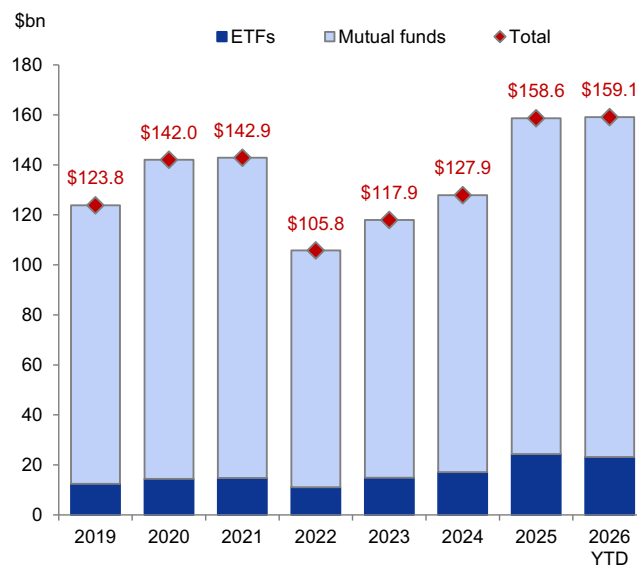
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 67: AUM in EUR IG-focused Mutual Funds and ETFs



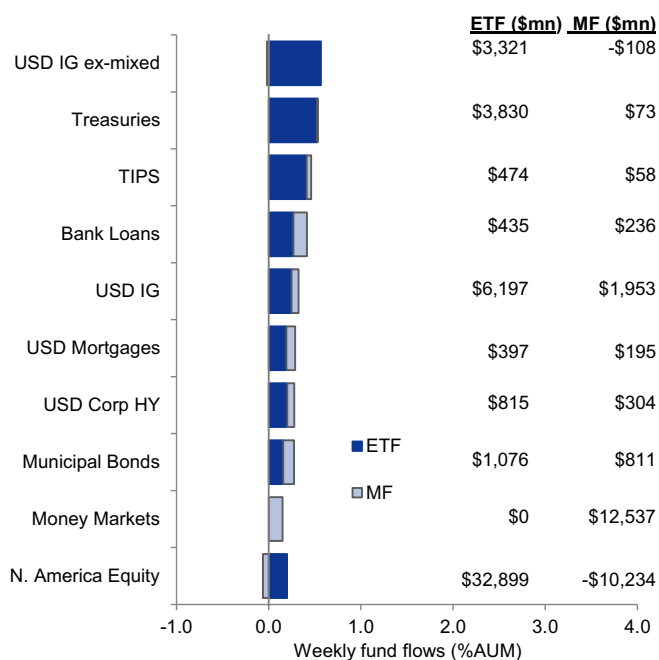
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 68: AUM in EUR HY-focused Mutual Funds and ETFs

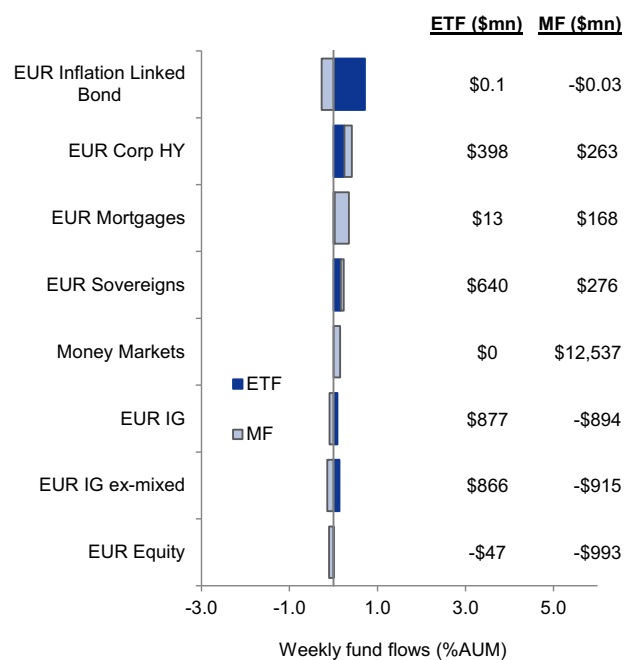


Source: EPFR, Goldman Sachs Global Investment Research

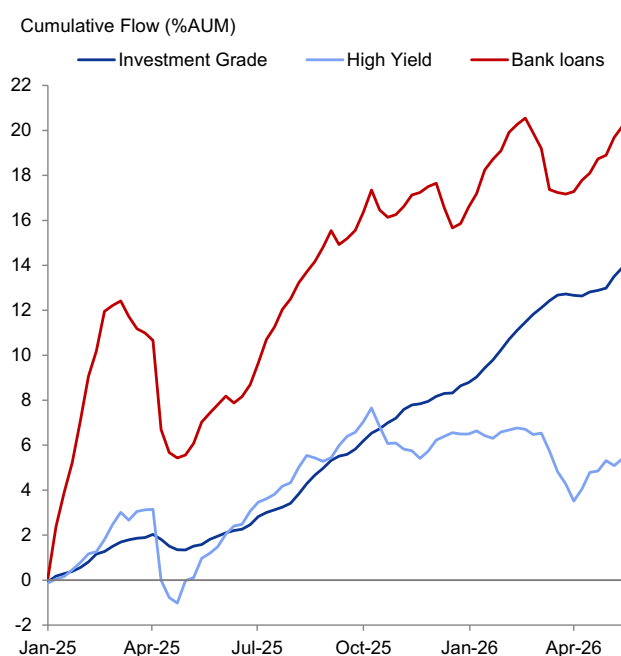
Flow dashboard: Mutual fund and ETF flows

Exhibit 69: Weekly mutual fund flows into US fixed income and equity markets

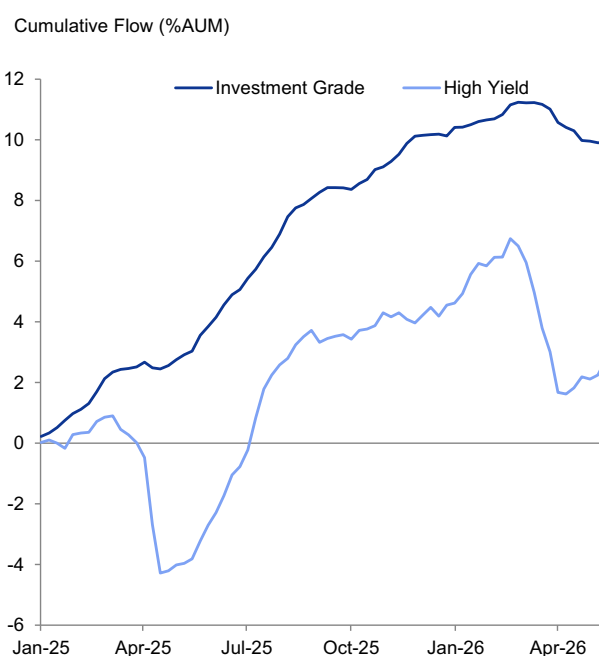
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 70: Weekly mutual fund flows into European fixed income and equity markets

Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 71: Year-to-date cumulative flows as a percentage of beginning of the year AUM for US credit funds

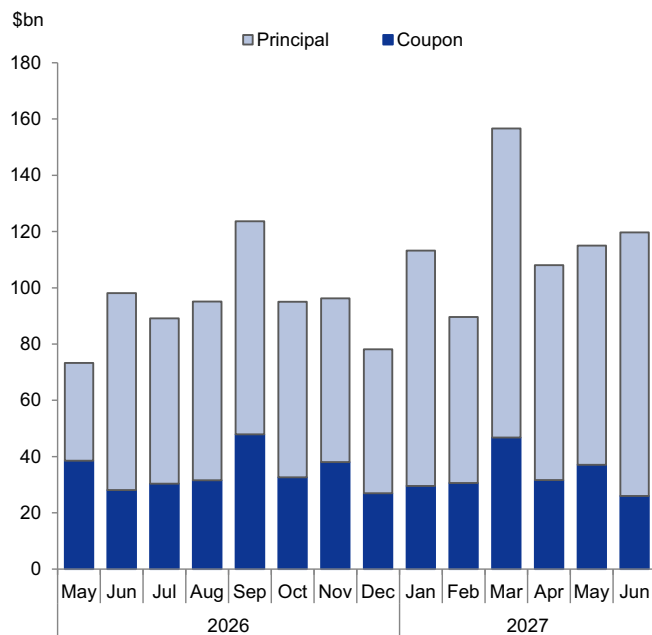
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 72: Year-to-date cumulative flows as a percentage of beginning of the year AUM for European credit funds

Source: EPFR, Goldman Sachs Global Investment Research

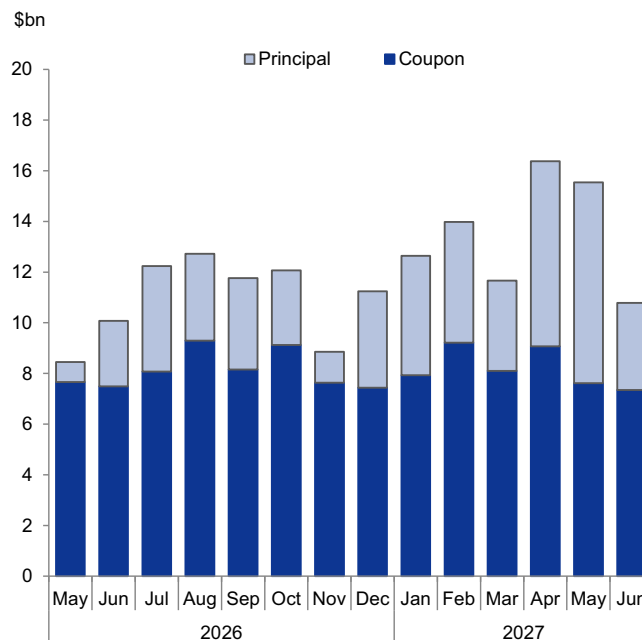
Flow dashboard: Upcoming coupon and principal payments

Exhibit 73: Our estimates of upcoming coupon and principal payments in the USD IG market



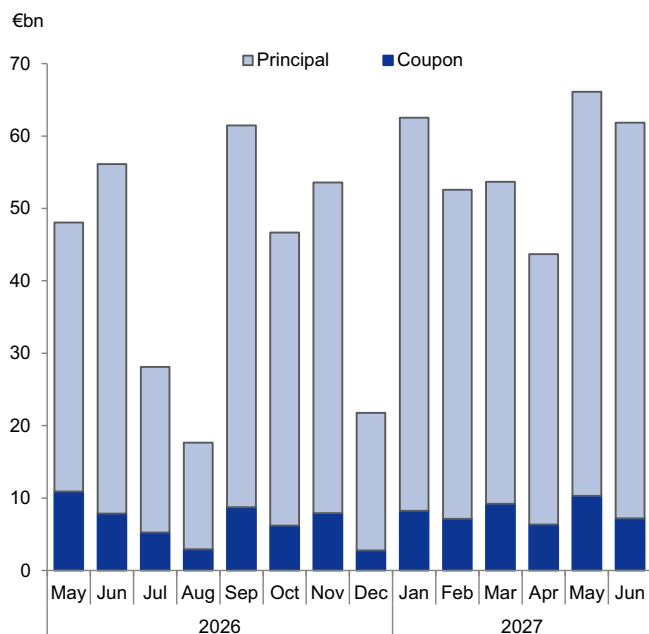
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 74: Our estimates of upcoming coupon and principal payments in the USD HY market



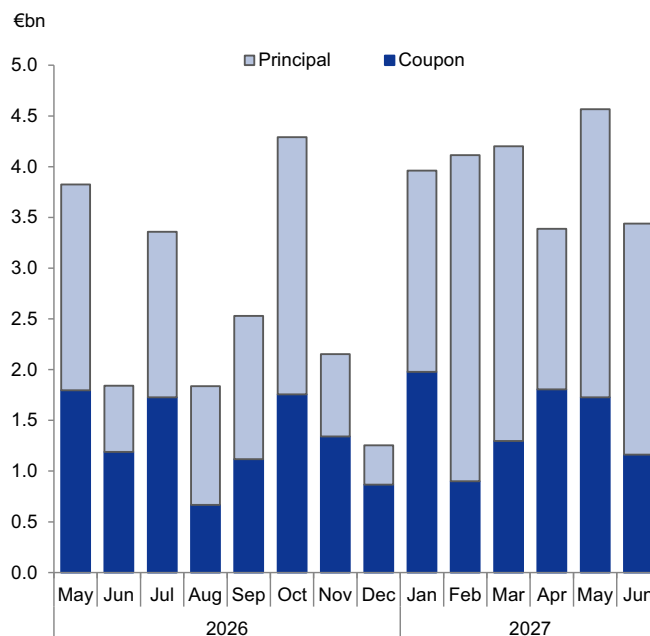
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 75: Our estimates of upcoming coupon and principal payments in the EUR IG market



Source: Bloomberg, Goldman Sachs Global Investment Research

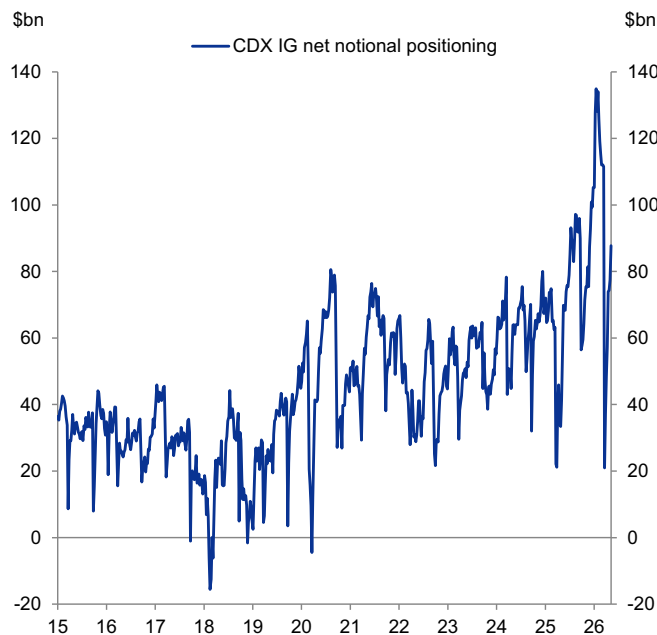
Exhibit 76: Our estimates of upcoming coupon and principal payments in the EUR HY market



Source: Bloomberg, Goldman Sachs Global Investment Research

Flow dashboard: Positioning in macro indices

Exhibit 77: Net notional positioning in the CDX IG 5-year on-the-run contract



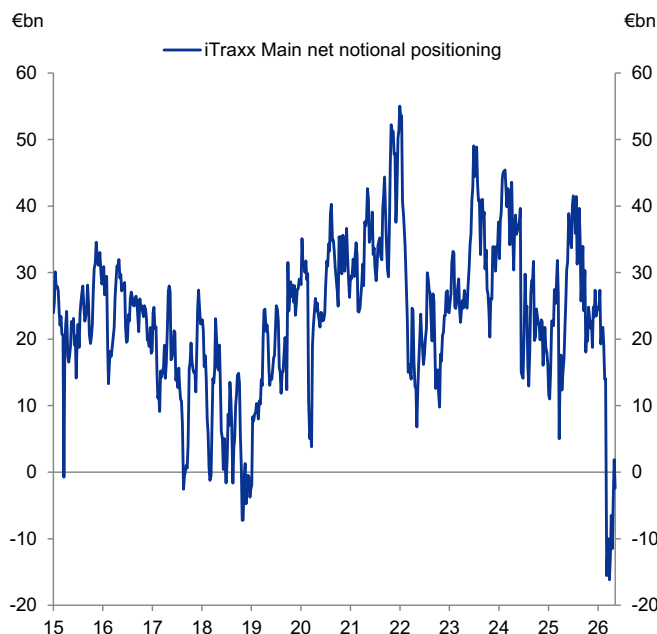
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 78: Net notional positioning in the CDX HY 5-year on-the-run contract



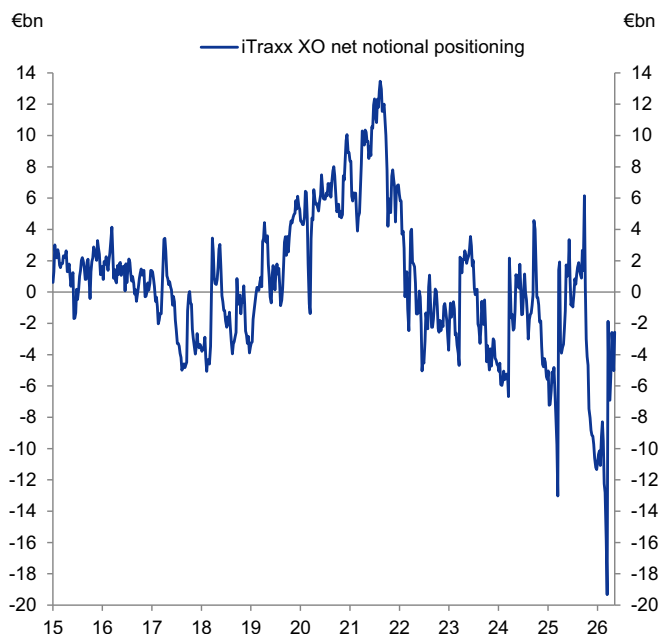
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 79: Net notional positioning in the iTraxx Main 5-year on-the-run contract



Source: DTCC, Goldman Sachs Global Investment Research

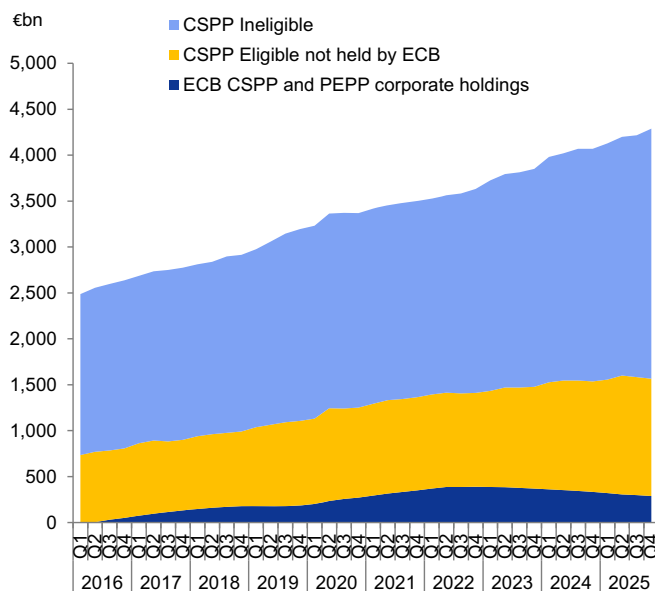
Exhibit 80: Net notional positioning in iTraxx over 5-year on-the-run contract



Source: DTCC, Goldman Sachs Global Investment Research

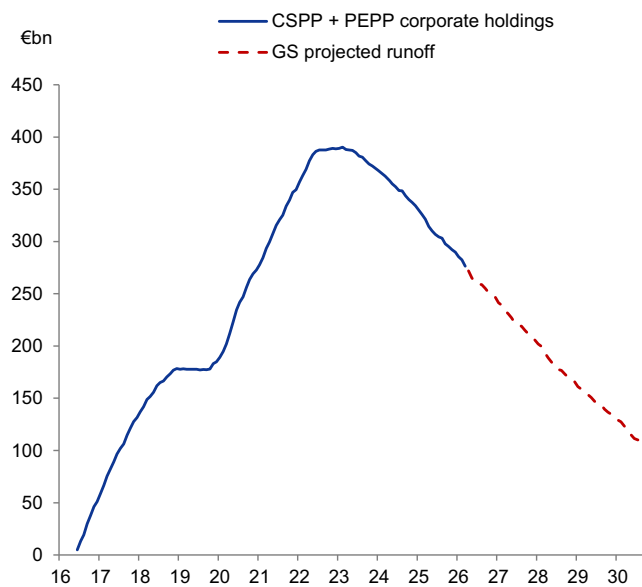
Flow dashboard: ECB QT

Exhibit 81: Ownership share of the ECB in the EUR IG market



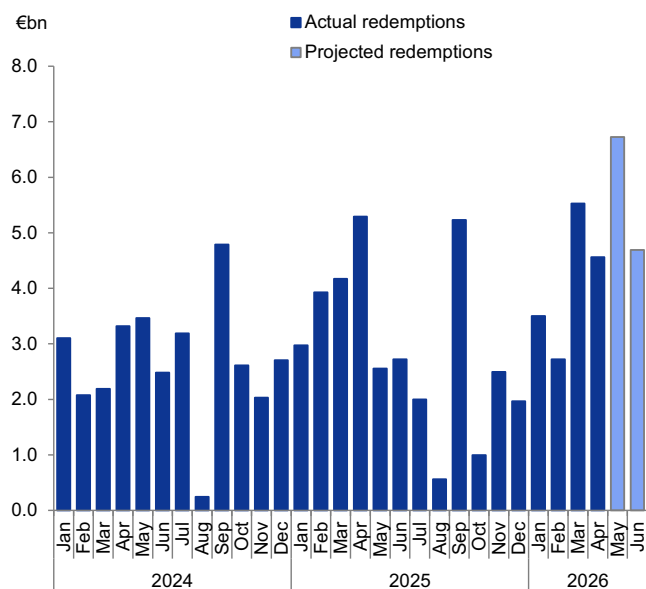
Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 82: Run-off projections for the ECB corporate bond holdings



Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 83: CSPP monthly run-off pace



Source: ECB, Goldman Sachs Global Investment Research

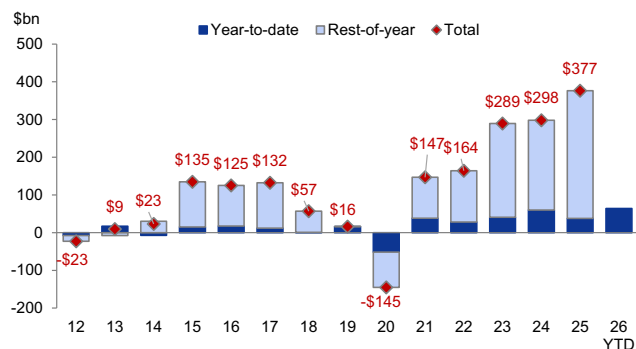
Exhibit 84: Performance of CSPP-eligible bonds vs. non-eligible bonds



Source: iBoxx, Goldman Sachs Global Investment Research

Foreign demand and hedging cost

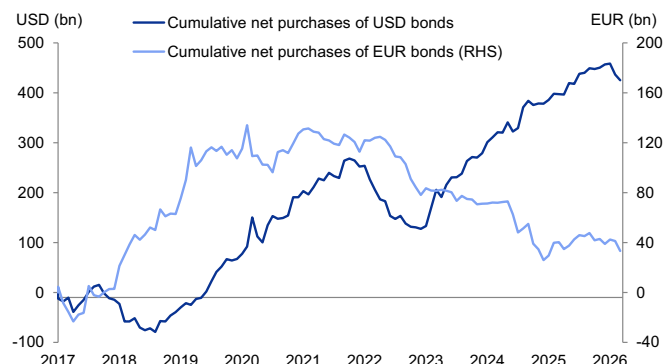
Exhibit 85: Net purchases of USD corporate bonds by foreign investors



Note: As of February 2026.

Source: Department of Treasury, Goldman Sachs Global Investment Research

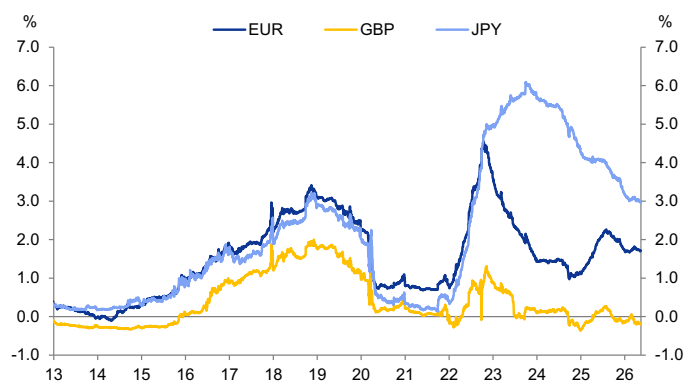
Exhibit 86: Cumulative net foreign purchases of USD and EUR bonds by Japanese investors



Note: As of February 2026.

Source: Bank of Japan, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 87: Annualized 3-month rolling cost of USD hedging against EUR, GBP and JPY

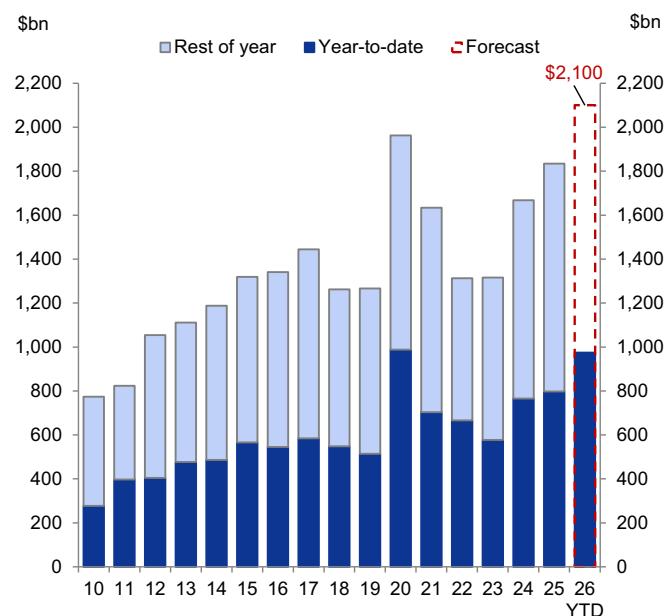


Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Primary market dashboard: USD gross issuance

Exhibit 88: USD IG gross issuance

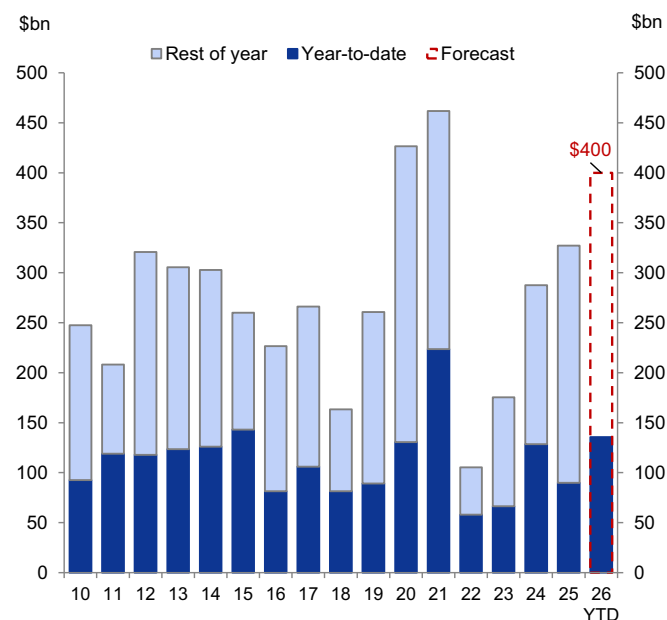
Historical investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 89: USD HY gross issuance

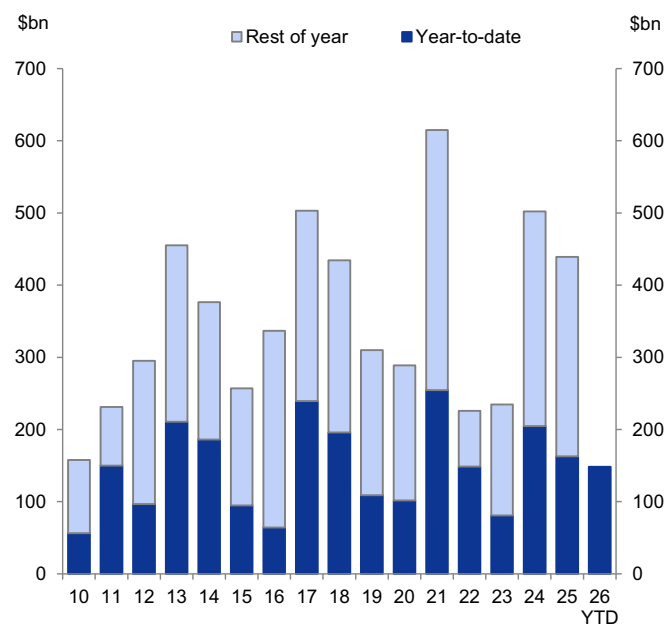
Historical USD high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 90: USD institutional leveraged loan gross issuance

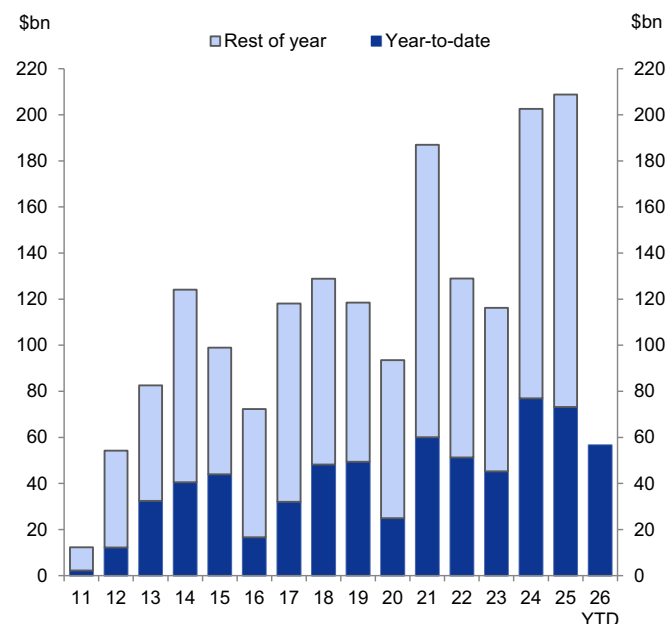
Historical leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 91: USD CLO creation

Historical USD CLO creation: Same period year to date vs. rest of the year

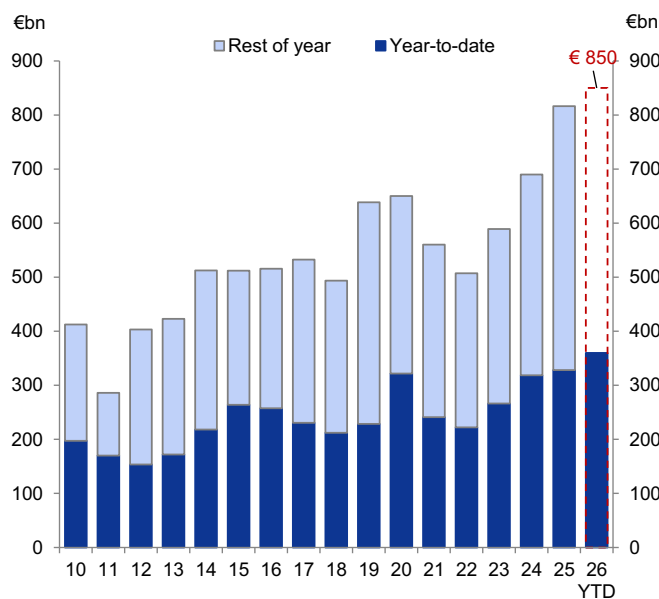


Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Primary market dashboard: EUR gross issuance

Exhibit 92: EUR IG gross issuance

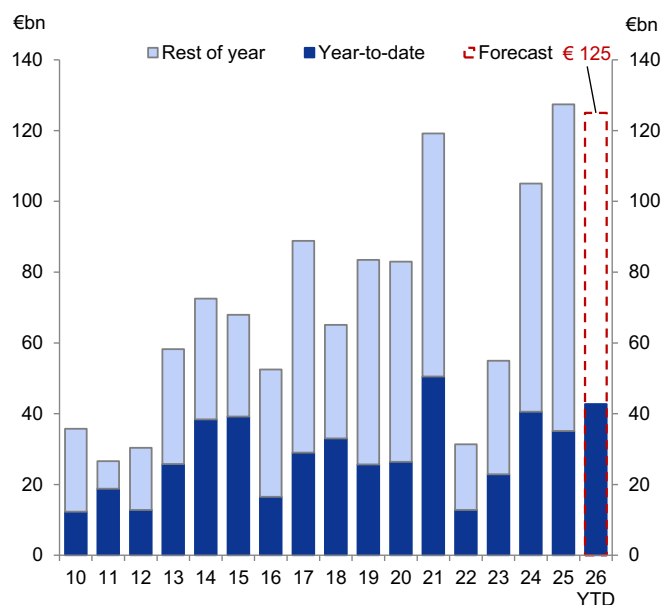
Historical EUR investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 93: EUR HY gross issuance

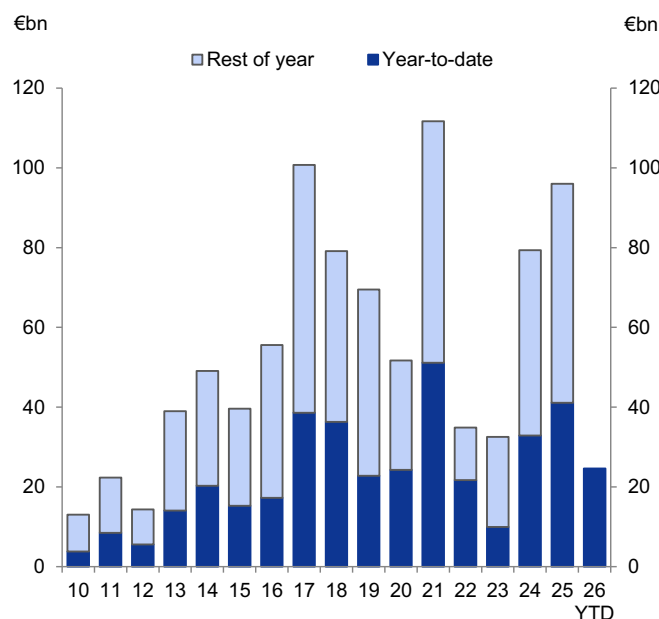
Historical EUR high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 94: EUR Leveraged loan gross issuance

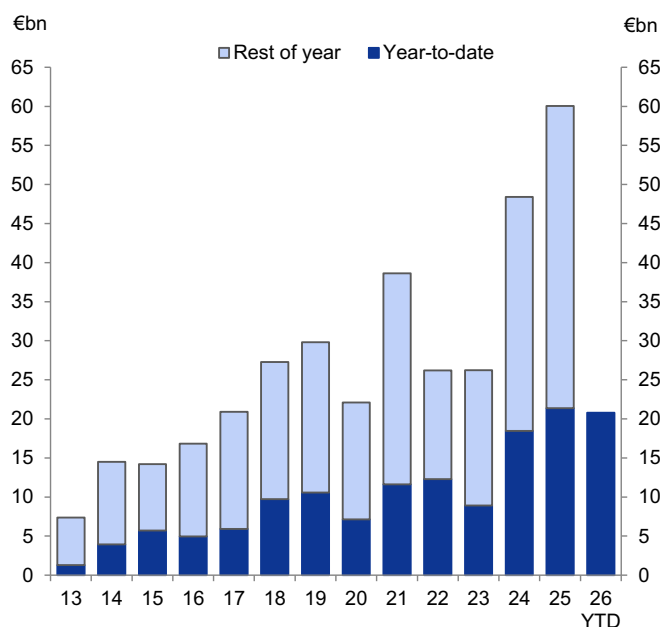
Historical EUR leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 95: EUR CLO creation

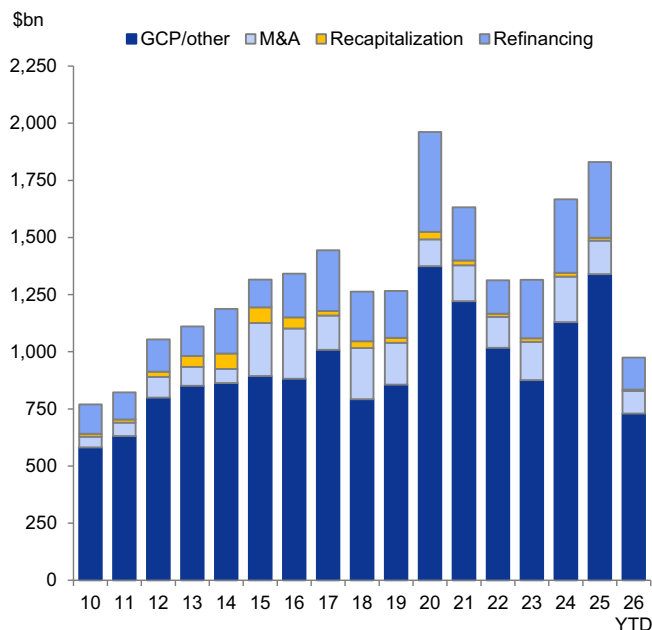
Historical EUR CLO creation broken: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

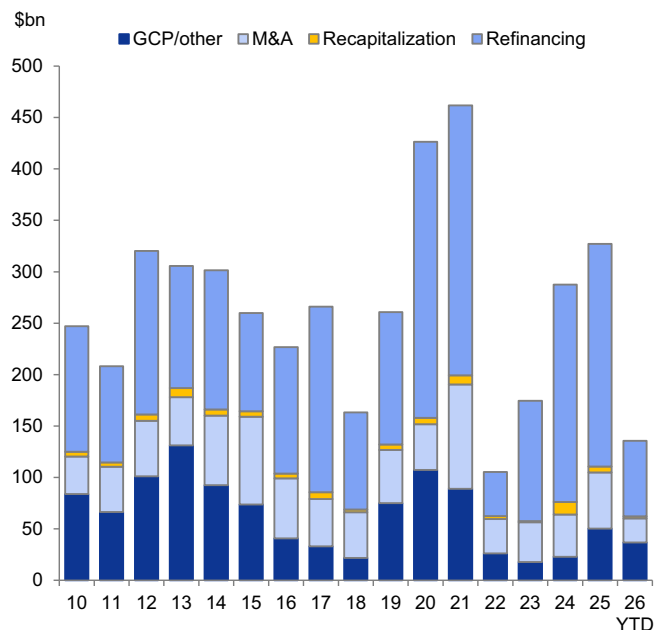
Primary market dashboard: Use of proceeds

Exhibit 96: Annual USD IG issuance volumes by use of proceeds



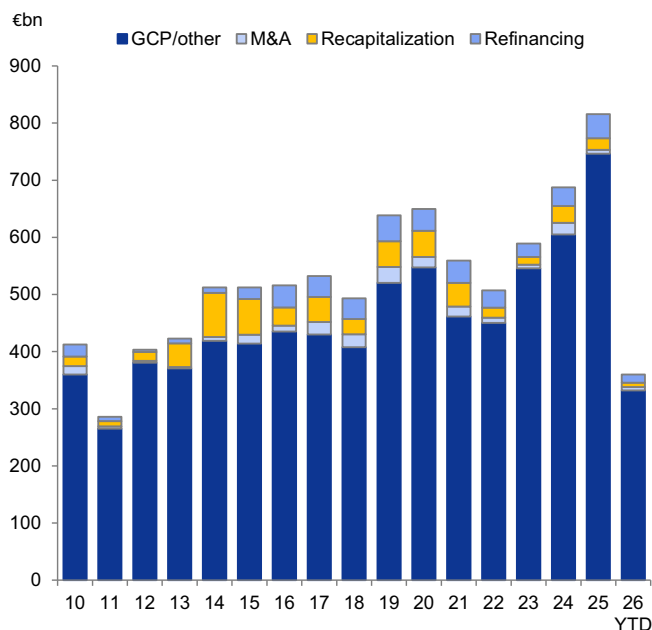
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 97: USD HY issuance volumes by use of proceeds



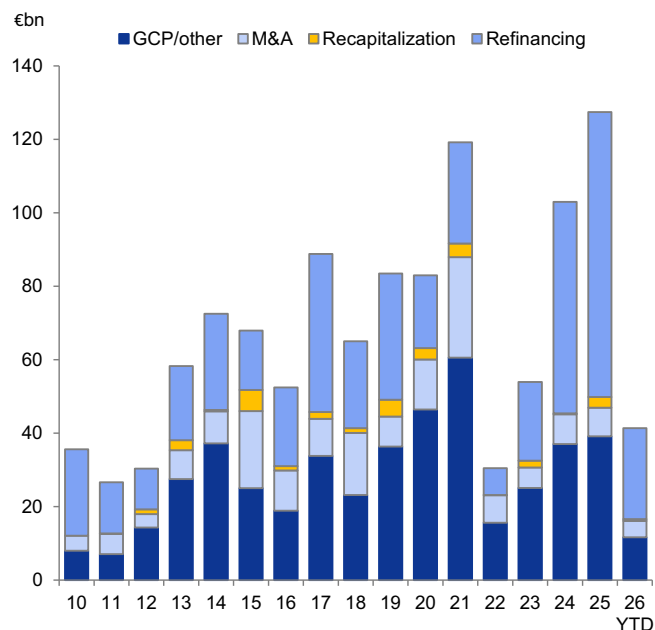
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 98: EUR IG issuance volumes by use of proceeds



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 99: EUR HY issuance volumes by use of proceeds

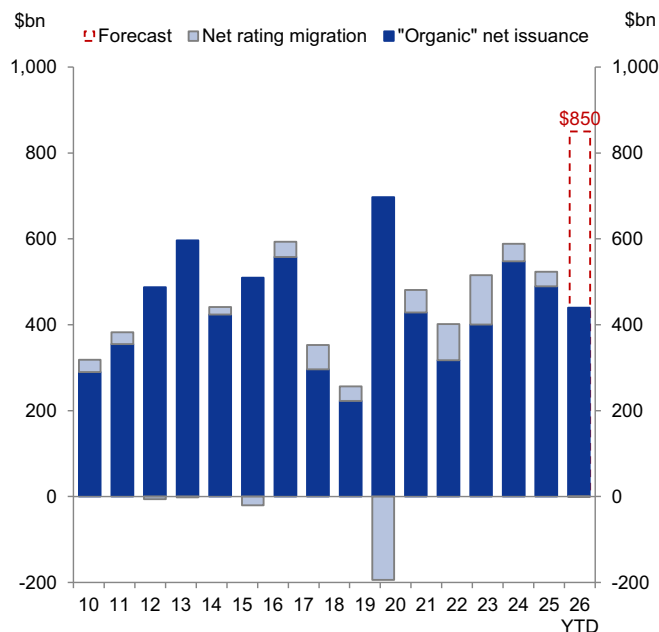


Source: Dealogic, Goldman Sachs Global Investment Research

Primary market dashboard: Net issuance

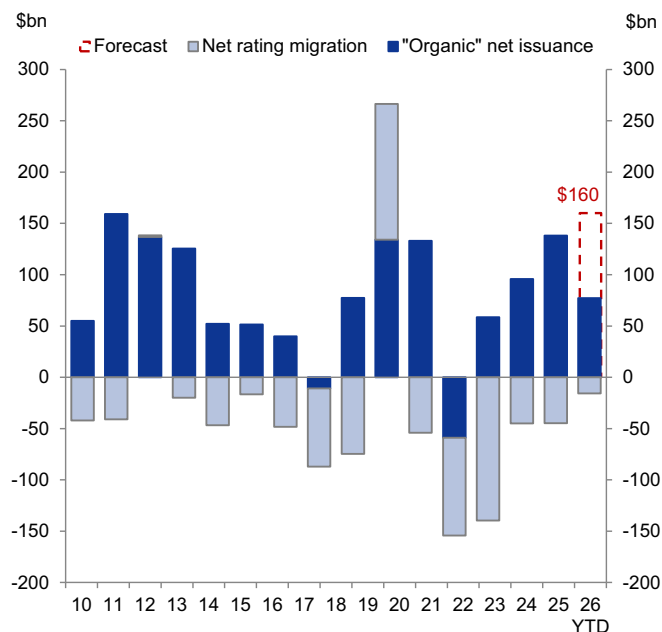
Note: Decomposition of the change in par value outstanding into net ratings migration vs. "organic" net issuance (gross issuance minus redemptions/matured bonds)

Exhibit 100: Annual USD IG net issuance



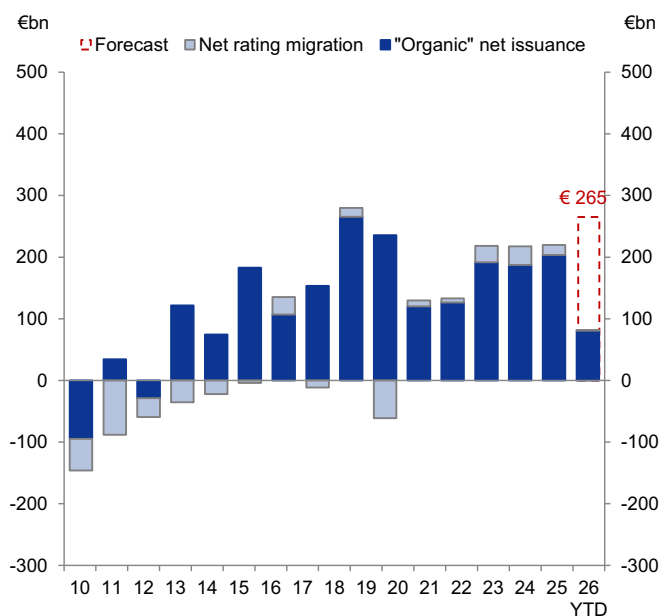
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 101: Annual USD HY net issuance



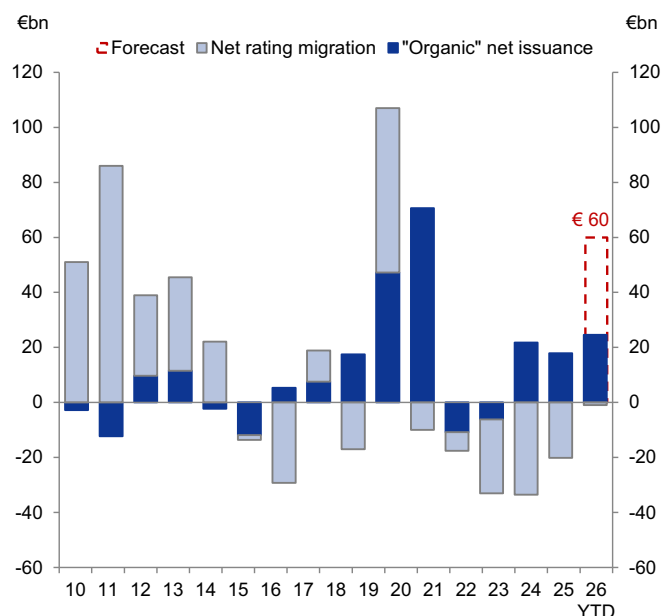
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 102: Annual EUR IG net issuance



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 103: Annual EUR HY net issuance

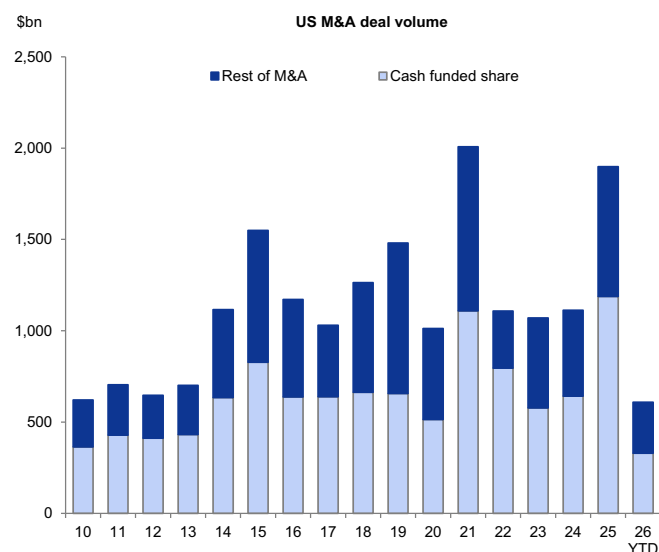


Source: Bloomberg, Goldman Sachs Global Investment Research

M&A volume

Exhibit 104: Annual US M&A deal volume

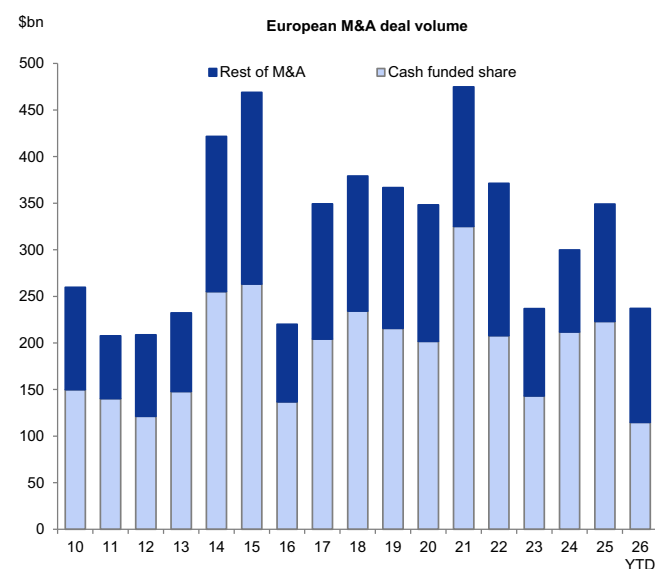
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 105: Annual European M&A deal volume

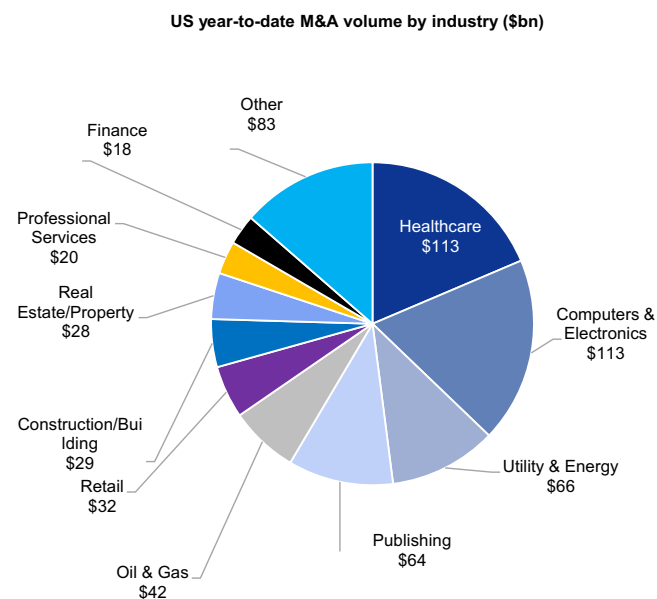
Includes strategic and sponsored deals



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 106: Year-to-date US M&A volume by sector

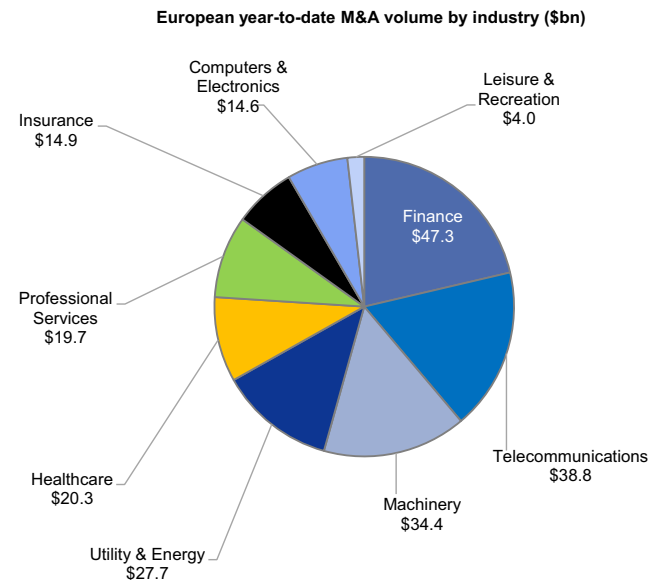
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 107: Year-to-date European M&A volume by sector

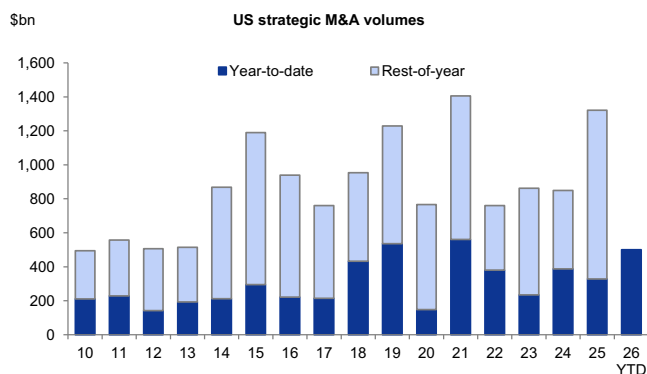
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

M&A volume (Strategic deals only)

Exhibit 108: Annual US strategic M&A volumes



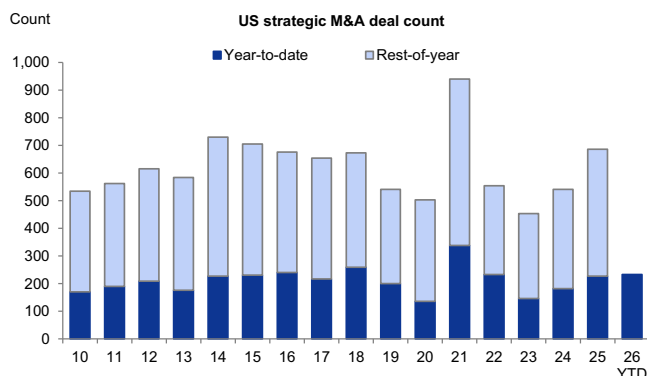
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 109: Annual European strategic M&A volumes



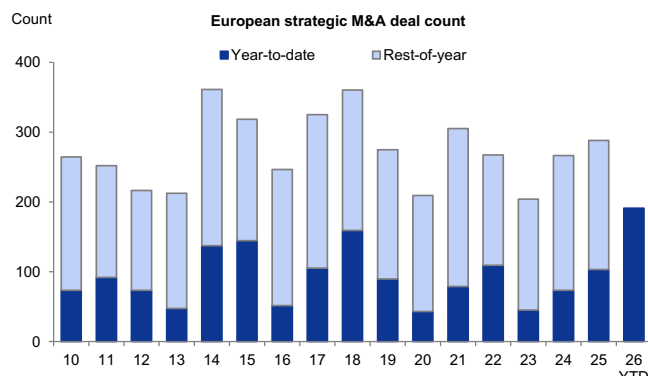
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 110: Annual US strategic M&A deal count



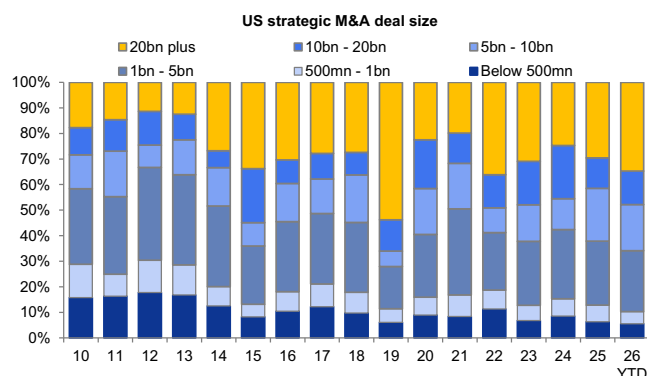
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 111: Annual European strategic M&A deal count



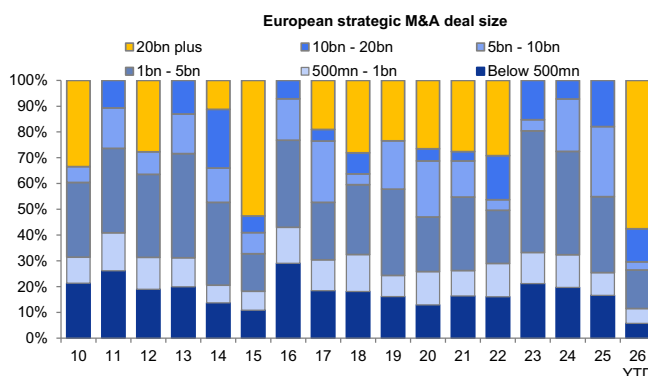
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 112: Annual US strategic M&A deal size



Source: Dealogic, Goldman Sachs Global Investment Research

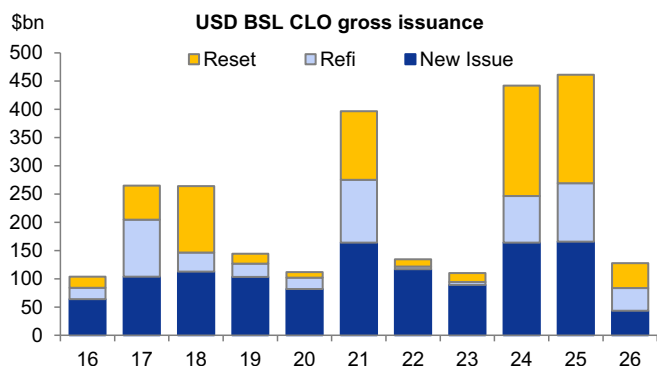
Exhibit 113: Annual European strategic M&A deal size



Source: Dealogic, Goldman Sachs Global Investment Research

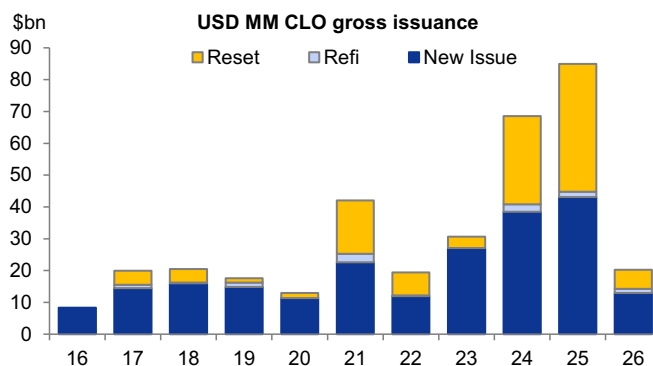
Primary market dashboard: CLO creation and refinancing

Exhibit 114: Annual USD Broadly syndicated loan CLO creation



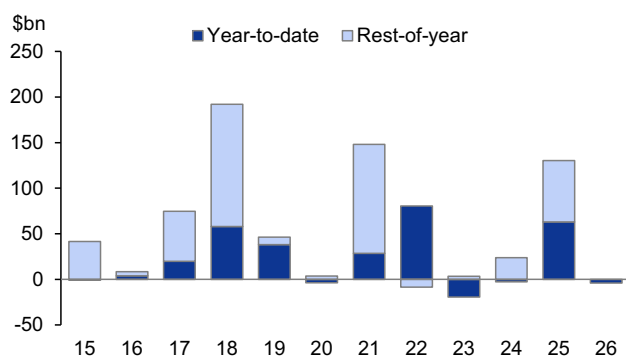
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 115: Annual USD Middle market CLO creation



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

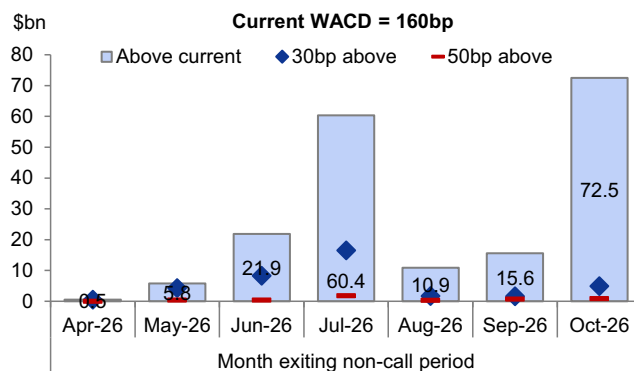
Exhibit 116: Annual USD leveraged loan net issuance



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 117: USD CLO refinance monitor

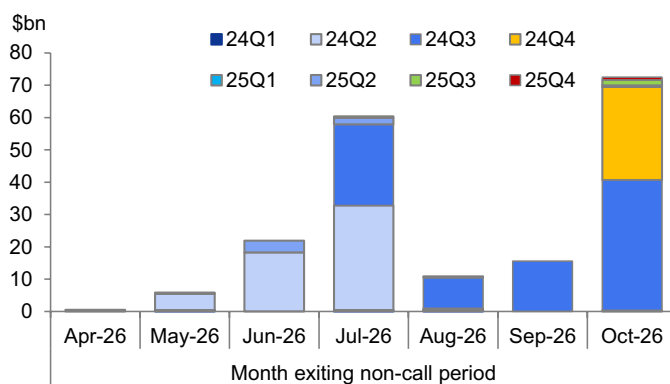
Upcoming notional above prevailing WACD



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 118: USD CLO refinance monitor

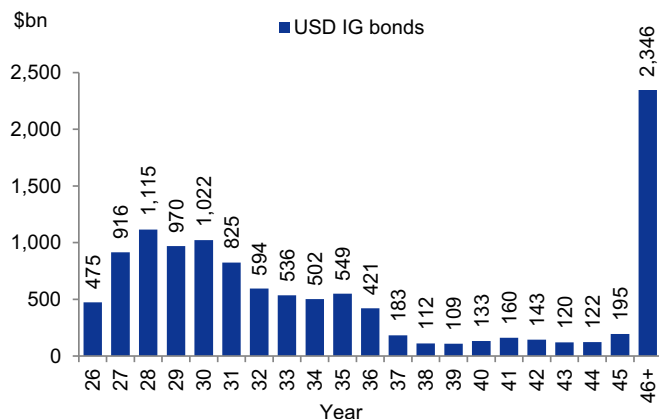
Notional above prevailing WACD by origination quarter



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

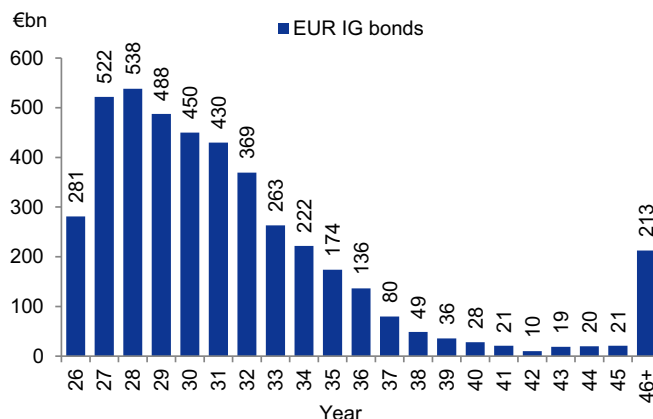
Primary market dashboard: Maturity walls

Exhibit 119: USD IG bond market maturity wall



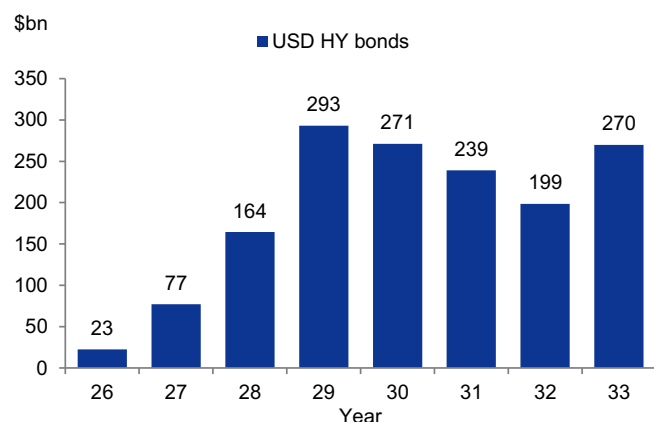
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 120: EUR IG bond market maturity wall



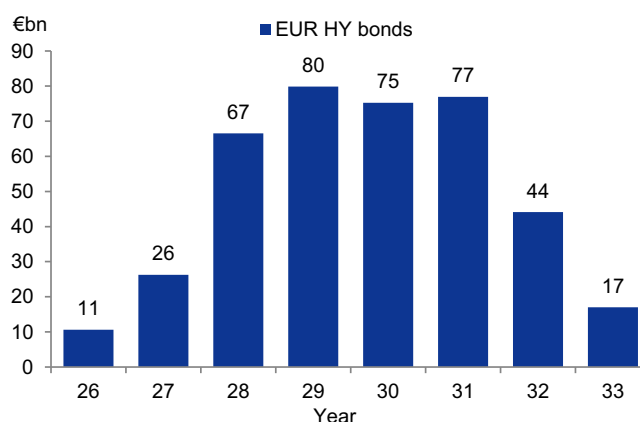
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 121: USD HY bond market maturity wall



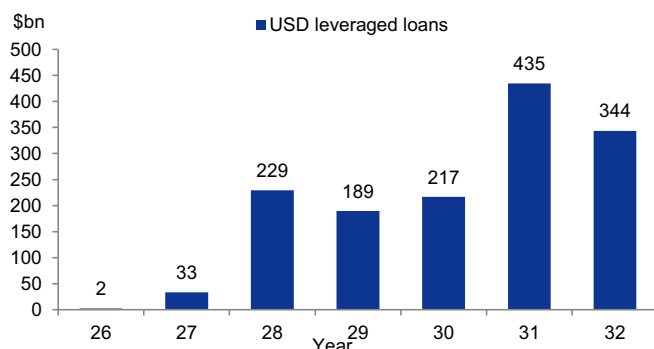
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 122: EUR HY bond market maturity wall



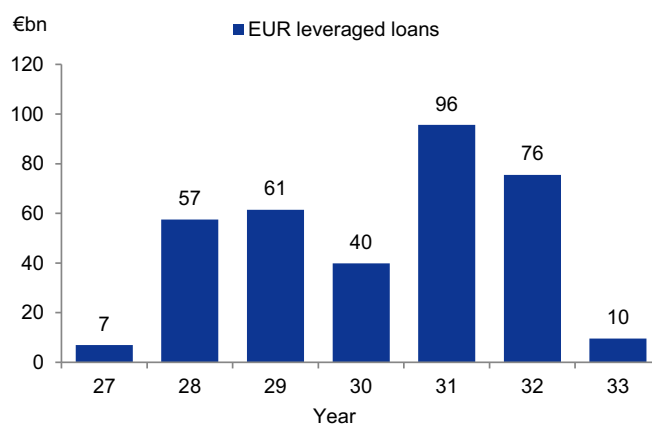
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 123: USD leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 124: EUR leveraged loan market maturity wall

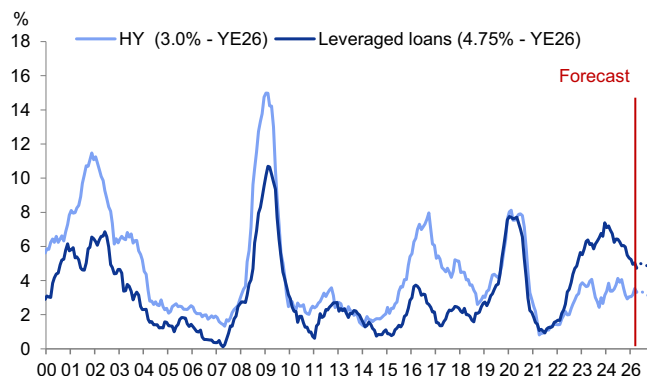


Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Default and ratings migration dashboard: Default trends

Exhibit 125: Default rates for US HY bond and leveraged loan issuers

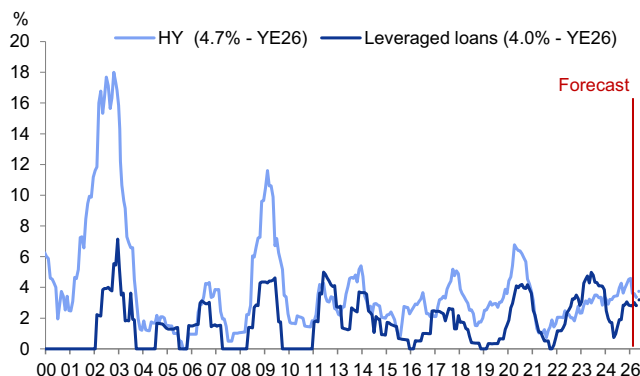
12-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 126: Default rates for European bond and leveraged loan issuers

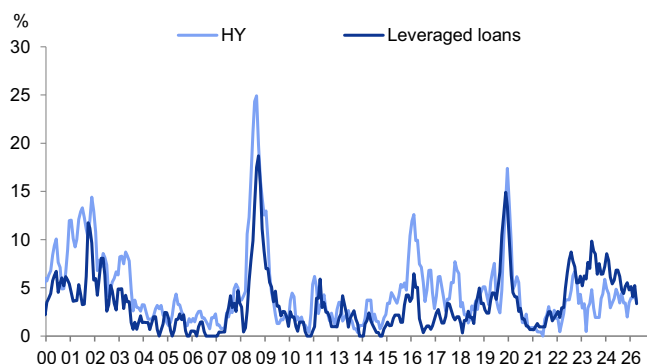
12-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 127: High frequency default rates for US HY bond and leveraged loan issuers

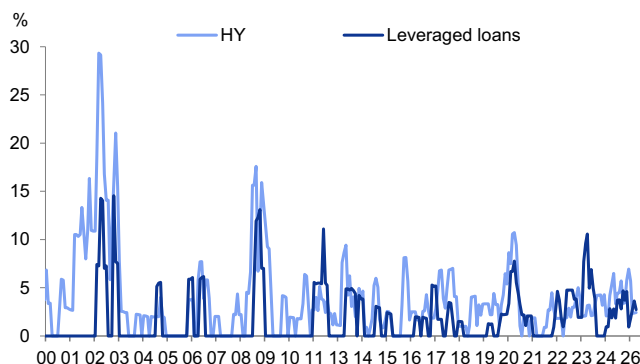
3-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

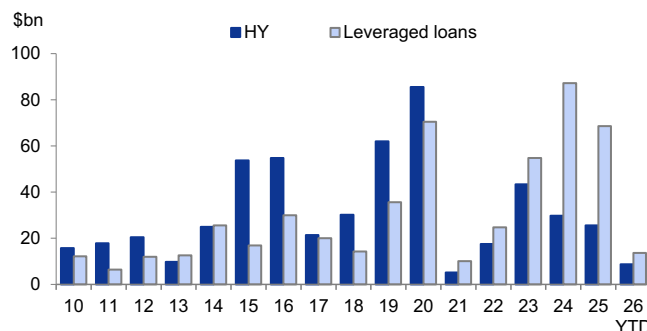
Exhibit 128: High frequency default rates for European HY bond and leveraged loan issuers

3-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



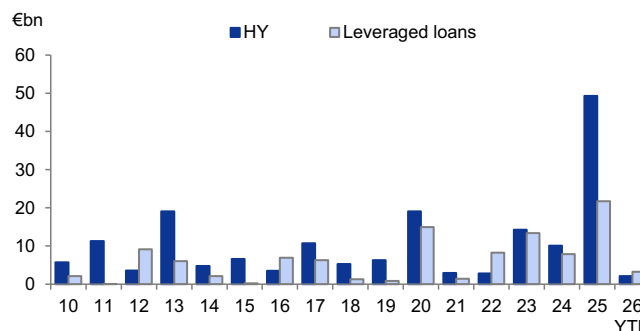
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 129: Annual par value amounts of defaulted USD HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 130: Par value amounts of defaulted EUR HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 131: Top 5 largest defaults in the USD HY bond and leveraged loan markets in the past 12 months

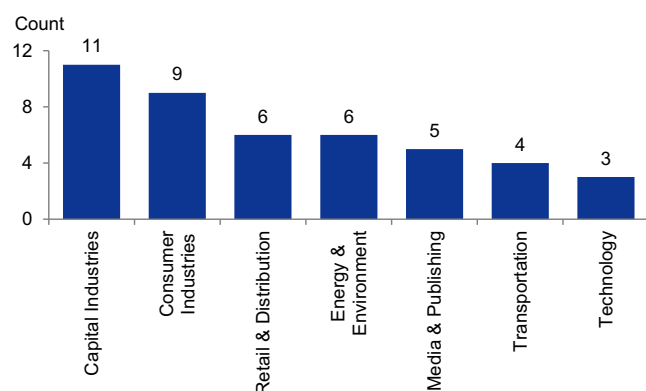
	Default date	Issuer	Industry	Default type	Notional amount (\$bn)
HY	1/29/2026	LABL, Inc.	Containers, Packaging, & Glass	Prepackaged Chapter 11	2.9
	11/20/2025	New Fortress Energy Inc.	Energy: Oil & Gas	Missed interest payment	2.7
	1/14/2026	Saks Global Enterprises LLC	Retail	Chapter 11	2.7
	8/8/2025	Saks Global Enterprises LLC	Retail	Distressed exchange	2.1
	6/30/2025	Radiate HoldCo, LLC	Telecommunications	Distressed exchange	1.9
Loans	9/28/2025	First Brands Group, LLC	Automotive	Chapter 11	4.7
	6/30/2025	Radiate HoldCo, LLC	Telecommunications	Distressed exchange	3.8
	5/30/2025	Quest Identity Intermediate Limited	High Tech Industries	Distressed exchange	3.2
	5/2/2025	Ivanti Software, Inc.	High Tech Industries	Distressed exchange	2.8
	2/17/2026	Vibrantz Technologies Inc.	Chemicals, Plastics, & Rubber	Distressed exchange	2.6

Source: Moody's, data compiled by Goldman Sachs Global Investment Research

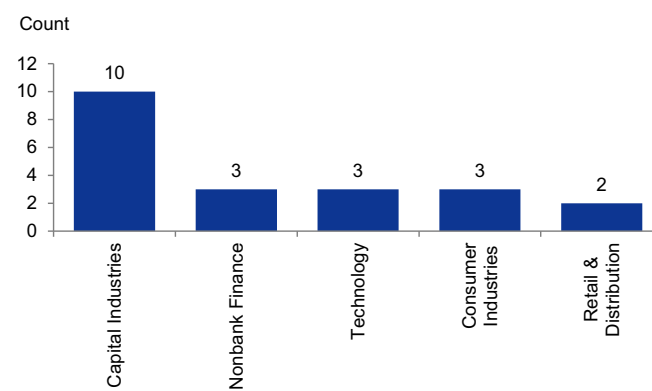
Exhibit 132: Top 5 largest defaults in the EUR HY bond and leveraged loan markets in the past 12 months

	Default date	Issuer	Industry	Default type	Notional amount (€bn)
HY	6/10/2025	Altice France SAS	Telecommunications	Suspension of payments	6.0
	6/16/2025	Altice France Holding S.A.	Telecommunications	Missed interest payment	2.1
	2/19/2026	Garfunkelux Holdco 3 S.A.	Finance	Distressed exchange	1.7
	6/25/2025	Garfunkelux Holdco 3 S.A.	Finance	Distressed exchange	1.7
	11/12/2025	Ardagh Group S.A.	Containers, Packaging, & Glass	Distressed exchange	1.4
Loans	6/10/2025	Altice France SAS	Telecommunications	Suspension of payments	3.5
	6/3/2025	Colisee Group	Healthcare & Pharmaceuticals	Missed interest payment	1.3
	1/29/2026	Eos Finco S.a.r.l.	Wholesale	Distressed exchange	1.3
	5/9/2025	Eos Finco S.a.r.l.	Wholesale	Distressed exchange	1.0
	11/4/2025	Klockner Pentaplast of America, Inc.	Containers, Packaging, & Glass	Prepackaged Chapter 11	0.8

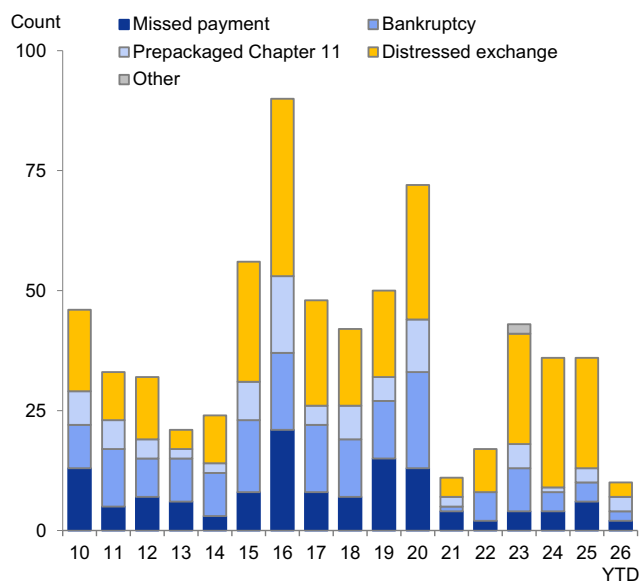
Source: Moody's, data compiled by Goldman Sachs Global Investment Research

Exhibit 133: Industry default count in the US

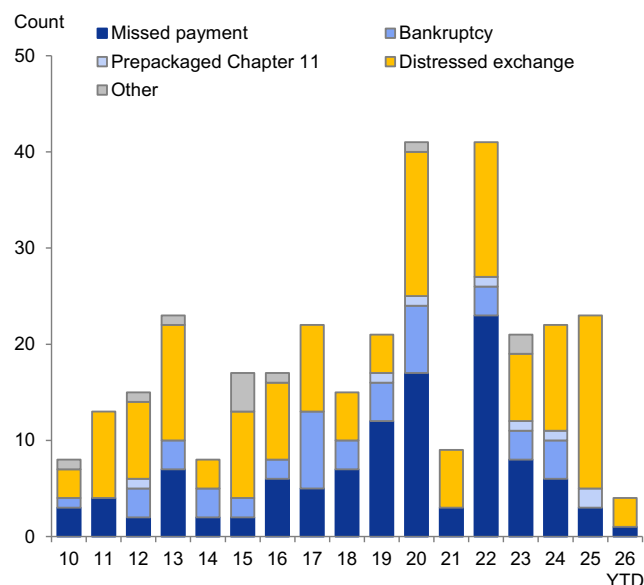
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 134: Industry default count in Europe

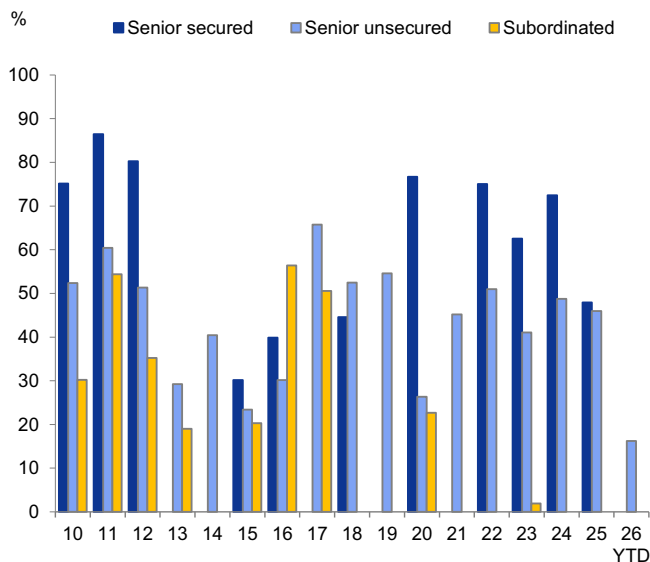
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 135: Composition of defaults in the USD HY bond market by type

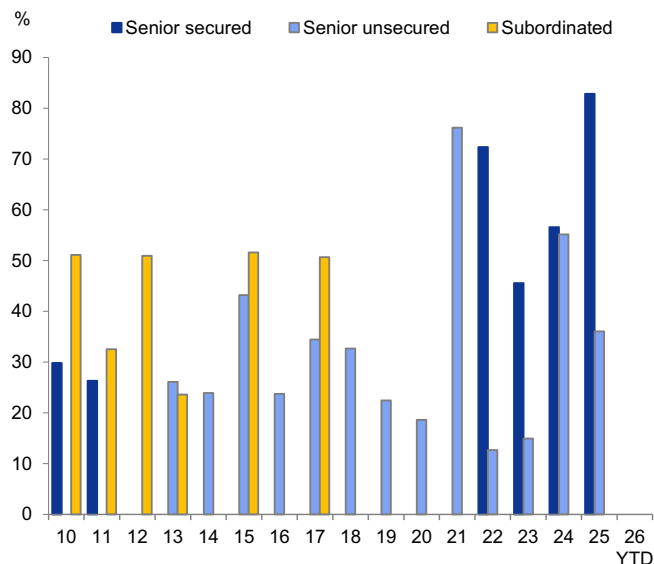
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 136: Composition of defaults in the EUR HY bond market by type

Source: Moody's, Goldman Sachs Global Investment Research

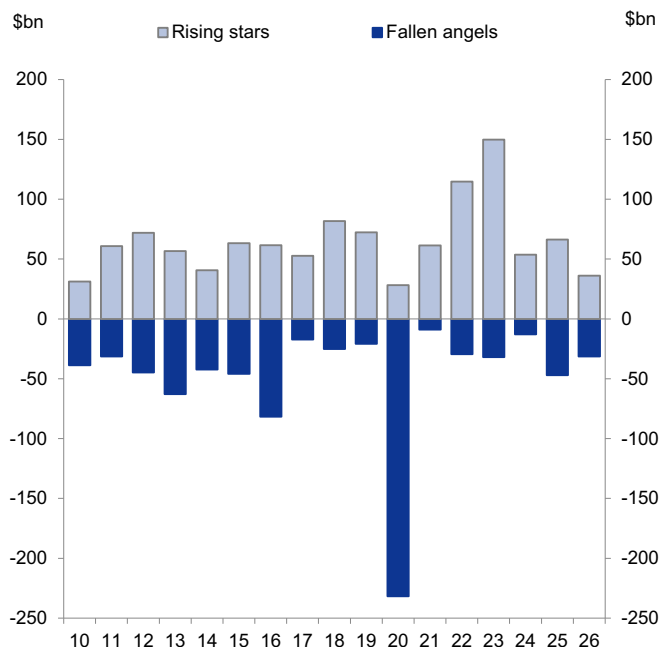
Exhibit 137: Annual average recovery rates by seniority for US issuers

Source: Moody's, Goldman Sachs Global Investment Research

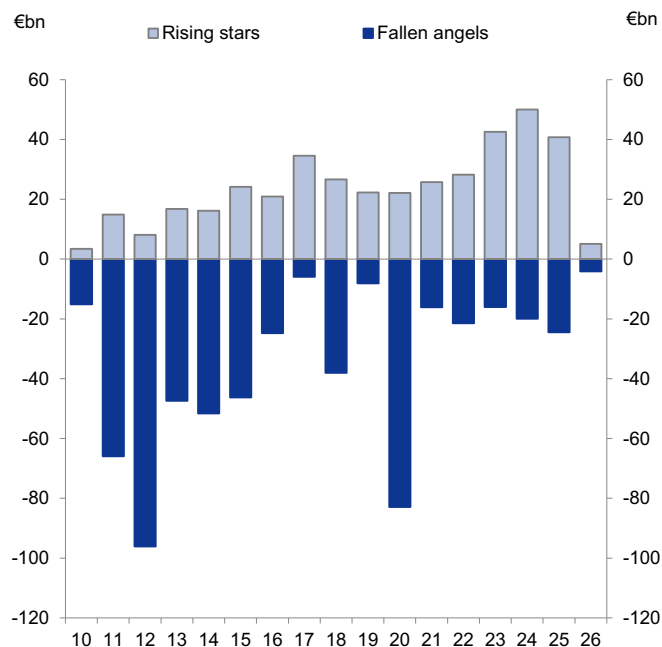
Exhibit 138: Annual average recovery rates by seniority for European issuers

Source: Moody's, Goldman Sachs Global Investment Research

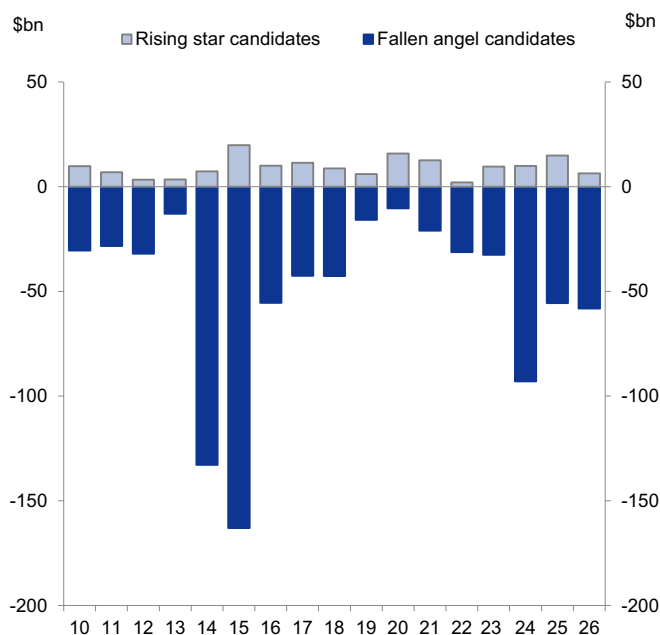
Default and ratings migration dashboard: Upgrades and downgrades

Exhibit 139: Fallen angels vs. rising stars in the USD market


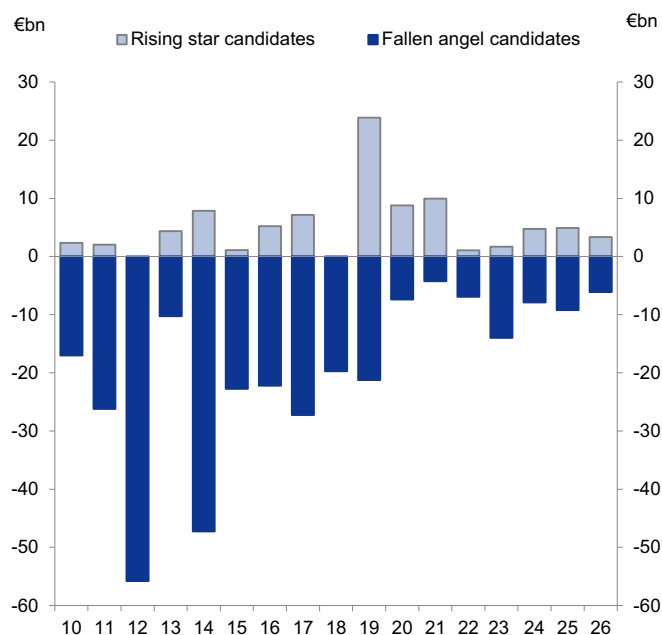
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 140: Fallen angels vs. rising stars in the EUR market


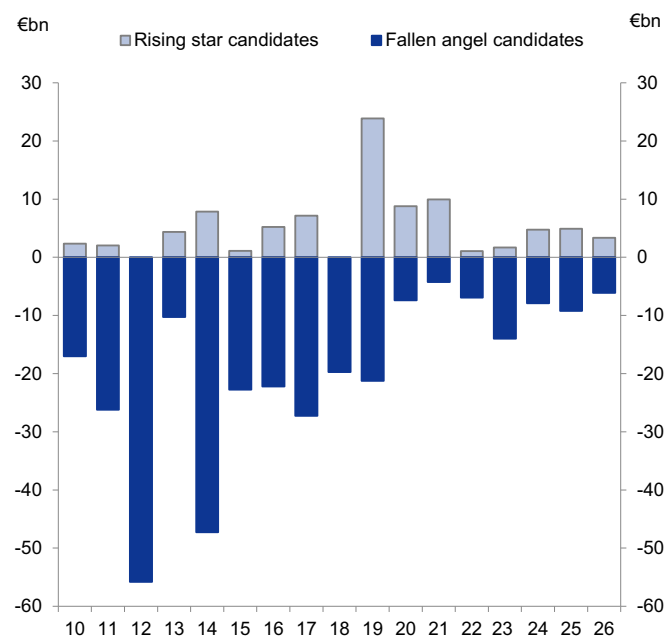
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 141: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the USD market


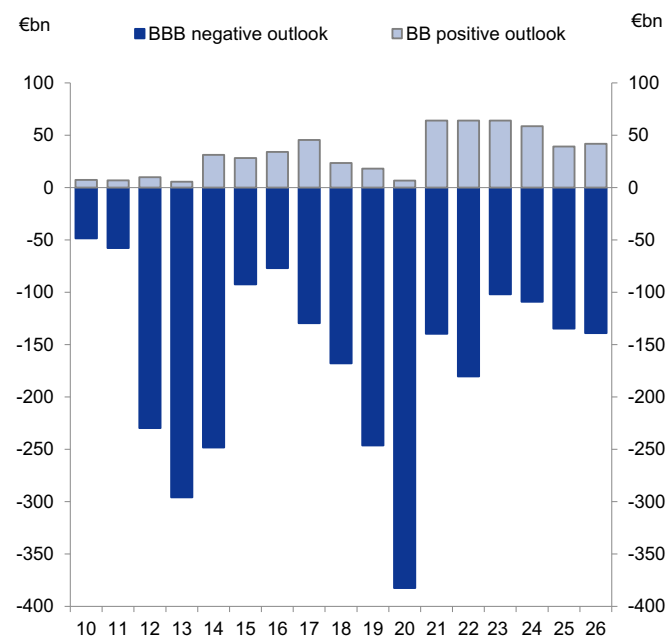
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 142: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the EUR market


Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 143: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the USD market

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 144: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the EUR market

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 145: Top 10 USD fallen angel issuers in the last 12 months

Company	Notional downgraded (\$bn)
Centene Corp	14.7
Paramount Global	13.0
Fs Kkr Capital Corp	3.8
Fmc Corp	2.5
Huntsman International L	1.5
Banco Santander Mexico	1.3
Pacificorp	0.9
Blackrock Tbp Cap Corp	0.7
Horizon Mutual Holdings	0.6
Ses Americom Inc	0.5

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 146: Top 10 USD rising star issuers in the last 12 months

Company	Notional upgraded (\$bn)
Visra Operations Co Llc	6.3
Medline Borrower Lp	4.5
Permian Resourc Optg Llc	3.6
Unicredit Spa	3.5
Frontier Communications	2.8
Frontier Communications	2.8
Constellation En Gen Llc	2.3
Standard Chartered Plc	1.5
Medline Borrower/Med Co	1.5
Midcap Financial Issr Tr	1.0

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 147: Top 10 EUR Fallen angel issues

Company	Notional downgraded (€bn)
Ses Sa	3.4
Lanxess Ag	2.2
Worldline Sa/France	2.2
Citycon Treasury Bv	1.7
Verallia Sa	1.6
Mediobanca Di Cred Fin	1.5
Unibail-Rodamco-Westfld	1.1
Adecco Int Financial Svs	0.5
Illimity Bank Spa	0.3
Solaben Luxembourg	0.1

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 148: Top 10 EUR rising star issuers in the last 12 months

Company	Notional upgraded (€bn)
Banca Monte Dei Paschi S	3.0
Terna Rete Elettrica	1.9
Aib Group Plc	1.7
Asr Nederland Nv	1.5
Unipol Assicurazioni Spa	1.3
Intesa Sanpaolo Assicura	1.1
Bper Banca	1.0
Piraeus Bank Sa	1.0
Czechoslovak Group	1.0
Abanca Corp Bancaria Sa	1.0

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

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